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Entrainment of Fine Particles in Froth Flotation

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Abstract

Entrainment has become a significant issue nowadays with fine grinding being used to liberate valuable minerals from low grade and complex ores to increase “true flotation”. Entrainment results in large amounts of fine gangue materials reporting to flotation concentrates along with recovered valuable minerals and reduces the final product quality. In froth flotation, entrainment is the dominant recovery mechanism for fine gangue mineral particles, and it is influenced by a number of factors in both the pulp and froth phase, including particle properties (e.g. particle size and density) and flotation operational variables (e.g. gas flowrate, froth depth, impeller speed and reagent dosage).

A number of models incorporating different factors have been developed to estimate entrainment recovery. Most of these models relate entrainment recovery to two parameters: the water recovery to concentrate, and the degree of entrainment, a classification factor to account for the degree of drainage of entrained particles with respect to the water in the pulp and froth phases of a flotation cell. However, there are still significant barriers to be overcome before they can be applied for routine estimation of entrainment in industrial applications. There is no consensus as to the variables which significantly affect the entrainment process and should therefore be included in the model as well as to the mechanisms involved.

This thesis investigated the effect of flotation operational factors and particle properties on the entrainment parameters (i.e. degree of entrainment and water recovery) and the mechanisms underpinning the observed effects. The objective of the thesis was to develop a comprehensive understanding of the key operating conditions and particle attributes influencing the entrainment of gangue minerals in flotation, which should enable the development of a more sophisticated predictive entrainment model by more accurately predicting the degree of entrainment and water recovery.

A two level factorial experimental program was adopted to investigate the effect of impeller speed, gas flowrate, froth height and specific gravity of gangue minerals on the degree of entrainment. Batch flotation experiments were performed in a 3.5 L conventional Agitair flotation cell using a feed which was a mixture of liberated chalcopyrite and one of two liberated gangue minerals, quartz and hematite. Results show that there was no direct correlation between the degree of entrainment and water recovery, indicating that the key drivers of water recovery were not the same as those that affected the degree of entrainment.

The degree of entrainment was significantly influenced by particle size, particle density, frother and the interaction between gas flowrate and froth height. The mechanisms underpinning the effect of these significant influencing factors were investigated. Results suggest that the classification of gangue mineral particles mainly occurs at the pulp/froth interface region. Whether a particle moves upward into the froth and subsequently follows the water phase depends on whether the drag force acting on the solids is greater than the apparent immersed weight of the particles which promotes particle settling. Particle size and density are two variables that affect apparent immersed weight and drag force. Frother, gas flowrate and froth height can alter the liquid velocity at the interface and hence the drag force, and therefore also have a significant impact on the degree of entrainment.

The results from the two level factorial experiment were also analysed with respect to water recovery. It was found that water recovery was significantly affected by (1) froth height, (2) gas flowrate, (3) impeller speed, (4) the interaction between impeller speed and particle density, and (5) the interaction between gas flowrate and froth height (in the order of reducing effects). These effects resulted from these factors affecting the water motion in a flotation cell: water drainage in the froth phase and water flow into the froth from the pulp phase. Moreover, the results show that water recovery could be modelled effectively using an exponential function, incorporating water drainage and water entering into the froth.

The outcomes of this work can provide plant operators with information they can use to minimise entrainment in their flotation operations as well as providing relationships that could be used to develop an improved predictive entrainment model.

Declaration by author

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Chapter 1 Introduction

Entrainment has become a significant issue nowadays with fine grinding required to liberate the valuable minerals from low grade and complex ores. Entrained gangue minerals that report to the concentrate with the recovered valuable minerals in the flotation process greatly reduce the quality of flotation concentrates. This thesis aims to develop a comprehensive understanding of the key operating conditions and particle attributes which influence the entrainment of gangue minerals in flotation. This study enables the development of improved empirical entrainment models which can be used to quantify the effect of entrainment on concentrate grade and provide flotation plants with a tool to identify and investigate strategies for minimizing adverse entrainment effects.

1.1 Problem statement

Froth flotation is a technique for physically separating minerals which utilises the difference in the surface properties of valuable and gangue minerals, and is widely used for processing various ores. Ideally, in a flotation cell, hydrophobic valuable mineral particles collide with air bubbles and form particle-bubble aggregates moving upwards against gravity to the froth from the pulp phase while hydrophilic gangue mineral particles report to the tailings (Yianatos et al., 2009).

With large quantities of difficult to treat low-grade ores being processed throughout the world nowadays, it is difficult for flotation operations to maintain flotation efficiency. To liberate valuable minerals from low-grade ores to increase “true flotation”, fine grinding is often used. By doing so, large amounts of fine gangue mineral particles are also generated. These fine gangue mineral particles report to the concentrate along with the recovered valuable mineral particles leading to significant deterioration in the concentrate grade. This phenomenon is known as entrainment.

Entrainment is the mechanism primarily responsible for the recovery of fine gangue mineral particles in flotation. It is a mechanical transfer process by which mineral particles suspended in water enter the flotation froth, move upwards, and finally leave the flotation cell with the mineral particles recovered by true flotation. The recovery of gangue materials by entrainment is determined by the water recovered as well as the degree of entrainment which is a classification function representing the degree to which entrained particles drain relative to water in the froth and the state of entrained solid particle suspension in the pulp (Johnson, 1972; Savassi et al., 1998; Zheng et al., 2005).

To date, a number of models have been proposed to estimate and predict entrainment with the objective of better controlling or predicting flotation performance (Bisshop an White, 1976; Moys, 1978; Ross and Van Deventer, 1988; Kirjavainen, 1992; Maachar and Dobby, 1992; Savassi et al., 1998; Neethling and Cillers, 2002a; Neethling and Cillers, 2009; Yianatos and Contreras, 2010). These models range from the empirical and semi-empirical to the fundamental, developed for either flotation columns or mechanical flotation cells. Of them, some were developed to directly predict the entrainment flow, while the others predict entrainment through the prediction of the degree of entrainment and water recovery (Zheng et al., 2006b). The following shows the current AMIRA P9 model used for predicting the degree of entrainment which has been widely used in industrial applications (Savassi et al., 1998):

$$ENT_i = \frac{2}{\exp\left[2.292\left(\frac{d_i}{\zeta}\right)^{adj}\right] + \exp\left[-2.292\left(\frac{d_i}{\zeta}\right)^{adj}\right]} \quad 1\text{-}1$$

$$adj = 1 - \frac{\ln(1/\delta)}{\exp(d_i/\zeta)} \quad 1\text{-}2$$

where d_i is the particle size (μm), ξ is the entrainment parameter defined as the particle size at which the degree of entrainment is 0.20, and δ is the drainage parameter which represents the relative drainage of the fine to the coarse particles.

However, a number of barriers remain and limit the applicability of these models for predicting entrainment in various industrial applications. The fundamental models include a large number of variables, many of which are difficult to measure accurately. In comparison, the empirical and semi-empirical models have proven to be more practical for routine prediction, as they are simple equations with easily-measured variables. However, there is no consensus as to the variables which should be included in these models, as evidenced by the different forms of equations used in the abovementioned entrainment models. It should be emphasized that a requirement for a good model is that it should describe the basic phenomena and account for the factors affecting the response significantly (Bisshop, 1974). It is not clear, however, which factors have a significant enough effect to warrant inclusion in a predictive entrainment model. Moreover, the underpinning mechanisms by which these significant factors affect entrainment have not, experimentally, been comprehensively studied. To develop a more sophisticated predictive and practical entrainment model, therefore, more work is required.

1.2 Thesis objectives

The objective of this study is to develop a comprehensive understanding of the key operating conditions and particle attributes affecting the entrainment of gangue minerals in flotation, which will enable the development of improved entrainment models that can be used to quantify the effect of entrainment on concentrate grade and provide flotation plants with a tool to identify strategies for minimizing adverse entrainment effects.

The thesis mainly focuses on the following two aspects:

- Identifying the significant operational variables and particle attributes affecting the degree of entrainment and water recovery;
- Investigating the mechanisms underpinning the effects of the identified variables to enable better models to be developed to predict the degree of entrainment and water recovery.

1.3 Thesis outline

Following this introduction to the problem, **Chapter 2** gives a detailed review of entrainment in froth flotation. The mechanisms of the entrainment process are reviewed, the potential factors affecting entrainment are discussed, and the methods for the quantification of entrainment and true flotation are summarised. Current entrainment models in the literature are also reviewed in terms of their suitability for use in industrial applications. The literature review provides an overall picture of the current status of studies in this area and pinpoints the gaps in knowledge which are presented at the end of this Chapter.

Chapter 3 describes the materials, reagents and experimental methodologies.

Chapter 4 identifies the significant operational variables and particle attributes affecting the degree of entrainment and concludes that particle density, frother and the interaction between gas flowrate and froth height (in addition to particle size) are the most significant factors affecting the degree of entrainment. This work led to the investigations in **Chapters 5**, **6**, **7** and **8**.

Chapter 5 investigates the mechanism by which particle size and particle density affect the degree of entrainment and develops a predictive and practical empirical *ENT* (the degree of entrainment) equation incorporating the proposed mechanism.

Chapter 6 investigates the mechanism by which frother type and concentration affect the degree of entrainment in froth flotation.

Chapter 7 investigates the mechanism by which gas flowrate and froth height affect the degree of entrainment in froth flotation.

Chapter 8 presents the observed effect of the investigated variables on water recovery, and proposes that water recovery can be effectively modelled using the mechanisms underpinning the effect: water drainage and water flow into a froth.

Chapter 9 concludes the thesis with a summary of the major contributions to knowledge arising from the thesis work, and makes recommendations as to the future work required to improve the modelling of entrainment.

Chapter 2 Literature review

A comprehensive and critical review of the literature was conducted to provide a summary of the current state of knowledge regarding the entrainment of gangue mineral particles in froth flotation as well as to justify the problem and research objectives stated in ***Chapter 1****.*

This chapter provides a review of the mechanisms of the entrainment process, discusses the potential significant factors affecting entrainment, summaries the methods for the quantification of entrainment, and reviews the current entrainment models in terms of range of application and practicality of use.

2.1 Introduction

The objective of this chapter is to summarise the findings of a comprehensive and critical review of the entrainment literature performed to identify the research gaps and develop the hypotheses for this thesis which aims to develop a comprehensive understanding of the key operating conditions and particle attributes that affect the entrainment of gangue minerals in flotation. Initially, a detailed description of the entrainment mechanism is provided. This is followed by a review of the literature on various factors that affect entrainment. The methods of determining true flotation and entrainment as well as the existing entrainment models are then presented and discussed. At the end of this chapter the literature on entrainment is summarized with respect to the objectives of this thesis.

2.2 Entrainment in froth flotation

In mineral processing, froth flotation is a technique used to separate and concentrate ores for economical smelting or further hydrometallurgical treatment (Crozier, 1992). Froth flotation involves three phases, i.e. solid particles, water and air bubbles. Selective wetting of the particles in the feed enables separation. Reagents are added to adsorb, either physically or chemically, with one of the types of particles in the feed, which will provide the thermodynamic requirement for the particles to bind to the surface of a bubble. Utilising the difference in the surface properties of minerals, flotation ideally results in the hydrophobic valuable mineral particles suspended in the pulp attaching to air bubbles and forming particle-bubble aggregates which move upwards against gravity into the froth and then flow into the concentrate launder. Most of the hydrophilic gangue mineral particles, however, remain in the suspension in the cell reporting to the tailings.

Froth flotation is extensively used for processing base-metal ores (e.g. sulphide ores and oxide ores). Nowadays, low-grade ores in large quantities are being processed using flotation. This shift to lower grades ores over time is due to various factors including major improvements in metallurgical techniques, a movement to high volume and lower cost extraction technologies as well as the economic advantages of extending the life of older mines over finding and establishing new mines (West., 2011). Low-grade ores are generally difficult to process because they are commonly complex and valuable minerals are often associated with gangue minerals at even fine particle sizes. To increase the recovery of valuable minerals in complex ores, fine grinding is required. By doing so, however, large amounts of fine gangue mineral particles are observed to report to the

concentrate along with the recovered valuable mineral particles, leading to deterioration in the concentrate grade.

Entrainment is the primary mechanism for the recovery of gangue minerals especially when processing ores with a large proportion of fine particles, although in industry there are other mechanisms such as the recovery via composite particles and slime coatings as well as entrapment which can also contribute to gangue recovery (Fuerstenau, 1980; Kirjavainen, 1996). Entrainment is a transfer process by which mineral particles suspended in water enter the flotation froth, move upwards, and finally leave the flotation cell with the valuable mineral particles recovered by true flotation. In a flotation cell, it occurs simultaneously along with true flotation. Unlike true flotation, entrainment is not chemically selective and it occurs without a direct attachment of particles to bubbles. As a result, both valuable and gangue minerals experience entrainment. Figure 2-1 describes the mass transfer throughout a flotation cell. The mass flows indicated in the figure are:

1) Transfer of valuable mineral particles to the froth from the pulp by true flotation.
2) Transfer of valuable mineral particles to the concentrate from the froth by true flotation.
3) Transfer of mineral particles to the froth from the pulp by entrainment.
4) Transfer of entrained mineral particles to the concentrate from the froth by entrainment.
5) Transfer of mineral particles from the froth to the pulp due to the drainage of detached particles and entrained particles.

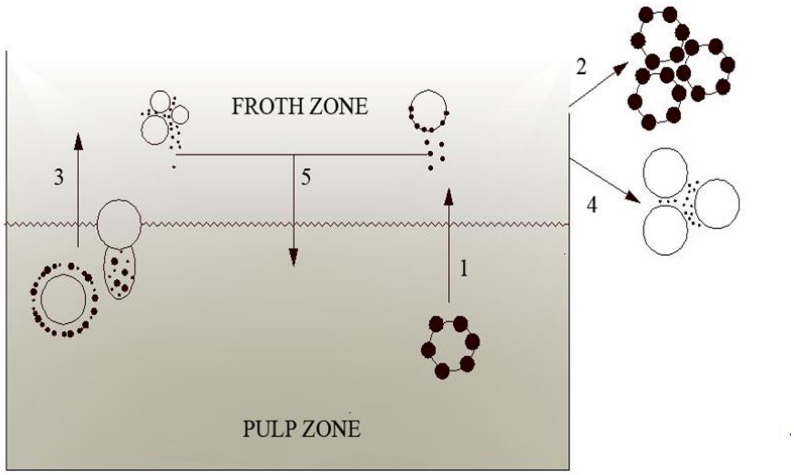

Figure 2-1 Mechanisms of transfer of fully liberated mineral particles in a flotation cell (after Savassi, 1998).

Since entrainment has a detrimental effect on the grade of the concentrate, a number of studies have been carried out to better understand the entrainment mechanisms, identify factors affecting

entrainment and develop models with an objective of predicting entrainment in a flotation cell. These will be critically reviewed in the following sections.

2.3 Entrainment mechanisms

Entrainment is generally considered as a two-step process. In step 1, mineral particles ascend upwards into the froth phase from the region below the pulp/froth interface, and in step 2, entrained particles in the froth are transferred to the concentrate in water.

Entrainment starts in the pulp phase. In conventional flotation cells, the pulp phase is usually divided into two regions, that is, a turbulent zone and a quiescent zone. In the turbulent zone, air is drawn into the flotation cell and turned into swarms of small bubbles dispersed in all directions within the whole turbulent zone by an impeller. These bubbles adsorb frothers and collide with valuable mineral particles rapidly. Then they ascend upwards to the quiescent zone, and ultimately become metastable. When these bubbles reach the region below the pulp/froth interface, hydrophilic gangue mineral particles start the transfer process to the froth via the entrainment mechanism.

To date, three mechanisms have been proposed to explain how mineral particles in the flotation cell travel across the pulp/froth interface from the pulp to the froth by entrainment. They are Boundary Layer Theory, Bubble Wake Theory and Bubble Swarm Theory.

In the Boundary Layer Theory, entrained mineral particles are transferred to the froth phase in the bubble lamella, i.e. the thin hydrodynamic layer of water surrounding the bubble (Gaudin, 1957; Moys, 1978; Hemmings, 1981; Bascur and Herbst, 1982), while in the Bubble Wake Theory, fine mineral particles suspended in the water are transferred to the froth phase in the wake of an ascending bubble (Smith, 1984; Yianatos et al., 1988). Figure 2-2 shows a schematic of how water and suspended mineral particles move from the region below the pulp/froth interface to the froth according to the Boundary Layer Theory and Bubble Wake Theory.

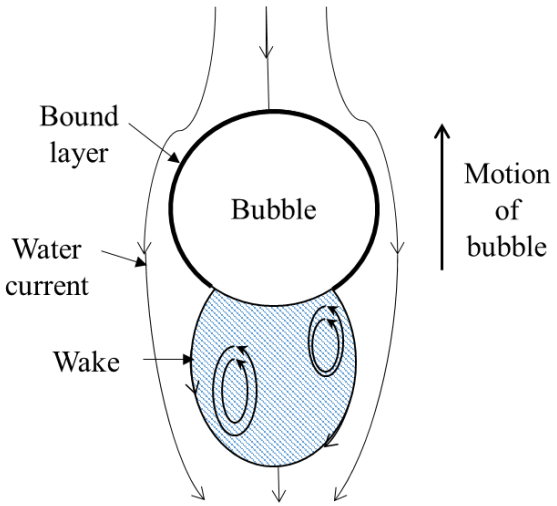

Figure 2-2 Schematic of the Boundary Layer Theory and Bubble Wake Theory (after Smith and Warren, 1989).

However, the Boundary Layer Theory and Bubble Wake Theory, alone, are not adequate to account for the total mass of mineral particles transported to the froth by entrainment. The Bubble Swarm Theory proposed by Smith and Warren (1989) provides an additional mechanism for transport. Figure 2-3 demonstrates how suspended mineral particles in the water transfer to the froth from the pulp phase based on the Bubble Swarm Theory. Figure 2-3(a) shows the bubbles rising up through the cell. Figure 2-3(b) indicates how the region below the froth/pulp interface is congested with bubbles as the bubbles slow down and crowd together. Because of drainage through the rising bubble swarm, some water and suspended mineral particles drop back while some of the water and suspended particles are squeezed upwards due to the buoyancy of the bubble swarm. Figure 2-3(c) demonstrates that, as each layer of bubbles is pushed up, another layer of bubbles will form. In this way, more mineral particles in the water are pushed up into the froth. It's likely that all three mechanisms contribute to the transfer of entrained gangue and valuable particles into the froth, providing a basis for the commencement of the entrainment process. The Bubble Swarm Theory, however, is likely to be the dominant mechanism.

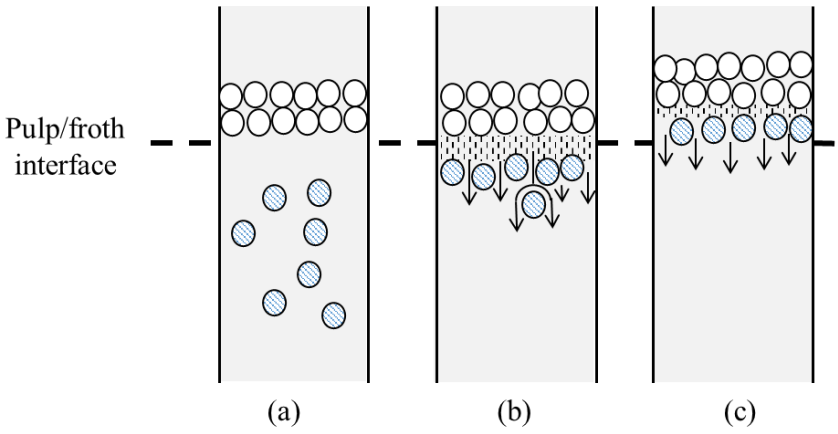

Figure 2-3 Schematic of the Bubble Swarm Theory in the flotation cell (after Smith and Warren, 1989).

Not all entrained mineral particles entering the froth are transferred to the concentrate due to drainage. When the solid particles suspended in the water enter the froth, some particles are entrained to the concentrate, whilst some experience drainage back to the pulp. Three mechanisms have been proposed by which water and solids drainage occur within the froth (Cutting, 1989). They can drain through the Plateau borders which exist within the froth structure. There can be rapid transfer of water and solid particles downwards in a local area of the froth due to froth collapse. Sedimentation can also be induced by shear activity. It should be noted that Plateau borders in the froth are particularly important for entrainment, as they contain most of the liquid present within the froth and provide the drainage channels for the entrained solids. They are formed by the thin water films (lamellae) meeting at 120°. In the Plateau borders, the unattached hydrophilic and hydrophobic particles move freely in the water and undergo settling (Neethling and Cilliers, 2002a). Figure 2-4 shows a schematic of a Plateau border and its associated lamellae in the froth.

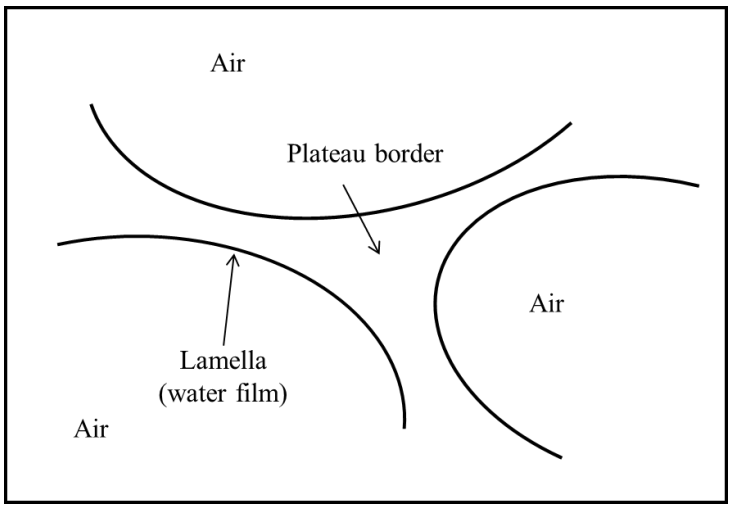

Figure 2-4 Schematic of a Plateau border and its associated lamella (after Ross and Van Deventer, 1988).

Drainage generally results in a decrease in the water and solid particles recovered by entrainment. The recovery of particles by entrainment in essence is a result of a trade-off between the transfer of suspended particles upwards into the froth phase and drainage of particles downwards across the pulp/froth interface to the pulp from the froth phase. The amount of materials entrained to the concentrate is determined by the net flow of water including suspended solid particles upwards.

Neethling and Cilliers (2002a,b) proposed that the motion of entrained materials in the froth is a consequence of a balance between liquid motion, particle settling and particle dispersion. They proposed that the profile of the gangue solids concentration within the froth can be represented by three distinct regions. There is a rapid decrease in the gangue solid concentration just above the pulp/froth interface, resulting from the rapid decrease in water content in this region. The solids that enter the froth phase are determined by the balance of solids dispersion and solids settling. After this initial decrease, in the middle regions of the froth, the balance of the liquid motion and the hindered solid settling determines the relative motion of the gangue particles, resulting in a steady reduction in the concentration of gangue particles as the height from the pulp/froth interface increases. In the upper region of a deep froth, the gangue particle concentration remains relatively constant, due to there being no relative downward motion of the solids with respect to the water in this region.

Experimental observations, however, suggest that the gangue distribution within the froth may follow a simpler trend. Figure 2-5 shows an example of the gangue composition measured by Szatkowski (1987) who investigated the effects of air flowrate, air bubble size distribution and size and mineral concentration on the ash content of the froth in coal flotation. Gangue concentration

changes significantly in the region of the pulp/froth interface, after which it remains relatively constant. This suggests that it is the interface where a majority of the drainage of the solids with respect to the water occurs. Above this region, solids follow the water phase which either reports to the concentrate or drains back to the pulp. The importance of the pulp/froth interface region as being the region where most of the entrained materials are rejected has been recognised by a number of researchers (Cutting et al., 1984; Szatkowski, 1987; Yianatos et al., 1987; Tao et al., 2000; Smith and Warren, 1989; Ata, 2001; Neethling and Cilliers, 2002a,b; Schwarz, 2004; Zheng, 2009).

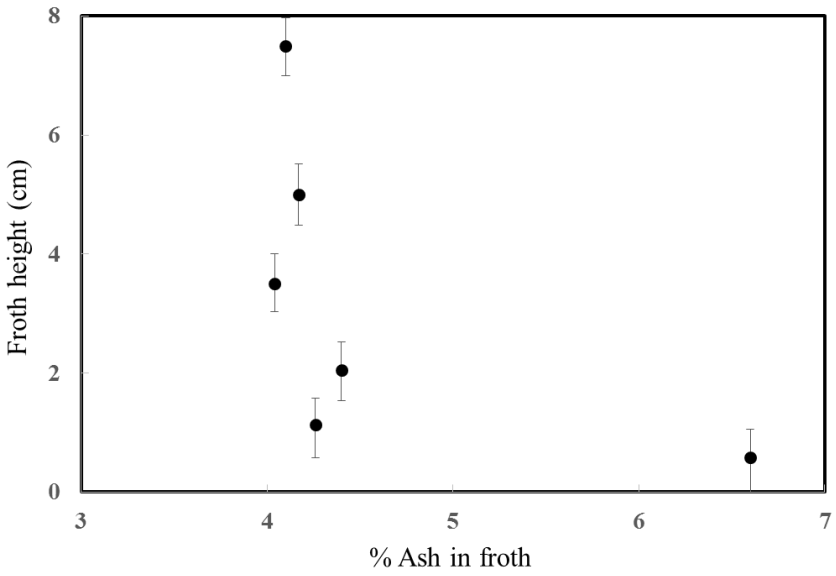

Figure 2-5 Gangue (ash) content measured at different froth heights within a froth produced using a superficial gas velocity of 15 cm/min and MIBC concentration of 11 ppm (error bar indicating uncertainty in height of sampler relative to base of froth) (after Szatkowski, 1987).

The resulting motion of solid particles at the interface may depend largely on the settling and drag forces acting upon the particles. The interface region, according to the Bubble Swarm Theory, is where water and suspended solid particles are being pushed upwards by rising air bubbles. The significant reduction in solids concentration in this region is an indication that a large proportion of the particles are settling through the water and returning to the pulp due to gravity. There will, however, be a net liquid flowrate upwards as more and more bubbles congest in the region and move upwards resulting in an upward drag force on the particles which counteracts the downward gravitational force on the particles due to their weight. This enables a proportion of the suspended solid particles to follow the water phase into the froth when the drag force overcomes the gravitational forces.

Equations are available for estimating the magnitude of the drag force acting on a solid particle in a fluid (Elger et al., 2013). If the liquid velocity is high and the flow is turbulent, the drag force is given by:

$$F_D = \frac{1}{2}\rho v^2 C_D A \qquad 2\text{-}1$$

where F_D is the drag force (N), ρ is the density of the fluid (kg/m^3), υ is the free-stream velocity measured relative to the objective (m/s), A is the cross-sectional area over which the force is applied (m^2) and C_D is the drag coefficient which is not a constant but varies as a function of liquid flowrate, the direction of the flow, object size, fluid density and fluid viscosity.

In laminar flow conditions, drag force will be proportional to the particle velocity relative to the fluid velocity, in the direction of the fluid motion:

$$F_D = bv \qquad 2\text{-}2$$

where b is a constant that depends on the properties of the fluid (i.e. fluid viscosity) and the dimensions of the object.

The drag force is a strong function of liquid velocity. In the literature, effective liquid velocity can be calculated using the liquid flowrate into the froth and overflowing the lip, and liquid holdup in the froth (Neethling and Cilliers, 2009):

$$v_l = \frac{Q_l}{A_c}\frac{1}{\varepsilon} \qquad 2\text{-}3$$

where Q_l is the volumetric liquid flowrate into the froth and overflowing the lip (m^3/s), A_c is the cross-sectional area of the froth flotation cell (m^2), and ε is the liquid content (i.e. liquid holdup).

It must be emphasized that the effective liquid velocity refers to the actual vertical liquid velocity in the froth rather than the superficial liquid velocity (Q_l / A_c), and it does vary with position in the froth phase (Neethling and Cilliers, 2009). Superficial liquid velocity, on the other hand, is constant through the froth. There is a changing amount of water flowing upward and downward at different heights within the froth but the net upward flow at every position at equilibrium must be the same.

As there is a great variation in liquid holdup from the pulp/froth interface to the froth surface, liquid velocity varies with position due to a change in water content in the froth (Neethling and Cilliers, 2009). It is the actual liquid velocity that results in the drag force acting on solid particles rather than the superficial liquid velocity.

The lowest liquid velocity occurs at the interface. Liquid holdup decreases with height in the froth due to drainage. According to the equations above, this will result in an increase in the liquid velocity as it rises within the froth and therefore the drag force increases on the particle. It is therefore likely that if the drag force on the particle is sufficient for it to rise at the interface region, it will continue within the water phase and only return to the pulp phase due to water drainage. This would result in a gangue particle concentration profile in the flotation cell as follows: from the bottom of the flotation cell to the region below the pulp/froth interface, there is classification of solids because of incomplete solids suspension in the pulp phase (Zheng et al., 2005). At the pulp/froth interface, there is a rapid change in the gangue concentration due to the balance of forces determining the relative drainage of solids to water in that region. After this change, the gangue particle concentration remains relatively constant, as observed in Figure 2-5. This is a plausible explanation for the observed gangue concentration profile observed in Figure 2-5. It should be pointed out, however, that this reasoning is purely speculative at this stage and more work is required to validate the mechanisms involved.

2.4 Principal factors affecting entrainment

2.4.1 Water recovery

Entrainment is strongly dependent on the water which is considered as the carrying medium to transfer the mineral particles into the concentrate in flotation processes. The paper by Jowett (1966) was the first to state that the recovery of mineral particles by entrainment was initiated by water currents. He also observed that the concentration of gangue in the pulp was correlated with that observed in the concentrate. After that, further research was carried out by a number of researchers to identify the relationship between the water recovery and recovery of solid particles by entrainment. Johnson (1972) conducted continuous pilot tests and full scale tests as well as laboratory experiments. Both the laboratory data and industrial data indicated that there was a direct correlation between the recovery of gangue mineral particles by entrainment and the water recovered to the concentrate. This conclusion is in agreement with Engelbrecht and Woodburn (1975) and Laplante et al. (1989). In general, the correlation between the recovery by entrainment

and the water recovery shows a linear trend for ultrafine particles and a parabolic trend for coarse particles. Figure 2-6 shows the data from a continuous pilot plant system on silica recovery (gangue) versus water recovery reported by Engelbrecht and Woodburn (1975).

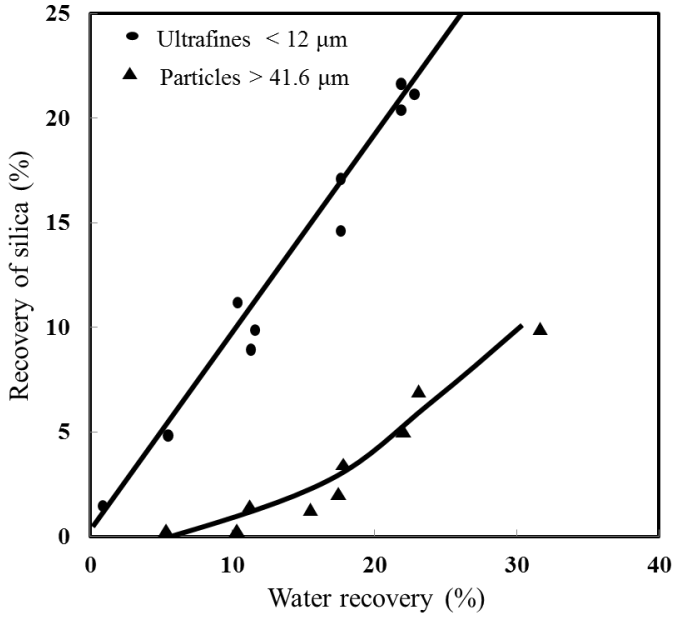

Figure 2-6 Recovery of sized silica by entrainment as a function of water recovery (after Engelbrecht and Woodburn, 1975).

Therefore an increase in the amount of recovered water results in more gangue particles reporting to the concentrate due to entrainment. An increase in gangue recovery will cause a decrease in concentrate grade and thus it is expected that there will be an inverse relationship between the water recovered to concentrate and concentrate grade. This is demonstrated by Yianatos and Contreras (2010) who investigated the effect of water recovery on the concentrate grade in an industrial 130 m3 flotation cell treating a copper ore (Figure 2-7). They showed that an increase in the water recovery led to a significant decrease in the concentrate copper grade, highlighting the importance of water recovery in determining the gangue recovery and concentrate grade.

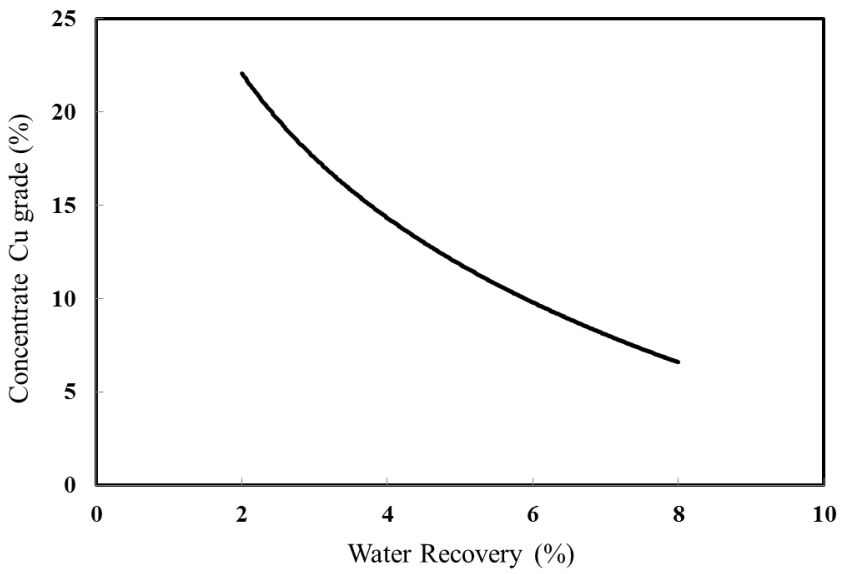

Figure 2-7 Concentrate copper grade as a function of water recovery (after Yianatos and Contreras, 2010).

It is worth noting that varying definitions of "water recovery" are used in the literature, e.g. "a fraction of total water recovered to the concentrate" and "the water flowrate of concentrate". Hence, much attention needs to be paid to definitions to avoid the possible confusion when the experimental results are considered and compared (Smith and Warren, 1989; Zheng et al., 2006a).

2.4.2 The degree of entrainment

The concentration of entrained materials in the water recovered to concentrate is not the same as that in the pulp of a flotation cell. This is because of a classification effect occurring in both the pulp and the froth phase which is often referred to as the degree of entrainment.

Johnson (1972) proposed a classification function to represent the degree of entrainment due to drainage in the froth phase:

$$ENT_i^f = \frac{(mass\ of\ free\ gangue\ per\ unit\ mass\ of\ water)_{concentrate}}{(mass\ of\ free\ gangue\ per\ unit\ mass\ of\ water)_{top\ of\ pulp}} \qquad 2\text{-}4$$

where ENT_i^f is the classification function representing the drainage in the froth phase, subscript *i* represents the particle size and superscript *f* represents the froth phase.

Similarly, Zheng et al (2005) introduced a classification function to describe the state of solids suspension in the pulp:

$$CF_i^p = \frac{(mass\ of\ free\ gangue\ per\ unit\ mass\ of\ water)_{top\ of\ pulp}}{(mass\ of\ free\ gangue\ per\ unit\ mass\ of\ water)_{tailings}} \qquad 2\text{-}5$$

where CF_i^p is the classification function and superscript *p* represents the pulp phase.

Taking the classification effects in both the pulp and the froth into account, the overall classification effect can be determined by (Zheng et al., 2006b):

$$ENT_i = CF_i^p \cdot ENT_i^f \qquad 2\text{-}6$$

where ENT_i is the degree of entrainment representing the overall classification effect in both the pulp and the froth (for particle size *i*).

ENT_i can be used directly for the estimation of gangue recovery by entrainment. The gangue recovery by entrainment is essentially the water recovered corrected by the degree of entrainment. There are two models of gangue recovery by entrainment that are commonly seen in the literature (Jowett, 1966; Engelbrecht and Woodburn, 1975; Laplante et al., 1989):

$$\mathrm{R}_{ent,i} = \frac{ENT_i \cdot R_w}{1 + R_w \cdot (ENT_i - 1)} \qquad 2\text{-}7$$

$$\mathrm{R}_{ent,i} = ENT_i \cdot R_w \qquad 2\text{-}8$$

where $R_{ent,i}$ is the recovery by entrainment and R_w is the water recovery.

It should also be noted that Equation 2-7 was derived from the definition of gangue recovery (based on the mass of the gangue in the concentrate and the feed), water recovery (based on the mass of the water in the concentrate and the feed) and *ENT* (based on the mass of solids and water in both the concentrate and the tailing). Hence, it can deliver a very precise value of gangue recovery by entrainment. Equation 2-8 is a simplified form of Equation 2-7. Figure 2-8 compares the gangue recovery calculated by Equation 2-7 and Equation 2-8 which shows that Equation 2-8 gives similar estimates of recovery by entrainment to Equation 2-7 when water recovery is below 30 % or for

ultrafine particles regardless of the water recovery. Thus the slope of the relationship between gangue recovery by entrainment and water recovery is a reasonable estimate of the degree of entrainment in most situations.

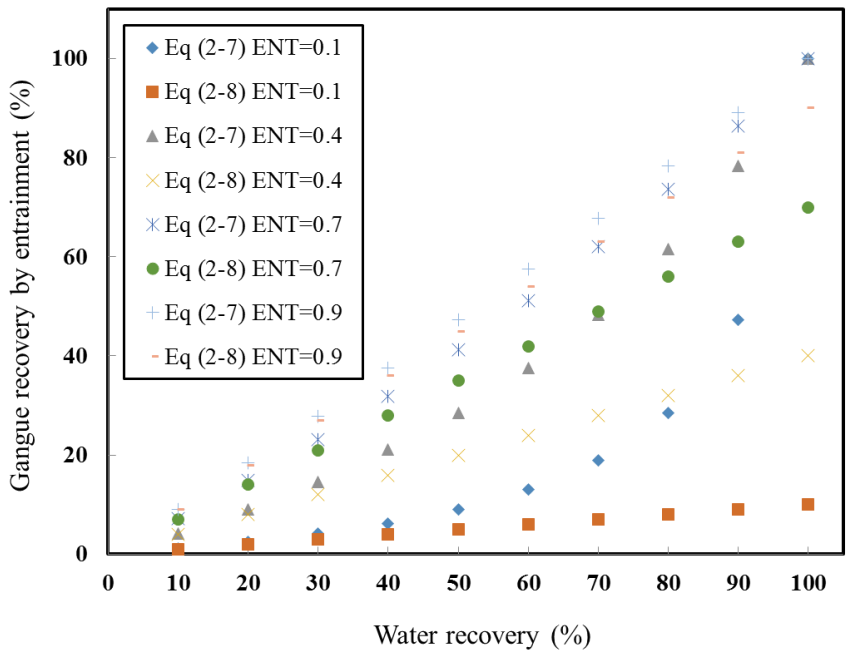

Figure 2-8 Gangue recovery at different water recoveries and *ENT* values.

2.5 Effect of flotation variables on entrainment

2.5.1 Solids percentage in the pulp

Solids percentage in the upper regions of the pulp has a marked effect on the entrainment recovery (Johnson, 2005; Zheng et al., 2006b; Yianatos et al., 2009). In the literature, the recovery of gangue minerals by entrainment is often only related to the water recovery and the degree of entrainment (without a pulp classification effect) because all the mineral particles are perfectly dispersed and mixed in the pulp phase in the small sized cells in which the testing has been performed. In the large full scale flotation cells being installed today, however, there has been found to be significant settling of coarse particles in the upper portion of the cells. When significant settling occurs in a flotation cell, the solids percentage in the pulp phase where entrainment starts undoubtedly becomes particularly important, as it determines the amount of solids entrained to the concentrate. Hence, the state of solids suspension in that region must be taken into consideration in the non-perfectly mixed

case. Table 2-1 shows the degree of solids suspension for mineral particles of different size fractions at the top of the pulp region in industrial flotation cells (Zheng et al., 2006b).

Table 2-1 Summary of solids suspension measurement results in industrial flotation cells at different mine sites.

Size fraction (μm)	-38	+38	+53	+75	+106	+150	+212	+300
CF **(Outokumpu 150m^3 tank cell at the Newcrest Cadia copper concentrator)**	**0.99**	**0.82**	**0.81**	**0.67**	**0.53**	**0.39**	**0.26**	**0.10**
CF **(Outokumpu 100m^3 tank cell at the Perilya Broken Hill lead/zinc concentrator)**	**0.82**	**0.43**	**0.25**	**0.12**	**0.063**	**0.034**	**0.014**	**0.0038**

In the pulp, the classification function CF_i^p represents the degree of solids suspension. Its value is known to decrease with an increase in particle size, which indicates that the coarse particles tend to settle and be located in the lower portion of a flotation cell while the fine particles are more likely to be uniformly dispersed in the pulp phase. Solids percentage in the pulp below the pulp/froth interface is also a function of the size of the flotation cell in use in addition to the particle size and other influencing variables (Zheng et al., 2005).

Figure 2-9 illustrates the effect of the solids percentage in the region below the pulp/froth interface on the recovery by entrainment (Johnson, 2005). The contribution of entrainment to the recovery of fine and intermediate particles is much larger in the perfectly mixed case than that when the pulp phase is not perfectly mixed. Generally, better suspension in the pulp often leads to more material entering the froth due to entrainment.

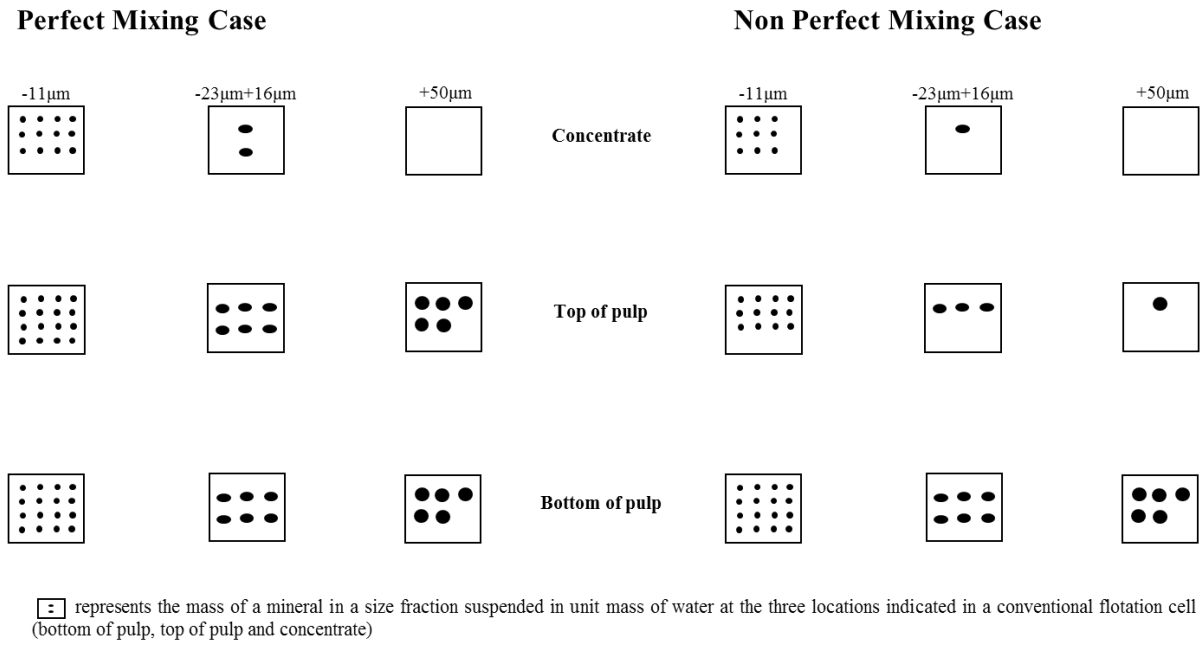

Figure 2-9 Recovery by entrainment as a function of solids percentage below the pulp/froth interface: three size fractions (-11 μm，-23+16 μm and +50 μm) are illustrated with the degree of entrainment (*ENT*) of 0.75, 0.33 and 0, respectively, (after Johnson, 2005).

What one can also conclude from Figure 2-9, is that an increase in the pulp density of the feed, which is likely to result in an increase in mass in the top regions of the pulp, would also have a marked effect on the recovery by entrainment. More solids would move into the froth phase and a greater proportion would be recovered to the concentrate.

This was demonstrated by Johnson et al. (1974) who performed pilot plant testing using pure silica (d50=74 μm) added at different ratios with water to produce different pulp densities. Their results are shown in Figure 2-10. The amount of silica entrained at a given recovery rate of water increased with an increase in the pulp density from 17 to 42%. It has also been demonstrated in a number of other studies that there is an increase in the recovery of non-floatable gangue at the same water recovery with increased pulp density (Jowett, 1966; Warren, 1985; Cilek, 2009).

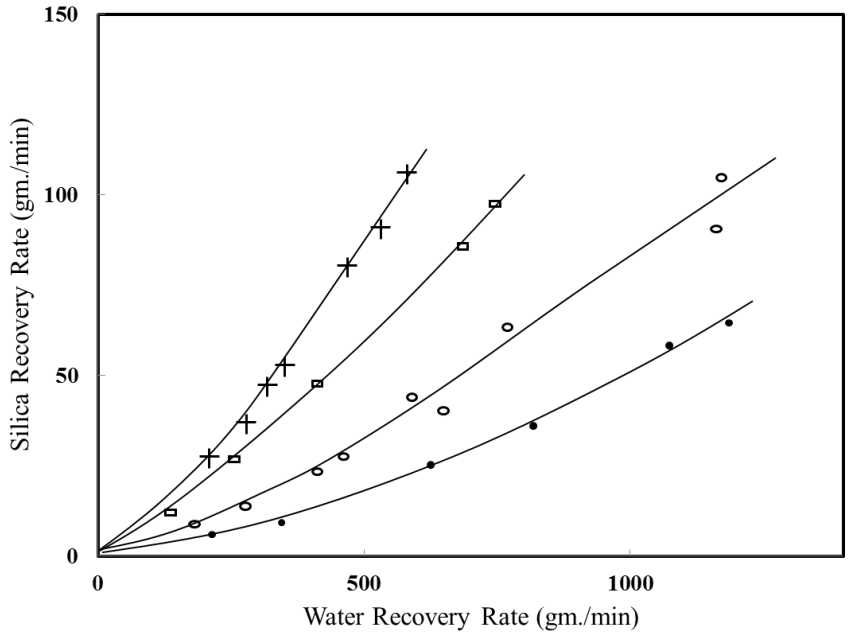

Figure 2-10 Relationship between silica recovery rate and water recovery rate at various pulp densities (cross, 42.0%; square, 33.5%; open circle, 24.0%; closed circle, 17.0%) (after Johnson et al., 1974).

2.5.2 Particle size

The particle size has a significant effect on entrainment. This effect is often demonstrated by using a recovery-size curve which is obtained by plotting the recovery from a particular size fraction against the average size of particles in that fraction. Unlike the recovery of hydrophobic particles by true flotation showing a maximum at an intermediate size, the recovery of hydrophilic minerals by entrainment increases significantly as the particle size decreases (Lynch et al., 1981; Smith and Warren, 1989; Nguyen and Schulze, 2004).

Entrainment also becomes an important mechanism for the recovery of floatable minerals as particle size becomes finer as shown in Figure 2-11. As particle size decreases, recovery by entrainment is increasing whereas the recovery by true flotation starts to decrease.

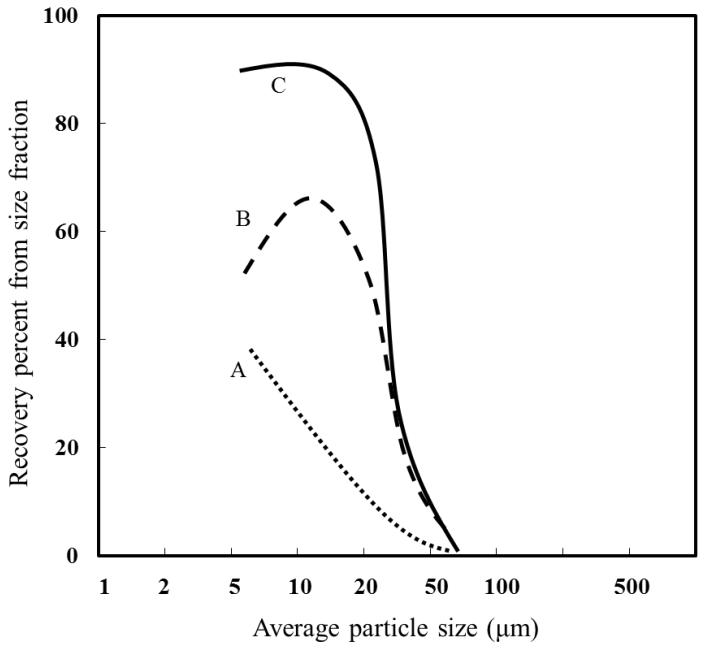

Figure 2-11 Recovery of siderite as a function of the average particle size (A: recovery by entrainment; B: recovery by true flotation; C: recovery by entrainment & true flotation) (after Trahar, 1981).

Mineral particles generally smaller than the size of 50 μm are known to be recovered by entrainment more easily (Savassi et al., 1998). In the pulp phase, fine particles are easily suspended in the water or the water film surrounding the bubbles in the region below the pulp/froth interface compared with coarse particles (refer to Figure 2-9), and hence they have more chance to travel up through the froth to the concentrate. Figure 2-12 shows the typical relationship between the degree of entrainment and particle size. The degree of entrainment is known to usually fall between zero and one; zero representing the condition of total drainage where there is no entrained recovery of particles regardless of the water recovered, and one representing the condition where there is no drainage and the concentration of particles in the water in the concentrate (due to entrainment) is the same as in the pulp phase (Runge, 2010). The degree of entrainment is strongly dependent on size with the coarser particles exhibiting a higher degree of drainage. Very fine particles usually follow the water phase with ENT_i=1. It indicates that the coarse particles drain more readily from the voids between the bubbles than the fine particles. Therefore, coarse particles are less entrained to the concentrate.

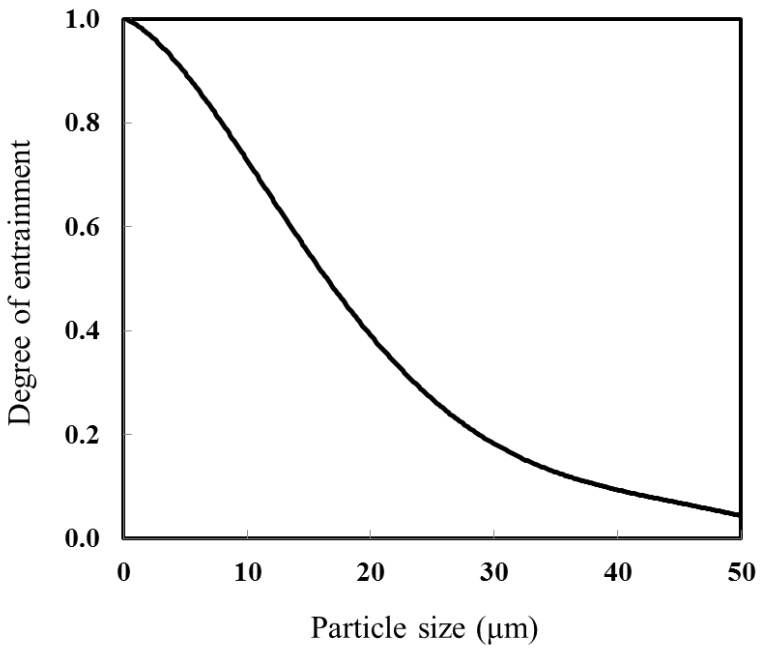

Figure 2-12 Relationship between the degree of entrainment and the particle size for hydrophilic silica in a laboratory flotation cell (after Bisshop and White, 1976).

2.5.3 Particle density

Particle density, also known as mineral specific gravity, is also a factor that affects entrainment (Johnson, 1972; Bisshop, 1974; Thorne et al., 1976; Maachar and Dobby, 1992; Drzymala, 1994). In the testwork performed by Bisshop (1974), seven pure minerals were used to study the effect of particle density on the degree of entrainment. Results showed that particle density ranked second only to particle size in terms of importance. Higher entrainment was achieved for the low-density minerals and this was attributed to the fact that they have a greater tendency to move with the water due to their lower sedimentation velocities. Smith and Warren (1989) used Stokes' Law to calculate the expected sedimentation rates for different density minerals in the pulp of a flotation cell, showing that those particles with higher density would settle more quickly and have less chance to be transported to the concentrate.

The degree to which density impacts on entrainment, however, is not always consistent. Maachar and Dobby (1992) measured the entrainment recovery of silica (SG=2.6) and galena (SG=7.5) in a laboratory column and showed that, although galena the heavier mineral exhibited a lower degree of entrainment than silica, the effect was much smaller than would be expected based on the large differences in their specific gravity (Figure 2-13). For fine particles (<5 μm), for example, the degree of entrainment is 0.74 for hydrophilic silica and 0.70 for hydrophilic galena. Tests conducted

by Drzymala (1994) in a Hallimond tube suggested a stronger effect of particle density on the degree of entrainment than that observed by Maachar and Dobby (1992). These observations indicate that particle density might interact with other factors which contribute or affect entrainment in a flotation cell. More work is required to investigate these effects.

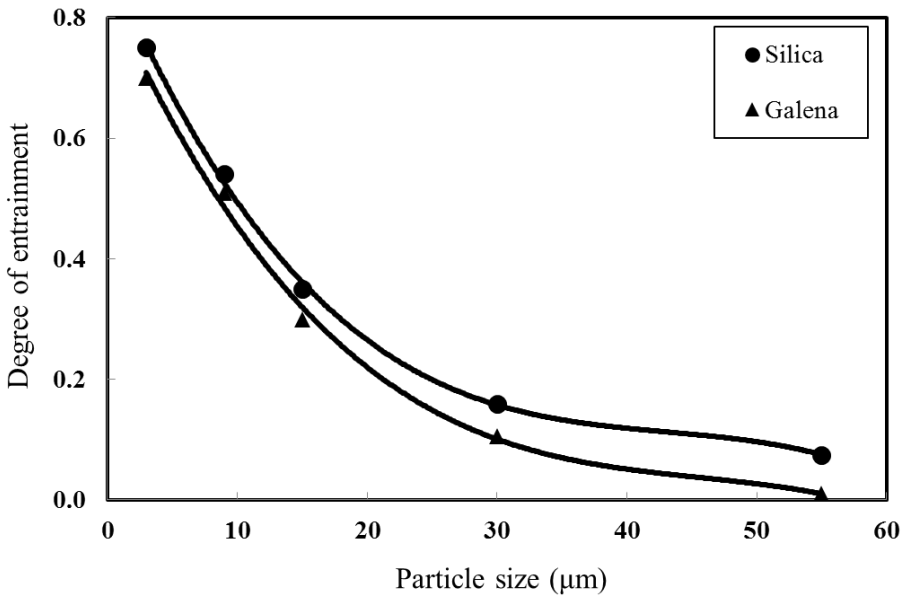

Figure 2-13 Effect of particle density on the degree of entrainment in a laboratory test (after Maachar and Dobby, 1992).

2.5.4 Impeller speed

In a flotation cell, the function of the impeller is to increase the chance of collisions between bubbles and mineral particles, and to make both solid particles and air fully dispersed and mixed in the pulp phase. However, a change in the impeller speed also changes entrainment behaviour. Cilek (2009) noted that in mechanical cells, impeller speed can be regulated to a range within which the recovery of gangue minerals by entrainment would be reduced but true flotation would be promoted. The effect of impeller speed on the recovery by entrainment can also be seen from other studies. Akdemir and Sönmez (2003) investigated the influence of various impeller speeds on the coal recovery by true flotation and ash recovery by entrainment. Results indicated an increase in recovery by entrainment as a result of increased impeller speeds.

In addition, the design of the turbulence generating system also has an effect on the gangue recovery in mechanical flotation cells. For example, a system which has a double-finger impeller and a stator consisting of radially arranged and relatively high-position blades may reduce the

recovery of gangue minerals by entrainment (Schubert, 1999). This is because such a system gives rise to high local energy dissipation rates and results in less turbulent pulp flow in the upper cell volume (Nguyen and Schulze, 2004; Cilek, 2009).

In general, excessive agitation leads to an increase in the recovery of fine gangue particles. The addition of energy not only provides the momentum for particles and bubbles to collide with each other, but also enhances the water recovery. It may also affect the state of solids suspension in the pulp which results in a higher solid concentration in the pulp water moving into the froth and hence the concentrate. Consequently more gangue minerals are entrained to the concentrate.

2.5.5 Gas flowrate

It has been found that gas flowrate has a significant effect on the water recovery and gangue recovery by entrainment (Zheng et al., 2006b). An increase in gas flowrate will inevitably carry over a greater amount of water with the suspended solid particles to the froth from the pulp. In the froth, however, gas flowrate also determines the extent of drainage of water including suspended solid particles by changing the retention time of the water in the froth (Zheng et al., 2006b). In the froth, it should be noted that too high a gas flowrate may cause an unstable froth and a possible change in the froth structure, and hence has a significant effect on both entrainment and true flotation (Nelson and Lelinski, 2000; Nguyen and Schulze, 2004).

Zheng et al. (2006b) investigated the effect of gas flowrate on the degree of entrainment using industrial experimental data. With an increase in gas flowrate at a particular froth depth, he not only observed an increase in water recovery but also, as shown in Figure 2-14, an increase in the degree of entrainment of a particular particle size. It indicates that gas flowrate changes the drainage of suspended particles relative to water in the froth. Zheng et al (2006b) attributed this result to the gas affecting the drainage process in the froth by influencing the froth rising velocity and the froth retention time.

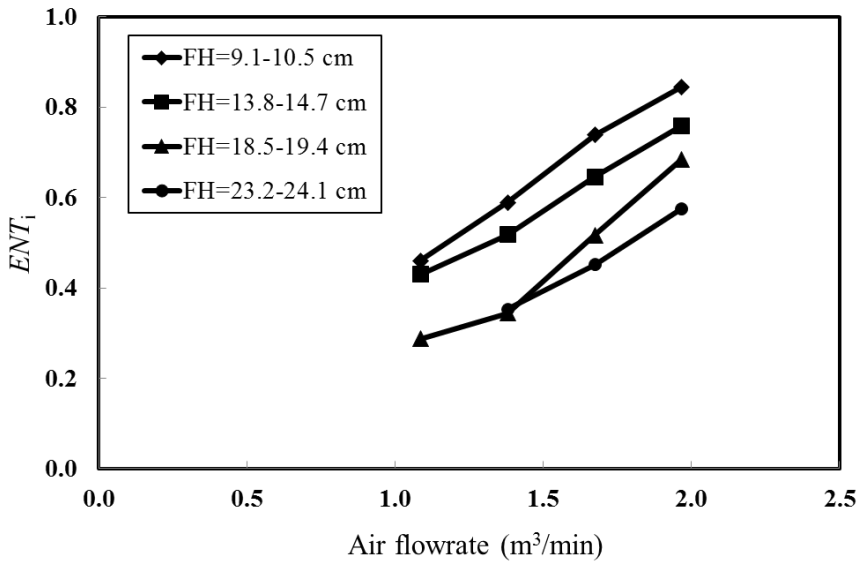

Figure 2-14 Effect of gas flowrate on the degree of entrainment of mineral particles below 5.3 µm at different froth heights in the Outokumpu 3 m^3 tank cell operated at the Xstrata Mt. Isa Mines copper concentrator (after Zheng et al., 2006b).

In summary, gas flowrate significantly affects entrainment by influencing both the water recovered to concentrate as well as the amount of suspended solid particles within the water phase.

2.5.6 Froth height

The recovery of gangue particles by entrainment is influenced greatly by froth height. This conclusion can be clearly demonstrated by the research work from Englebrecht and Woodburn (1975) and Cutting et al. (1981) who quantified the water content in the froth from the base of a froth column to the top. Their experimental results suggested that a decrease in the froth height caused a wetter froth where more water including the suspended gangue mineral particles went into the concentrate.

Zheng et al. (2006b) performed a detailed study of the effect of froth height on entrainment using data obtained from an Outokumpu 3 m^3 tank cell operating at the Xstrata Mt. Isa Mines copper concentrator. An increase in froth height resulted in a decrease in entrainment due to both a reduction in water recovery as well as a decrease in the degree of entrainment. Figure 2-15 shows the relationship between the degree of entrainment of three selected particle sizes and froth height, for tests performed using the same gas flowrate, obtained during this work.

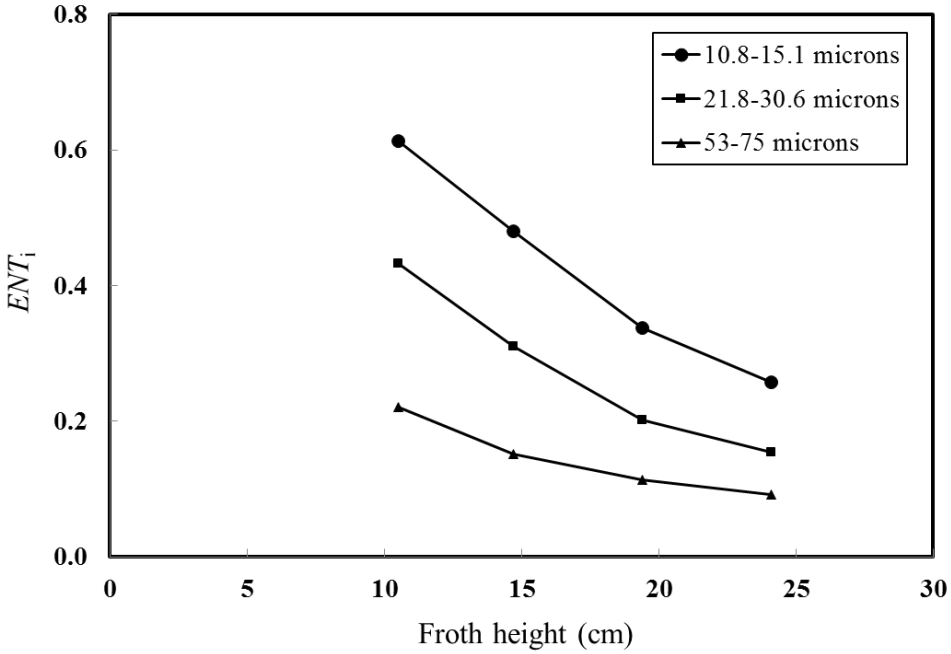

Figure 2-15 The degree of entrainment of different size fractions as a function of froth height in the Outokumpu 3 m^3 tank cell operated at the gas flowrate of 1.66 m^3/min at the Xstrata Mt. Isa Mines copper concentrator (after Zheng et al., 2006b).

Generally a decrease in recovery of gangue minerals occurs when there is an increase in froth height. It is believed that the increase in froth height results in an increase in froth residence time which promotes drainage of water including the solid particles (Zheng et al., 2016b). As a result, the following general principle applies with respect to entrainment and froth height: the deeper the froth, the lower the recovery of minerals by entrainment.

2.5.7 Froth residence time

Froth residence time, or the time particles and bubbles exist within the froth phase, consists of two time periods, i.e. the time for the froth to move vertically from the base of the froth phase to the level of the concentrate launder lip and the time for the froth to move horizontally to the edge of the flotation cell. However, froth retention time in the literature generally takes only the vertical froth transportation into account for modelling, and it is defined as (Szatkowski, 1987):

$$\lambda_{air} = \frac{H_f}{J_g} \qquad 2\text{-}9$$

where λ_{air} is the froth residence time of gas, H_f is the froth height and J_g is the superficial gas (i.e. air) velocity which is the ratio between the volumetric gas flowrate and cross-sectional area of the flotation cell.

As shown in Equation 2-9, froth residence time is determined by froth height and superficial gas velocity. It has been confirmed that an increase in the froth height and a decrease in the gas flowrate result in a lower recovery of gangue minerals by entrainment. By extension, longer froth residence times will undoubtedly lead to a decrease in the recovery of gangue minerals by entrainment.

In the froth, the drainage process plays a very important role in determining the extent of gangue recovery by entrainment. An increase in froth residence time is believed to provide a longer time for water including suspended particles to drain back to the pulp phase from the froth phase. Besides its effect on drainage rate, a change in the froth residence time can also result in a change in froth properties such as the thickness of the lamellae and diameter of the Plateau borders, the froth viscosity and the rate of bubble coalescence and bursting which affect the drainage process in particular areas within the froth (Zheng et al., 2006b). As a result, froth residence time has a significant effect on entrainment recovery.

2.5.8 Rheology

Rheology is an important factor that is reported to affect entrainment, although to date its effects are ill understood. Slurry viscosity is generally the rheological parameter investigated when studying the effects of rheology on entrainment. Tests carried out by Kirjavainen (1992) using granular and flaky minerals at different slurry densities and only a frother showed that the recovery by entrainment was influenced by slurry viscosity, and accordingly, a mathematical model was proposed to include a viscosity parameter.

A viscous pulp will affect the degree of solids suspension in the pulp phase. It affects the amount of energy imparted to the slurry and the slurry flow patterns as well as the degree of gas dispersion within the vessel which significantly changes solids suspension (Shabalala et al, 2011). It also reduces the degree of sedimentation of particles due to an increase in the drag force of the fluid.

The viscosity in the froth phase is another parameter which can affect entrainment. Unlike the rheology of liquid or liquid-solid systems, the rheology of a flotation froth is difficult to measure. This is because the shearing force produced by the rheometer leads to bubble rupture in the froth

phase (Shi and Zheng, 2003). To overcome this problem, Shi and Zheng (2003) used a vane style rheometer instead of the standard bob and cup to measure viscosity in industrial scale flotation tests. Their results revealed that the viscosity of the froth was linearly correlated with the grade of hydrophobic minerals but inversely correlated with the grade of the hydrophilic minerals. They concluded that high froth viscosity, in their study, led to an increase in froth residence time which decreased the water hold-up in the froth phase and provided a longer time for water and entrained quartz to drain back from the froth to the pulp phase (Shi and Zheng, 2003). Therefore, the high froth viscosity resulted in a decrease in the recovery of suspended quartz particles by entrainment and hence a lower grade of quartz in the concentrate. Figure 2-16 shows the relationship between the froth viscosity and the grade of hydrophilic mineral (quartz) obtained in their study.

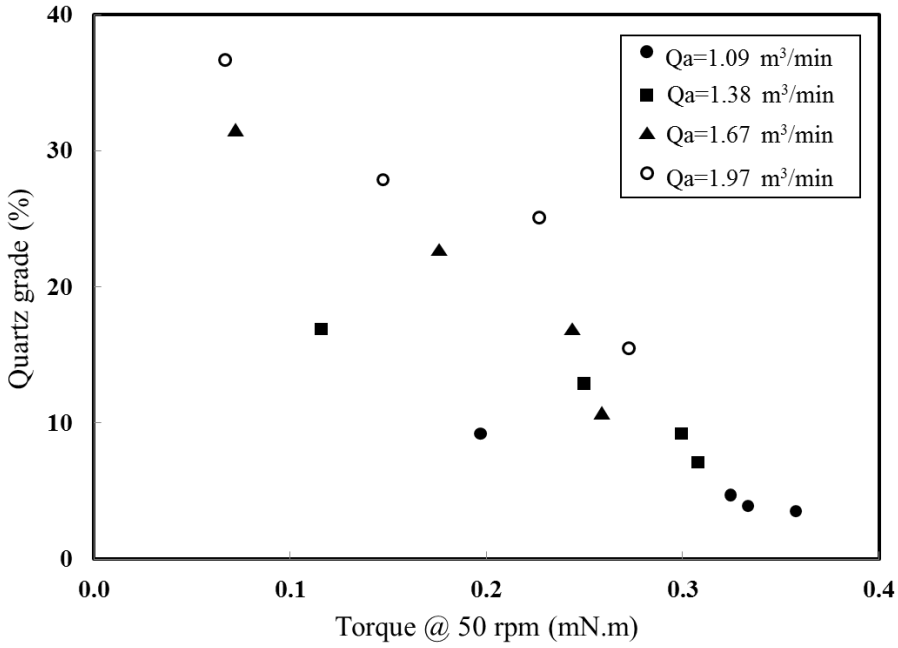

Figure 2-16 Froth rheology reading (a representation of viscosity) versus the quartz concentrate grade for various air flowrates measured 15 cm from the froth launder (after Shi and Zheng, 2003).

The important effect of rheology within the froth on entrainment has been acknowledged in a number of studies (Nentura-Medina and Cilliers, 2002; Neethling and Cilliers, 2003; Nguyen et al., 2006; Neethling and Cilliers, 2009). It is known that entrainment recovery is a strong function of water recovery. Neethling and Cilliers (2003) modelled the behaviour of a flotation froth and showed that water motion is a result of force balance that includes gravity, viscosity and surface tension. Nguyen et al (2006) also investigated the water drainage process in a column froth and found that water recovery was controlled by the interfacial shear viscosity at the gas-liquid interface.

A change in the froth rheology would affect the transportation of water in the froth and hence change the entrainment recovery of solid particles.

In addition, literature also shows that Plateau border dimensions are determined by the slurry viscosity within the froth (Nentura-Medina and Cilliers, 2002; Neethling and Cilliers, 2003). It is well known that plateau borders are channels where the majority of recovered water and entrained solids are transported. The rheology within the froth can significantly vary the transportation of water and solids in that region, resulting in a change in the hindered settling of solids and thus entrainment recovery (Nentura-Medina and Cilliers, 2002; Neethling and Cilliers, 2003; Neethling and Cilliers, 2009). As evidenced from these studies, rheology within the froth affects both the transportation of entrained solids and water, resulting in changes in the entrainment recovery.

2.5.9 Frother

Variation in frother (type or dosage rate) has been observed to result in significant variation in the amount of gangue materials reporting to the concentrate due to entrainment (Rahal et al., 2001; Melo and Laskowski, 2007; Comley, 2012; Wiese and Harris, 2012; Kracht et al, 2016). In froth flotation, frothers are commonly added to provide a stable froth. They are heteropolar surface-active compounds which are generally composed of a hydrophilic polar group and a hydrophobic hydrocarbon chain. This enables the successful absorption of frothers onto air/water interfaces, lowering the surface energy or altering the visco-elastic property of air bubbles, reducing bubble bursting and coalescence, and enabling a stable froth to form (Fruhner et al., 1999). This stable froth phase facilitates the separation of valuable minerals from unwanted gangue minerals by providing time for water drainage without an excessive amount of bubble coalescence.

Frothers are normally classified by their properties and behaviours in the pulp solution. Neutral frothers (i.e. alphatic alcohols, cyclic alcohols and natural oils, alkoxy paraffins, polypropylene glycol ethers, polyglycol ethers and polyglycol glycerol ethers) have been widely used in the flotation of base-metal ores, oxidic minerals, and industrial minerals as they are functional in both acidic and alkaline pulps, and of these, MIBC and polyglycol ethers (e.g. Dowfroth 250) accounts for nearly 90 percent of all frother used in metallic ore flotation (Khoshdast and Sam, 2011).

The performance of frothers in the flotation process is determined by its chemical structure together with the frother dosage (Farrokhpay, 2011). Rahal et al. (2001) undertook a study to determine the relationship between entrainment recovery and water recovery using five industrial frothers of

different chemical composition (i.e. Interfroth 60, MIBC, Mintech frother 110, 255 and 400) and found that increasing frother addition from 2.5 to 8.4 g/t resulted in significantly more water and entrainment recovery. Other studies have made similar observations (Melo and Laskowski, 2007; Wiese and Harris, 2012; Kracht et al., 2016). Moreover, frothers were also found to change the degree of entrainment, with frother dosage increasing the concentration of gangue materials in the water reporting to the concentrate (Rahal et al., 2001; Kracht et al., 2016). Therefore, variations in the entrainment recovery due to frother is a result of frother changing both water recovery and the degree of entrainment. This is generally attributed to the fact that frother results in a change in frother structure which influences the degree of drainage of the water and suspended solids.

It is worth mentioning that froth structure is very complicated, and is greatly influenced by a number of other factors apart from frother, including water content, gas rate, bubble size, the content of hydrophobic particles, and the concentration and type of other reagents (Savassi, 1998). It is therefore difficult to experimentally distinguish and quantify the effect of frother on entrainment from the effect of other variables.

2.5.10 Particle hydrophobicity

Entrainment is potentially affected by particle hydrophobicity. Particle hydrophobicity is known to influence froth stability as it influences the rate of bubble coalescence and bubble bursting which occurs within the froth structure (Savassi, 1998). Ata et al. (2004) conducted experiments in a continuous flotation cell using a hematite ore and methylated glass spheres with a varying degree of hydrophobicity. Their results showed that the degree of the hydrophobicity of the particles had a significant effect on the structure of the froth, which in turn influenced the extent of drainage of hydrophilic particles in the froth.

2.5.11 A summary of the variables that affect entrainment

Many flotation variables have been shown to affect the amount of gangue minerals recovered by entrainment, either by increasing the amount of water recovered to concentrate or by increasing the concentration of gangue in the water (i.e. degree of entrainment). These include particle size, percent solids, particle density, impeller speed, gas flowrate, froth height, froth residence time, rheology, frother dosage and particle hydrophobicity. Various reasons have been proposed to explain why these variables affect entrainment but in most cases the mechanisms involved have not yet been substantiated by experimental evidence.

2.6 Entrainment measurement methods

Entrainment is a non-selective process in flotation, and both hydrophobic and hydrophilic mineral particles experience entrainment. To classify the contribution of entrainment and true flotation to the recovery of minerals, a number of methods for the measurement of entrainment and true flotation have been developed in recent years, and many of these methods have already been reviewed and discussed by Trahar (1981), Warren (1985), Ross (1989), Savassi (1998) and Zheng et al. (2006b).

2.6.1 Method of Trahar (1981)

The method proposed by Trahar (1981) was to measure the recovery of solid particles and water in two batch flotation tests. One test was carried out with a collector and a frother, and the other was carried out with only a frother. Other operating conditions were all kept the same in both tests.

The evaluation of true flotation and entrainment by Trahar (1981) is based on two principal assumptions. One is that, at a constant water recovery, the amount of solid particles reporting to the concentrate by entrainment in both tests would be the same, i.e. there is the same level of entrainment at a given water recovery in both tests, and the collector does not affect entrainment, while the other is that true flotation only occurs in the presence of collector. Thus, a quantitative evaluation of the relative contributions of true flotation and entrainment can be obtained from the calculation of measured recoveries. Figure 2-17 shows a schematic for the method of Trahar (1981).

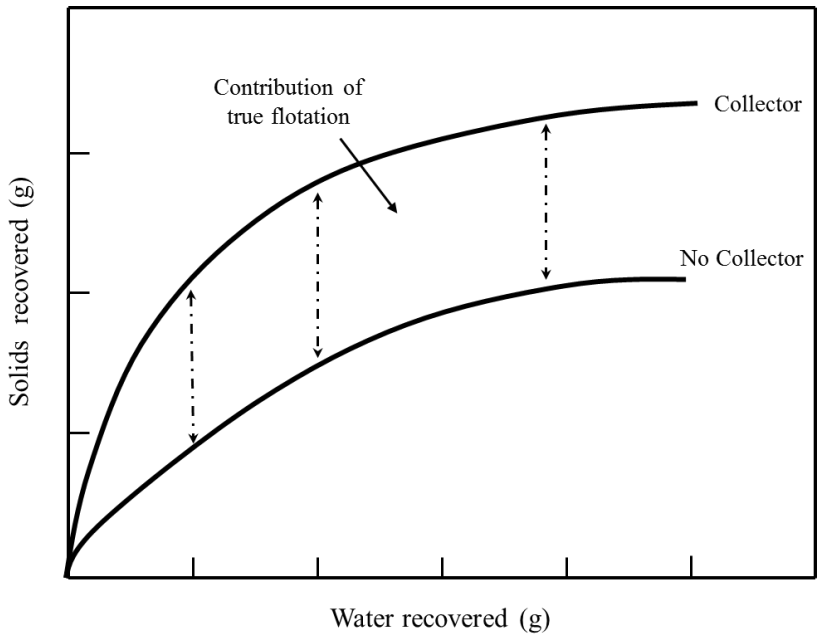

Figure 2-17 Determination of true flotation and entrainment by the method from Trahar (1981) (after Ross, 1989).

There are various potential problems that can occur when using this method which may result in inaccurate estimates of the recovery by entrainment and true flotation. It cannot be guaranteed that all the solid particles will only be recovered by entrainment without collector. Minerals, especially ultrafines, can often exhibit true flotation behaviour even in the absence of collectors (Trahar, 1981). Frother can also sometimes have a collecting power. In this scenario, the method would result in an overestimated value of recovery by entrainment in the test without collectors, and thus would underestimate the proportion of the recovery by true flotation.

The other problem is that froth characteristics will be different because of the presence of attached mineral particles in the system (Ross, 1989; Subrahmanyam and Forssberg, 1988). These particles usually stabilise the froth, resulting in changes in froth structure. This may affect the degree to which particles drain with respect to water in the froth and therefore the entrainment recovery. Taken together, the application of the method of Trahar (1981) is not recommended due to the difficulty in decoupling entrainment from true flotation in flotation tests (Ross, 1989; Subrahmanyam and Forssberg, 1988).

2.6.2 Method of Warren (1985)

An alternative method for estimating the contribution by entrainment was developed by Warren (1985). It involves performing a number of batch flotation tests at an identical flotation time but where froth height and froth removal rate are varied to change water recovery. Other chemical conditions are kept the same. Warren (1985) applied this technique to a deslimed (+6 um) sulphide rich cassiterite ore, an ultrafine fraction (-6 um) of a cassiterite ore and a bituminous coal. For each of the systems studied, he observed a linear relationship between mineral recovery and the mass of water recovered. Figure 2-18 shows the correlations between recoveries of ultrafine cassiterite and ultrafine gangue and the weight of water recovered after 15 min flotation of the ultrafine (-6 μm) cassiterite ore.

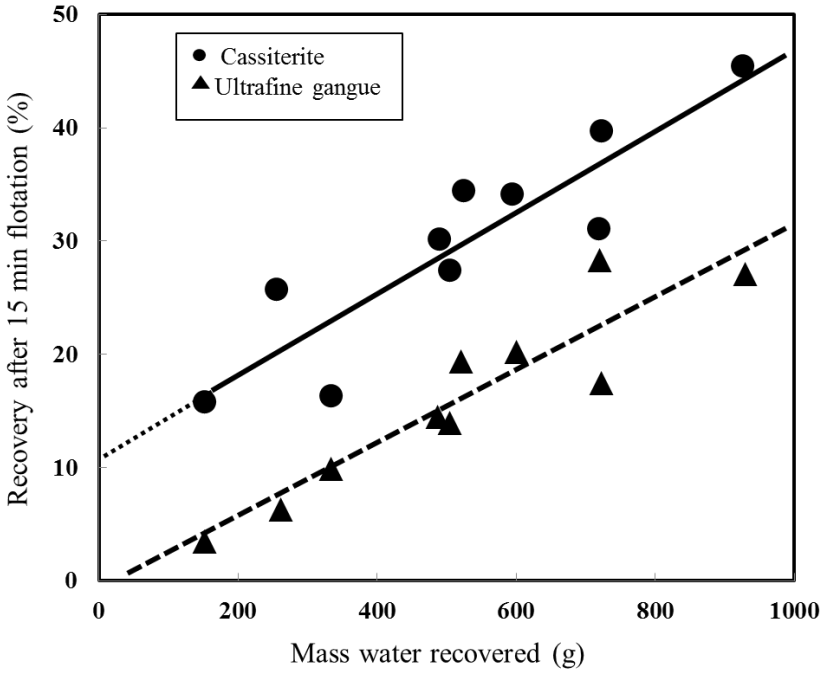

Figure 2-18 Determination of true flotation and entrainment by the method of Warren (1985).

The recovery by true flotation and entrainment can be calculated using the following equations:

$$R_M = F_M + e_M \cdot W_{water} \qquad 2\text{-}10$$

$$R_G = e_G \cdot W_{water} \qquad 2\text{-}11$$

where R_M and R_G are the cumulative recoveries of valuable and gangue minerals, respectively, W_{water} is the mass of water recovered after a given time, F_M refers to the intercept of the extrapolated line on the mineral recovery axis, an indication of recovery by true flotation, e_M and e_G are the degree of entrainment for the valuable and gangue minerals, respectively.

The method of Warren (1985) has several disadvantages (Ross, 1989). Firstly, this method is time-consuming and lacks efficiency due to the fact that it requires extensive batch flotation testing. Secondly, the degree of entrainment, e_G in Equation 2-10, is likely to be different during the duration of the batch flotation test. Therefore, the recovery by entrainment is often calculated inaccurately by applying this method. Ross (1989) recommended that this method be ideally applied over the flotation time that is just long enough for the recovery of the most easily floatable solid particles, but not for all of the floatable particles.

2.6.3 Method of Ross (1989)

The method proposed by Ross (1989) is used for the quantification of mineral particles recovered by true flotation and entrainment in batch flotation tests. In this method, only one batch flotation test is involved. A transfer factor X is defined to describe the differential classification of entrained solid particles and water during flotation. The transfer factor X is expressed as:

$$X(t) = \frac{E(t) \cdot C_W(t)}{W(t) \cdot C_M(t)} \qquad 2\text{-}12$$

where $E(t)$ and $W(t)$ are, respectively, the cumulative masses (in grams) of entrained solids and water that are recovered after a specific time interval t, and $C_W(t)$ and $C_M(t)$ are the corresponding concentrations (in grams per litre of pulp) of water and solid species in the pulp.

Similarly, $Y(t)$ was proposed to represent the relationship between the total mass of solid particles (recovered by both true flotation and entrainment) and water recovery in the batch flotation test. The relationship is expressed as:

$$Y(t) = \frac{R(t) \cdot C_W(t)}{W(t) \cdot C_M(t)} \qquad 2\text{-}13$$

where *R(t)* represents the total mass of solid particles recovered by both true flotation and entrainment after a time interval *t*. Thus, the recovery by true flotation (*F(t)*) can be obtained by subtracting the contribution by entrainment (*E(t)*) from the total mass of solids (*R(t)*), as shown in Equation 2-14 (see Figure 2-19):

$$F(t) = R(t) - E(t) \qquad 2\text{-}14$$

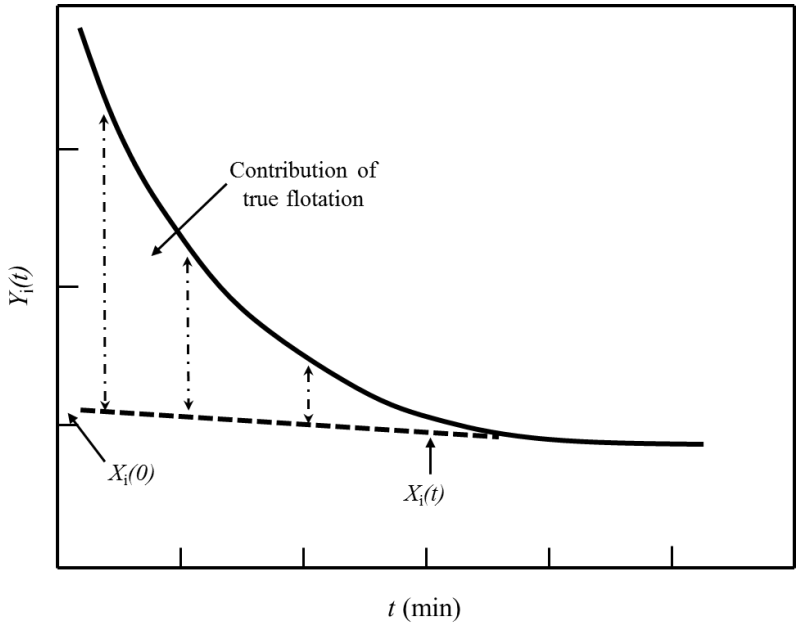

Figure 2-19 Determination of true flotation and entrainment by the method of Ross (1989).

Two assumptions are made to enable the assessment of solid particles recovered by true flotation and entrainment using this technique. One is that only entrained particles are recovered at the end of the batch flotation test, while the other is that the transfer factor decreases linearly with flotation time.

The method of Ross (1989) is easy and simple, as it involves only one batch test. It can be applied over any flotation time, because the changes in the characteristics of the froth and the composition of the pulp with flotation time are taken into consideration (Ross, 1989).

The method of Ross (1989) was developed based on a batch flotation test where flotation occurs sequentially as a function of time. At the end of a batch test, there are few floatable solid particles in a batch flotation cell. Therefore, this method can perform well in the batch flotation of typical

rougher feed. This is because in a low grade sample a majority of the gangue is recovered by entrainment, especially towards the end of the flotation process.

Caution should be used, however, when using this method when floating high grade samples or re-floating a concentrate (i.e. cleaner tests). Generally a much greater proportion of the recovered gangue may be as gangue-valuable mineral composites in a cleaner test, which result in relatively high floatable gangue recovery even at the end of the batch flotation tests. In such tests, *ENT* values are often greater than 1 which is not possible if gangue recovery is just via entrainment. To overcome this problem, mineral recovery-size-liberation data could be used to enable calculations based on just the liberated gangue in the system. Unfortunately this type of data is expensive and therefore usually not available when analysing results from routinely performed batch flotation tests.

2.6.4 Entrainment tracer

An entrainment tracer is a gangue mineral which is fully liberated, fully dispersed and non-floatable. The entrainment tracer is known to be recovered to the concentrate only by entrainment. Based on the degree of entrainment of an entrainment tracer, the degree of entrainment of different mineral species can be determined, and the mass transfer of all other mineral species in the flotation system can be calculated by a set of equations proposed by Savassi (1998) (see Equations 2-15 and 2-16). Thus, the contribution of entrainment and true flotation to the recovery of mineral particles can be determined. It should be noted that the degree of entrainment (*ENT*) in these equations is usually calculated on the basis of the tailings.

$$ENT_m = f(\rho, ENT_{tracer}) \qquad 2\text{-}15$$

$$R_{ent} = \frac{1 - R_{overall}}{1 - R_{water}} \cdot ENT_m \cdot R_{water} \qquad 2\text{-}16$$

where ENT_m and ENT_{tracer} are the degree of entrainment for different species and the entrainment tracer, respectively, ρ is the particle density, R_{ent} is the recovery of minerals by entrainment, $R_{overall}$ is the recovery by both entrainment and true flotation, and R_{water} is the water recovery.

Equation 2-15 requires that the degree of entrainment of different mineral species can be calculated based on the particle density and the degree of entrainment of the entrainment tracer. As outlined in **Section 2.5.3**, the relationship between particle density and the degree of entrainment is still a

matter of conjecture. Often, for practical purposes, the measured degree of entrainment of the tracer is used as an estimate of the degree of entrainment of the other minerals in the system ignoring the differences in density.

It should be emphasized that, intentionally adding a liberated tracer to a flotation system (which does not already exist in the ore) can always allow assessment of the contribution of true flotation and entrainment to the recovery of mineral particles. Furthermore, using an existing non-floating mineral in the ore as a tracer also becomes a realistic option if a significant proportion of the gangue is fully liberated in the sizes of interest or when sized liberation data is available to enable an analysis of only the liberated gangue. The entrainment tracer method has been applied by a number of researchers. For example, Vianna (2004) calculated the contribution of entrainment and true flotation to the recovery of valuable galena mineral particles in flotation tests at 1, 2, and 3 cm of froth height both on an un-sized basis and a size-by-size basis with quartz as an entrainment tracer.

In flotation the entrainment tracer method is the most common methodology used to estimate the recovery by entrainment, because this method is suitable for the estimation of the entrainment without any change in the operating conditions of the flotation tests. Further, it not only allows for the estimation in batch flotation but also is applicable for measurement in continuous flotation cells. These advantages over the other methods outlined above, therefore, make the entrainment tracer measurement technique more practical and more widely applied.

To perform entrainment assessment for real ores where liberation in the finer sizes is incomplete or in applications where a significant amount of the gangue in the feed is floatable (e.g. cleaners), liberation data can be used to isolate and calculate the recovery of only the liberated gangue in the system. Liberation information does, however, suffer from stereological error which results in some of the gangue minerals identified as liberated actually being locked with floatable mineral species. This results in erroneously higher entrainment values. This is why it is better to study entrainment using gangue which is known to be fully liberated.

2.7 Modelling of entrainment

To date, a number of models have been proposed to characterise and predict entrainment. These entrainment models currently available in the literature can be split into two categories: those that predict entrainment flow directly and those that estimate the degree of entrainment and water recovery which are then used to predict entrainment flow.

2.7.1 Direct estimation of entrainment flow

(1) Maachar and Dobby (1992)

Maachar and Dobby (1992) conducted tests to establish the relationship between water recovery (with respect to the feed) and entrained solids recovery in a laboratory flotation column. Their research work was undertaken in two steps: in step 1, the recovery of water was quantified, and in step 2, the recovery of entrained solids (hydrophilic galena and silica) was calculated. In their study, wash water was added to minimise the recovery of the water including the suspended solid particles reporting to the froth from the pulp region, and entrainment was prevented by maintaining a net downward flow of water (J_B) through the froth. It was found that feed water recovery was a strong function of J_B and the feed rate, and the degree of entrainment was related to the particle size and particle density. Thus, an empirical model was proposed for the estimation of entrainment flow:

$$E_S = \exp\left(-0.0325 \cdot \Delta\rho\right) \cdot \exp\left(-0.063 \cdot d_P\right) \cdot R_{FW} \qquad \text{2-17}$$

$$R_{FW} = 2.58 J_F^{-1} \exp\left(-13.1 \cdot J_B\right) \qquad \text{2-18}$$

where E_S is the recovery by entrainment, R_{FW} is the water recovery from the feed, $\Delta\rho$ is the specific gravity difference between the solids and the water, d_P is the particle size (in μm), J_F is the superficial feed rate (the ratio between the volumetric flowrate and the cell cross-sectional area, in cm/s), and J_B is the bias water flowrate in the froth (the difference between the superficial velocities for the wash water flow and the concentrate water flow, in cm/s).

It should be noted that the model of Maachar and Dobby (1992) was proposed on the basis of work performed in a column flotation cell. It relates the entrainment flow to the water recovery and particle attributes. Further validation and investigation are required before it can be applied with confidence to other flotation environments.

(2) Moys (1978)

Moys (1978) conducted experiments in a flotation cell to investigate the effect of various mechanisms (true flotation, entrainment and drainage) on the gradients of mineral concentration in the froth. Results showed that the entrainment mechanism was related to the state of solids suspension in the flotation cell and the drainage rate in the froth. Based on the entrainment

mechanism, a mathematical model of entrainment involving a large number of parameters was proposed:

$$m_{ei} = \alpha_R \cdot m_{ei}(z_W) \qquad 2\text{-}19$$

$$\alpha_R = Q_c / G \qquad 2\text{-}20$$

$$m_{ei}(z) = m_{ei}(0)\exp\left[-\frac{k_{ei}}{G}\int_0^z A_c(z')dz'\right] \qquad 2\text{-}21$$

$$k_{ei} = k_q + \alpha_e(\rho_i - 1.0) \qquad 2\text{-}22$$

where m_{ei} is the entrained mass flowrate (kg/s), z_W is the concentrate weir height (m), α_R is the fraction of upward-moving streams which enter the concentrate stream, Q_c is the volumetric flowrate (air plus slurry) of the froth flowing over the concentrate weir (m^3/s), G is the gas flowrate (m^3/s), z is the height above the froth/pulp interface (m), k_{ei} is the rate constant for transfer of entrained particles into the downward-flowing stream (s^{-1}), $A_c(z)$ is the cross-sectional area of the cell at the froth height z (m^2), k_q is the rate constant for transfer of entrained water to the downward-flowing stream (s^{-1}), α_e is a parameter used for determining k_{ei}, while ρ_i is the particle specific gravity.

The model of Moys (1978) includes a large number of parameters, some of which are difficult to estimate or measure, such as the entrainment flow into the froth from the pulp across the pulp/froth interface. Hence, this method may not be directly applied to predict entrainment in industrial applications. However, the model provides insight into the flotation process, and reveals that parameters such as the gas rate, product removal rate, particle specific gravity, froth height and water recovery rate are of importance in influencing the recovery by entrainment.

(3) Neethling and Cilliers (2002a)

The model proposed by Neethling and Cilliers (2002a) is based on fundamental descriptions of the behaviour of a flotation froth. This model considers the froth to consist of three phases (liquid, gas and solid) and involves modelling of bubble motion, liquid motion and solids motion. It was proposed in their model structure that the motion of particles attached to the bubbles follows the

path of the bubbles, while that of unattached particles is likely to follow the liquid in the froth. The unattached particles move relative to the liquid by means of the following mechanisms: hindered settling, geometric dispersion and Plateau border dispersion. The resulting model was proposed to predict the gangue recovery by entrainment:

$$F_{y,Entrained} = -f(Q, Q_{Air}) \times \frac{\varepsilon_F}{\varepsilon_P} \times \frac{\partial C_{S_i}}{\partial y} + (Q - v_{Settling,i} \times \varepsilon_F) \times \frac{C_{S_i}}{\varepsilon_P} \qquad 2\text{-}23$$

where $F_{y,Entrained}$ is the vertical flux of the gangue at a point y (kg/(s m^2)), $f(Q, Q_{Air})$ is the dispersion function, dependent on the air and water flux, among other variables, ε_F is the fractional slurry content of the froth, ε_P is the fractional water content of the Plateau borders, Cs_i is the concentration of solids of class i within the Plateau boarders (kg/m^3), Q is water flux (kg/(s m^2)), while $v_{Settling,i}$ is the hindered settling velocity of particle type i (m/s).

This mathematical model asserts that the vertical flux of the gangue at any point is the outcome of the (usually) upward motion of the liquid, the downward settling of the particles under gravity and the dispersion of solids. Although this fundamentally based model is not suitable for routine use in predicting entrainment, it provides the mechanisms by which solid particles move in a froth which is particular valuable information that can potentially be used to develop a simplified fundamental model, i.e. a semi-empirical model.

2.7.2 Estimation of entrainment using classification functions and water recovery

Entrainment recovery can also be estimated using the degree of entrainment and water recovery. According to Equations 2-7 and 2-8, the recovery by entrainment can be obtained if the degree of entrainment and the water recovery are known. In the literature a number of models have already been proposed to estimate the relative drainage of solids to water in the froth, the degree of solid suspension in the pulp and water recovery.

2.7.2.1 Degree of entrainment modelling

(1) Ross and Van Deventer (1988)

The research work conducted by Ross and Van Deventer (1988) aimed to investigate the mass transport of mineral particles and water in a flotation column froth. In their study, three size fractions of a sulphide ore (-212+150 μm,-75+38 μm and -38 μm) were tested. During the tests, samples of froth were collected at different froth heights, with a slide used to separate the pulp and froth phases to enable the ratio of the mass of solids and water in the froth to that in the pulp to be determined. A model was developed to describe the behaviour of mineral particles and water in the cleaning zone of the froth. The degree of entrainment is expressed as:

$$X_i = 1 - 0.429\left[\log\left(d_i\right) - 1\right]\left[\rho_s - 1\right] \qquad 2\text{-}24$$

where X_i is the degree of entrainment of the ith size fraction, d_i is the particle diameter (μm), and ρ_s is the density of the particle (g/cm^3).

The model of Ross and Van Deventer (1988) is simple, and it is related to the particle size and the particle density. However, the equation used in the model does not satisfy the boundary conditions of the system. When particle size is less than 10 microns, the model estimates that the degree of entrainment will be greater than 1, which means that the concentration of gangue particles in the water in the concentrate is greater than in the pulp phase which is not an expected result in froth flotation.

(2) Bisshop and White (1976)

A series of steady-state tests involving minerals of different particle density such as quartz, fluorite, hematite, scheelite, cassiterite as well as NBHC tailings (a mineral composite sample of flotation tailing from New Broken Hill Consolidated Limited) was conducted by Bisshop and White (1976) in a continuously operated pilot-scale flotation cell at the Julius Kruttschnitt Mineral Research Centre. In their research work, a semi-empirical model was proposed, relating the degree of entrainment to the residence time in the froth phase, particle size as well as the densities of entrained mineral particles and the pulp. The model is given by:

$$ENT_i = \frac{1+\alpha \cdot \tau_{fw}}{1+\beta_i \cdot \tau_{fw}} \qquad 2\text{-}25$$

$$\alpha = k_1 \cdot (\rho_m - \rho_p) \qquad 2\text{-}26$$

$$\beta_i = \alpha \cdot \exp(k_2 \cdot d_i) \qquad 2\text{-}27$$

where ENT_i is the degree of entrainment of the *i*th size fraction on the basis of the tailings, α and β_i are the drainage parameters of water and entrained particles, respectively, τ_{fw} is the residence time of entrained materials in the froth (froth volume divided by the volumetric flow of entrained materials in the concentrate) (sec), ρ_m and ρ_p represent the particle density of the entrained solids and the pulp (g/cm^3), respectively, d_i represents the entrained particle size (μm), while k_1 and k_2 are empirical parameters.

The empirical parameters k_1 and k_2 in the equations are determined by the experimental system and operating conditions used in the given system. In the model of Bisshop and White (1976), the degree of entrainment is directly related to the froth residence time of entrained materials (solids and/or water) in the froth, which can be calculated from the froth volume divided by the volumetric flow of entrained materials in the concentrate. In general, the volumetric flow of entrained materials in the concentrate is a prediction by the proposed model. Therefore, the model developed by Bisshop and White (1976) is difficult to use for prediction when the volumetric flow of entrained material is unknown (Savassi, 1998).

Zheng et al. (2006b) fitted the model of Bisshop and White (1976) to the experimental data for silica obtained from operating an Outokumpu 3 m^3 tank cell at the Xstrata Mt. Isa Mines copper concentrator. Results showed that the model fitted the experimental data well over the entire range of particle sizes. However, Zheng et al. (2006b) stated that the fitted curve was prone to resulting in inconsistent trends between the degree of entrainment and particle size.

(3) Kirjavainen (1992)

Kirjavainen (1992) investigated the entrainment of hydrophilic particles in a continuous laboratory flotation system (flotation products were returned immediately back to the cell by a small peristaltic pump), and proposed a mathematical model for the degree of entrainment. The flotation

experiments were carried out using two minerals (quartz and phlogopite) at different slurry densities with only frother added. A number of factors including the particle mass, particle shape, water recovery rate, and slurry viscosity, etc. were related to the degree of entrainment in the flotation tests, and these factors were incorporated into the developed model:

$$P = \frac{w^{0.7}}{w^{0.7} + b\Psi\eta^{-0.5\Psi} m^{0.5\Psi-0.4}} \qquad 2\text{-}28$$

where P is the entrainment factor (the ratio of the recovery of gangue and water), w is the water recovery (kg/m^2/s), m is the particle mass (pg), η is the slurry viscosity (mPa.s), ψ is a dynamic shape factor, while b is a constant.

(4) Savassi et al (1998)

The model developed by Savassi et al. (1998) for the degree of entrainment in a conventional flotation cell, as shown by Equations 1-1 and 1-2 in **Chapter 1**, was found to fit a number of industrial data sets well. The model relates the degree of entrainment to the particle size only, and it was derived using a hyperbolic secant function modified to comply with the following boundary conditions (Savassi et al., 1998):

$$d_i \to \infty \Rightarrow ENT_i \to 0$$
$$d_i \to 0 \Rightarrow ENT_i \to 1$$
$$d_i \to \zeta \Rightarrow ENT_i \to 0.20$$

This model currently only relates the degree of entrainment to particle size. Savassi et al. (1998) clearly stated that, because of the complex nature of froths, that the parameters δ and ζ would change with cell operating conditions and for more accurate prediction of entrainment recovery, auxiliary parameter predictive equations would need to be developed.

Zheng et al. (2006b) also used Savassi's model to fit the experimental data for silica obtained from operating an Outokumpu 3 m^3 tank cell at the Xstrata Mt. Isa Mines copper concentrator. Results showed that this model fit the experimental data well only in the fine particle size and mid-size range, but the entrainment of the relatively coarse particles was underestimated, as the model purposely forced the line towards zero at the coarse end of the particle size range. Zheng et al. (2006b) noticed that the actual *ENT* values were larger than that predicted by Savassi's model at the

coarse end of the particle size range, and attributed it to the "entrapment" mechanism by which coarse particles are not able to drain from the froth due to their large particle size. Generally, entrapment occurs when the thickness of the froth lamellae and Plateau borders reduces to a value similar to or less than the particle size, and the free drainage of the particles ceases.

(5) Yianatos and Contreras (2010)

Flotation tests were conducted by Yianatos and Contreras (2010) in an industrial 130 m^3 flotation cell in a rougher bank of a copper flotation circuit, and a dimensionless entrainment factor EF_i related to the mean particle size was proposed:

$$EF_i = \exp\left(-0.693\left(\frac{d_{p,i}}{\delta}\right)^{\phi}\right) \qquad 2\text{-}29$$

where EF_i is the entrainment factor representing the gangue/water recovery ratio, $d_{p,i}$ is the mean particle size of class i (μm), parameter δ (in microns) corresponds to the mean particle size at EF_i=0.5 for each data set, while ϕ is the drainage parameter depending on the mineral characteristics and cell operating conditions (i.e. cell design and water recovery).

The dimensionless entrainment factor EF_i complies with the following boundary conditions, with the smallest particles recovered at the same rate as water (EF_i=1), while the coarsest particles are not recovered ($EF_i=0$):

$$EF_i = 1.0; \frac{d_{p,i}}{\delta} = 0$$
$$EF_i = 0.5; \frac{d_{p,i}}{\delta} = 1$$
$$EF_i \to 0; \frac{d_{p,i}}{\delta} \to \infty$$

This model is very similar to that proposed by Savassi et al. (1998). Parameter δ in Equation 2-29 was shown to be related to the water recovery but the form of the relationship changed with flotation cell design and operating conditions of the flotation system. However, no specific equation was proposed by the authors relating the parameters ϕ and δ to any of these variables.

(6) *Neethling and Cilliers (2009)*

The degree of entrainment has been fundamentally modelled by Neethling and Cilliers (2009) incorporating the effect of both particle size and density. The model is based on an assumption that solid motion in a froth is a consequence of a balance between liquid motion, particle settling and particle dispersion (Neethling and Cilliers, 2002a,b). Mass flux of solids is a function of solid concentration and can be described by Equation 2-30:

$$F_s = (-D_{Axial}\frac{dC}{dy} + (v_l - v_{set})C)\varepsilon \qquad 2\text{-}30$$

where F_s is the mass flux of solids (kg/(s m^2)), as a function of solids concentration C (kg/m^3), D_{Axial} is the axial dispersion coefficient which can be approximated from the gas velocity (m^2/s), liquid velocity and Péclet number, v_l is the water velocity (m/s), v_{set} is the average solid particle settling velocity through the randomly-oriented Plateau borders (m/s), ε is the liquid content and y is the vertical position of solid particles in the froth (m).

At the pulp/froth interface, it is assumed that solid dispersion and solid settling determine the amount of solids entering the froth from the pulp (i.e. the solid flux and liquid velocity in Equation 2-30 are assumed to be relatively small). In this region:

$$C \approx C_{Pulp}\exp(-\frac{v_{set}}{D_{Axial}}y) \qquad 2\text{-}31$$

Liquid velocity at any vertical position within the froth is determined by the superficial net liquid velocity and liquid holdup at that position, as shown by Equation 2-3. At higher levels within the froth, liquid velocity increases due to a decrease in the liquid holdup as a consequence of water drainage. Due to the decrease in the liquid holdup with height in the froth, there is a critical point (y_{crit}) where the liquid velocity equals the settling velocity of a solid particle. Below that critical point, settling velocity is dominant while above the critical point, liquid velocity is dominant.

It is further assumed that in the middle region of the froth, that Equation 2-31 can be used to calculate solid concentration up to the critical froth height as settling velocity is the dominant mechanism. Above this critical point the solid content becomes constant as particle settling

becomes negligible. The overflowing solid content can then be estimated using that at the critical height:

$$ENT \approx \exp(-\frac{v_{set}}{D_{Axial}} y_{crit}) \qquad 2\text{-}32$$

A series of equations were also proposed by Neethling and Cilliers (2009) to determine the critical point for the estimation of *ENT* based on a number of assumptions. When the liquid velocity at the interface is dominant, *ENT*=1.

Based on the above model, the gangue concentration will decrease significantly at the pulp/froth interface and then be observed to continually decrease until a critical point in the froth is reached, after which gangue concentration will remain constant. As outlined in **Section 2.3**, a significant change in gangue concentration is often observed at the interface but evidence to date suggests that no significant decrease is observed after this point.

Equation 2-31 was developed based on the assumption that liquid velocity at the interface is very small in comparison to the other terms involved. This may not be a valid assumption. No experimental verification has yet been provided to validate the underlying assumptions associated with this model.

This model in its current form is impractical to use routinely for entrainment prediction. In this model, the critical froth height (y_{crit}) and the axial dispersion coefficient (D_{Axial}) are difficult to estimate which cannot be easily calculated from the prevailing data collected from industrial applications. It is an indication of the difficulty associated with the application of this fundamental model as it requires the measurement of parameters not usually performed in a flotation test.

Nevertheless, the study conducted by Neethling and Cilliers (2009) considers the fundamental forces acting on the particle and makes assumptions to model the degree of entrainment. This is a worthwhile approach to consider when developing semi-empirical models which can be used for more routine use. It is worthy of additional investigation.

(7) Classification effect in the pulp

The contribution of Zheng et al. (2005) to the modelling of entrainment is the development of a classification function to account for the state of solids suspension in the pulp phase. Using experimental information, it was shown that the classification function could be related to particle size using the following relationship:

$$CF_i^P = \phi \cdot e^{-\lambda d_i} \qquad 2\text{-}33$$

where CF_i^P, the ratio of the mass of mineral in size i per unit mass of water at the top of the pulp zone to the mass of mineral in size i per unit mass of water in the tailing, is the classification function describing the degree of solids suspension in the pulp phase, d_i is particle size (µm), and both ϕ and λ are constants.

The classification function can be used in conjunction with water recovery and the degree of entrainment across the froth phase to estimate the entrainment flow. It enables the effects of incomplete solids suspension to be incorporated into the overall entrainment calculation. Most researchers often relate entrainment directly to the water recovery and the degree of entrainment in the froth, assuming the pulp phase is perfectly mixed. As shown in Table 2-1, suspension is incomplete in larger industrial flotation cells. The development of CF_i^P eliminates the need to assume perfect mixing in the pulp enabling classification effects in both the pulp and froth to be incorporated in an entrainment model. Nonetheless, it is worth mentioning that more work is still required to enable prediction of CF_i^P as the constants in the model would be expected to change with flotation conditions e.g. gas flowrate, impeller speed and potentially pulp viscosity.

2.7.2.2 Modelling of water recovery

Water recovery plays an important role in determining the gangue recovered to the concentrate by entrainment. A number of models have been proposed to describe and assess water recovery. In the literature, some water recovery models are linked to the solids recovery (King, 1973; Alford, 1990), while some other models relate the water recovery to the cell operating conditions such as the froth residence time (Savassi, 1998; Gorain et al., 1998; Harris, 2000; Moys, 1984). There are also models developed based on the fundamental mechanism of water transfer in a flotation cell (Neethling et al., 2003). These models were reviewed and evaluated by Zheng at al. (2006a) with the data from an Outokumpu 3 m^3 cell operating in the Xstrata Mt Isa copper concentrator.

It was found that none of the current empirical models to date could be used to accurately predict the change in the water recovery with a change in the feed or operating characteristics of a flotation cell due to the fact that the water reporting to the concentrate is very sensitive to the froth structure. It is worth noting that the fundamental water recovery model of Neethling et al (2003) potentially can be employed to estimate the water recovery for prediction purposes. However, the techniques required to accurately measure physical variables in the model such as air recovery and bubble size at the froth surface limits its application (Zheng et al., 2006a). Further research with the aim of developing a semi-empirical water recovery model that can accurately and more practically be used for prediction would therefore be worthwhile.

2.7.3 Summary of entrainment modelling

The modelling of entrainment ranges from the direct estimation of entrainment flow to the indirect estimation using the water recovery and the classification functions in both the froth and pulp. These entrainment models were mostly developed from entrainment studies performed in flotation columns and only a few developed in mechanical flotation cells. Because of the effect of flotation cell design on entrainment, the use of these models in different types of flotation cells, especially the empirical models, needs further validation.

The different variables affecting entrainment have been selectively studied by different authors and various models have then been developed. The fundamental models include a large number of variables, some of which are difficult to measure or estimate. In comparison, the empirical and semi-empirical models have proven to be more practical for routine prediction, as they are simple equations with easily-measured variables. However, there is no consensus as to the variables which should be included in these models, as evidenced by the different forms of equations used in the abovementioned entrainment models. It is not clear, however, which factors have a significant enough effect to warrant inclusion in a predictive entrainment model. Moreover, the underpinning mechanisms by which these significant factors affect entrainment have not, experimentally, been comprehensively studied. The challenge remains to develop an entrainment model that can be applied, practically, over a wide range of applications. To achieve this goal, there is a need to further investigate the factors that affect entrainment to determine those that are significant enough to be included in an entrainment model and the mechanisms involved.

2.8 Summary of literature review

In froth flotation, entrainment results in a large amount of fully-liberated fine gangue mineral particles reporting to the concentrate along with valuable minerals recovered by true flotation, which leads to deterioration in the concentrate grade. Many factors in both the pulp and the froth phase (e.g. particle size and density, solid concentration in the pulp, gas flowrate, froth height, impeller speed, froth residence time, frother, particle hydrophobicity as well as rheology) are reported to affect the transfer of gangue mineral particles from the pulp to the concentrate. These factors result in a change in the gangue recovery by entrainment by either affecting water recovery and/or the degree of entrainment. As yet, however, there has been no comprehensive study of entrainment that statistically investigates and compares the impact of the influencing factors on entrainment recovery (i.e. the degree of entrainment and water recovery).

Motion of entrained solid particles in the water phase of a flotation froth will be a consequence of the various forces acting on the particle. Experimental observations have shown that the gangue solid concentration above the pulp/froth interface remains relatively constant, suggesting that entrainment classification in the froth may occur predominantly at the pulp/froth interface and it is the forces in this region that are the predominant factors. According to the Bubble Swarm Theory, water and suspended solid particles are pushed upwards by rising bubbles in the pulp/froth interface region. In this scenario, entrained particles will be subjected to an upward drag force by the water as it moves up with the bubbles which is counteracted by a downward tendency for solid settling due to gravity. However, more experimental work is required to investigate the validity of this mechanism.

To quantify entrainment in a flotation cell, a number of methods have been proposed. The use of a fully liberated gangue mineral tracer to quantify entrainment has many advantages over other methods. Fully liberated gangue can be assumed to be recovered only by entrainment and therefore provides an accurate estimate of the degree of entrainment.

A number of entrainment models, ranging from the empirical and semi-empirical to the fundamental have been developed for both mechanical flotation cells and flotation columns. They incorporate a range of different influencing variables. It is concluded, however, more work is required to improve the current empirical models by including the factors that have a significant enough effect. The fundamental models are impractical to apply routinely and are yet to be sufficiently validated using experimental information in a range of different systems. There is a need for the development of a more sophisticated practical entrainment model for use in process

modelling and optimisation. Ideally this new model would incorporate the significant variables and mechanisms of entrainment process.

2.9 Research questions and hypotheses

A good model should describe the basic phenomena and account for the factors affecting the response significantly. It is not clear, however, what factors have a significant effect enough to warrant inclusion in a predictive entrainment model. For example, many studies have shown that the degree of entrainment is influenced by particle density (Johnson, 1972; Bisshop, 1974; Thorne et al., 1976; Maachar and Dobby, 1992; Drzymala, 1994). However, the extent to which it affects the degree of entrainment is unresolved.

In addition, the underpinning mechanism by which these significant factors affect entrainment is also unsolved. Literature proposes that the motion of gangue mineral particles are subject to a number of forces, such as drag force and gravitational force. Despite this, there is still lack of experimental evidence demonstrating the mechanism by which the key variables of the flotation process affect the entrainment parameters. It should also be noted that proposed entrainment mechanisms have been studied mostly in flotation columns. It is unknown whether the information obtained from these studies is useful to interpret the behaviour of gangue mineral particles in mechanical flotation cells.

As indicated from this literature review, it is evident that numerous studies on many aspects of entrainment have been carried out over the last few decades. However, there is still a need to further understand the entrainment process, especially in terms of the mechanisms involved and to determine the main variables which significantly affect the process.

The research questions of the thesis are:

- What are the most significant variables affecting the entrainment parameters (i.e. the degree of entrainment and water recovery) in froth flotation?
- What are the mechanisms underpinning the effects of these major influencing variables?
- Can entrainment be better predicted using these identified effects and mechanisms?

Based on the literature review, the following hypotheses have been formulated with the objective of answering these research questions:

- There are some other operational variables and particle attributes that significantly affect the degree of entrainment, apart from particle size.

- Flotation variables affect the degree of entrainment by changing the forces acting on the solid particles at the interface which determine the relative motion of solid particles to water.
- An improved understanding of the key variables affecting entrainment and their underpinning mechanisms will enable the development of improved entrainment models.

Chapter 3 Experimental

Batch laboratory flotation experiments are performed in this thesis to (1) investigate what particle properties and flotation operational variables have a significant effect on entrainment parameters and (2) explore the mechanisms involved. This chapter describes the ore sample and reagents used in this thesis and details the experimental procedure and measurements performed during these experiments.

3.1 Introduction

The objective of this thesis was to investigate the significant factors that affect entrainment and to explore the mechanisms involved. To perform this investigation it was decided to perform batch laboratory flotation tests using pure minerals. Tests either involved floating an artificial ore created by mixing a floatable mineral with a non-floating gangue mineral or entrainment only tests where only a non-floating gangue mineral was present in the flotation cell. As outlined in **Chapter 2**, entrainment is best characterised using a non-floating tracer mineral. By using pure minerals in the tests, it is known that the gangue mineral recovery is primarily by entrainment and cannot be due to true flotation. Thus entrainment recovery and entrainment parameters can be measured accurately.

The materials, reagents and the methodologies involved in the experiments are presented in the following sections. The repeatability of the test procedure is also outlined.

3.2 Mineral samples and reagents

3.2.1 Minerals

The non-floating gangue mineral used in the flotation tests was either quartz (SG=2.65), ilmenite (SG=4.70) or hematite (SG=5.26). In the artificially created ore tests, one of these gangue minerals would be mixed with pure chalcopyrite (SG=4.2). All these pure minerals were purchased from Geodiscoveries.

3.2.2 Reagents

5.0 wt.% sodium hydroxide solution which was prepared using solid sodium hydroxide supplied by EMD Millipore Corporation, was used to adjust the pH of the flotation slurry prior to flotation.

Sodium ethyl xanthate (SEX), which was supplied by Qingdao Lnt Chemical Ltd., was used as collector. Its chemical structure is shown in Figure 3-1:

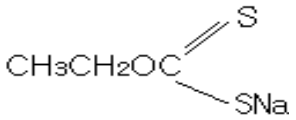

Figure 3-1 Chemical structure of Sodium ethyl xanthate.

The frother used in the flotation experiments was either Dowfroth 250, a polypropylene glycol-type frother, or Methyl Isobutyl Carbinol (MIBC). They were supplied by Dow Chemical Company and Recochem Inc., respectively. Table 3-1 shows some properties of these two frothers. Note that foaming ability of a frother is described by the dynamic foamability index (DFI) and the critical coalescence concentration (CCC). The CCC value is the minimum frother concentration required to prevent coalescence in the pulp phase of a flotation cell.

Table 3-1 Characteristics of the frothers used in the experimental program including their CCC values and experimentally determined DFI values (after Laskowski et al., 2003).

Frother	Chemical formula	CCC (ppm)	DFI ($s \cdot dm^3/mol$)
Dowfroth 250	$CH_3(-OC_3H_6-)_4OH$	8.7	208,000
MIBC	$CH_3CHCH_2CH(OH)CH_3$	11.2	34,000

All the reagent solutions used in this study were prepared daily prior to the flotation tests using Brisbane tap water. Brisbane tap water was used throughout the tests. The composition of this water is shown in Table 3-2 (Liu and Peng, 2014).

Table 3-2 The composition of Brisbane tap water (mg/L) (after Liu and Peng, 2014).

Na^+	K^+	Ca^{2+}	Mg^{2+}	Si^{4+}	Cl^-	SO_4^{2-}	CO_3^{2-}
48.6	3.16	25.4	17.3	2.98	105	42.3	58

3.3 Experimental methodology

3.3.1 Feed preparation

On receipt of the samples from Geodiscoveries, the pure chalcopyrite, quartz, ilmenite and hematite samples were crushed to below 3.35 mm using a laboratory jaw crusher and then split to create samples of 53 g, 1530 g, 1530 g and 1530 g, respectively. These weights were chosen so that when

one sample of chalcopyrite was mixed with one sample of a gangue mineral, it would result in a 1 wt.% copper feed grade and approximately 37 wt.% solids in the flotation cell. Chalcopyrite samples were stored in a freezer after preparation to avoid oxidation.

Prior to a particular flotation test, the minerals to be used were ground prior to being mixed in the flotation cell. When chalcopyrite was added to the test, it was ground with water (35 wt.% solids) in a rod mill operated at a speed of 76 rpm for the time required to obtain 80% particles passing 106 μm, a size similar to that in typical flotation plants. The gangue minerals, however, were ground dry in a Bond ball mill for the time required to obtain a P80=70-80 μm. This finer grinding was chosen to result in a substantial quantity of -50 μm particles which typically are recovered by entrainment. The bond mill was chosen to grind the gangue because it resulted in a similar shape of product size distribution whereas the rod mill resulted in significant changes in the shape of the size distribution as different gangue minerals were ground. A similar size distribution was desirable as entrainment measured for each of the different gangue mineral tests would then solely be due to density rather than any particle size effect.

Grinding time calibration tests established the time chalcopyrite and each gangue mineral needed to be ground to produce the target P80 size. Because of the difference in the hardness of the various minerals, different grinding times were required. The size distributions determined by sieving the three gangue minerals after grinding for the target grind time are shown in Figure 3-2. It can be observed that the size distribution of quartz, ilmenite and hematite produced by grinding was very similar.

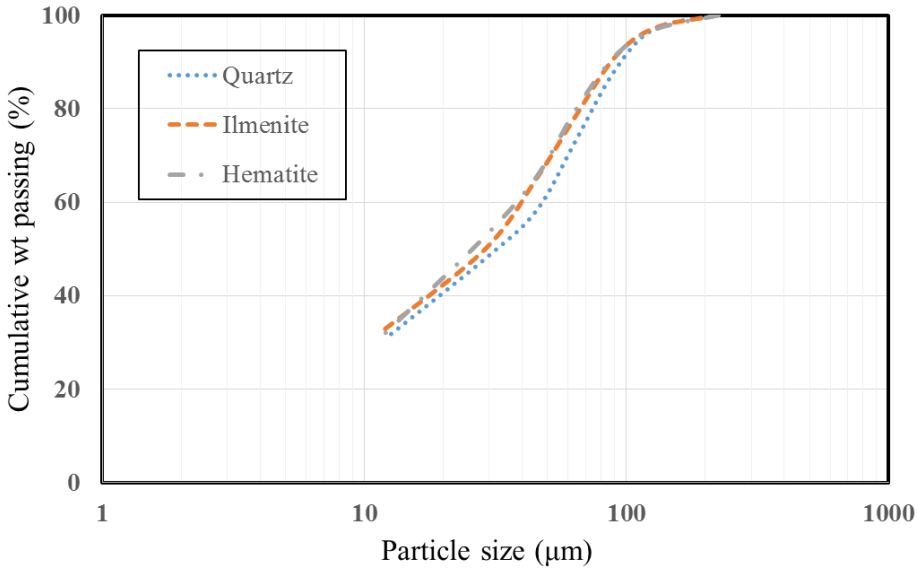

Figure 3-2 Particle size distributions of the quartz, ilmenite and hematite after grinding.

The bond mill used to grind the gangue was 0.305 m by 0.305 m with rounded corners and a smooth lining, and was operated at 70 rpm for grinding. The charge consisted of 285 balls, weighing about 20 kg in total. The rod mill was 0.260 m by 0.203 m, with a charge consisting of four stainless steel rods, weighing 2307 g. Figure 3-3 shows the grinding mills used in this thesis.

a) Rod mill b) Bond ball mill

Figure 3-3 The rod mill and Bond ball mill used in the laboratory tests.

3.3.2 Flotation

3.3.2.1 Flotation apparatus and test procedure

Flotation tests were performed either with a feed which consisted of chalcopyrite mixed with a gangue mineral or using a gangue mineral alone. The first step of the flotation procedure was to grind the mineral(s) used in the test for the target grind time, after which the sample(s) was transferred to a 3.5 L transparent conventional Agitair flotation cell with a movable lip to enable the froth height to be changed without altering the effective pulp volume (3.1 L) (see Figure 3-4). Brisbane tap water was then used to make up the volume in the cell to 3.1 litres. The flotation was conducted with the impeller set to 1200 rpm. The pH of the pulp slurry was adjusted to 9.5 by adding a 5 wt.% sodium hydroxide solution. When chalcopyrite was present in the flotation feed, 50 g/t collector was added to the slurry, while in the gangue only entrainment tests only frother was used. Reagents were then allowed to condition for 1 minute.

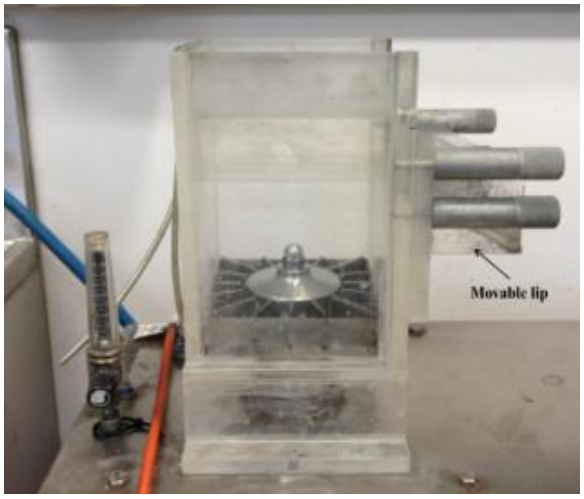

Figure 3-4 Side view of the plastic flotation cell with a movable concentrate lip.

After conditioning, the froth height, gas flowrate and impeller speed were then set to those required for the particular experiment. Once air was introduced into the flotation cell, concentrates produced from the particular experiment were collected over selected time periods. The froth was scraped every 10 s. The froth scraper was designed to only remove the froth above and 1 mm below the cell launder lip to minimise any disturbance to the underlying froth structure. Brisbane tap water was added to the test periodically to maintain the pulp level. This added water was dosed with frother at the same concentration to that established at the beginning of the test with the aim of achieving a relatively constant concentration of frother in the cell throughout the experiment. On completion of

the test, all concentrates (or a combined concentrate) and tailings were weighed wet, filtered, dried and then re-weighed. Samples collected during the chalcopyrite flotation tests were assayed for copper.

3.3.2.2 Calculation of gangue recovery, water recovery and ENT

As gangue minerals used for the experiments were fully liberated and should be hydrophilic in the presence of the xanthate collector, they would be recovered into the concentrate by entrainment only. Recovery of a gangue mineral by entrainment was determined by comparing the mass of the gangue mineral recovered into the concentrate to that in the feed. The degree of entrainment was determined by the mass of gangue mineral per unit mass of water in the concentrate divided by that in the tailing, as shown by Equation 3-1.

$$ENT_i=\frac{(mass\ of\ free\ gangue\ per\ unit\ mass\ of\ water)_{concentrate}}{(mass\ of\ free\ gangue\ per\ unit\ mass\ of\ water)_{tailing}} \qquad 3\text{-}1$$

The gangue concentration in the tailings of a flotation cell decreases with flotation time in a batch flotation test as solid particles and water are continually removed and additional water is added to maintain a constant pulp volume. The overall degree of entrainment in this thesis is calculated by assuming the tailing concentration is that in the cell at the completion of the test. The degree of entrainment across a stage of flotation is calculated assuming the tailing concentration is that at the completion of each stage of flotation.

In the literature, water recovery is calculated using two methods: the percentage of water in the cell recovered into the concentrate, or the water flowrate to the concentrate (Zheng et al., 2006a). In the current study, the first calculation method was used. However, it is worth noting that the water in the cell used in this calculation included both the water initially added to the flotation cell as well as the frother-dosed water which was added to maintain the pulp level. Caution is required when comparing the water recovery values presented in this thesis with those presented elsewhere as they may have been calculated in different ways.

3.3.2.3 Flotation reproducibility

The error associated with the experimental procedure was assessed using the results of four replicate flotation tests performed using an artificial ore created by mixing chalcopyrite and quartz.

The cell was operated at a gas flowrate of 11.5 L/min, a froth height of 1.5 cm, an impeller speed of 1000 rpm and a Dowfroth 250 concentration of 10 ppm.

The reproducibility of copper recovery, entrainment recovery of quartz and water recovery achieved in these replicate tests is shown in Figure 3-5. After 10 min flotation, the relative error of copper recovery, quartz recovery and water recovery are 3.4%, 3.3% and 5.5% at a 95% confidence level, respectively. Note that only three tests were chosen for the demonstration of the reproducibility of copper recovery. A low copper recovery found in the other test is an outlier and is not included in the reproducibility demonstration. This degree of error is considered sufficiently low to enable the effect of different variables on the entrainment parameters to be determined with statistical confidence.

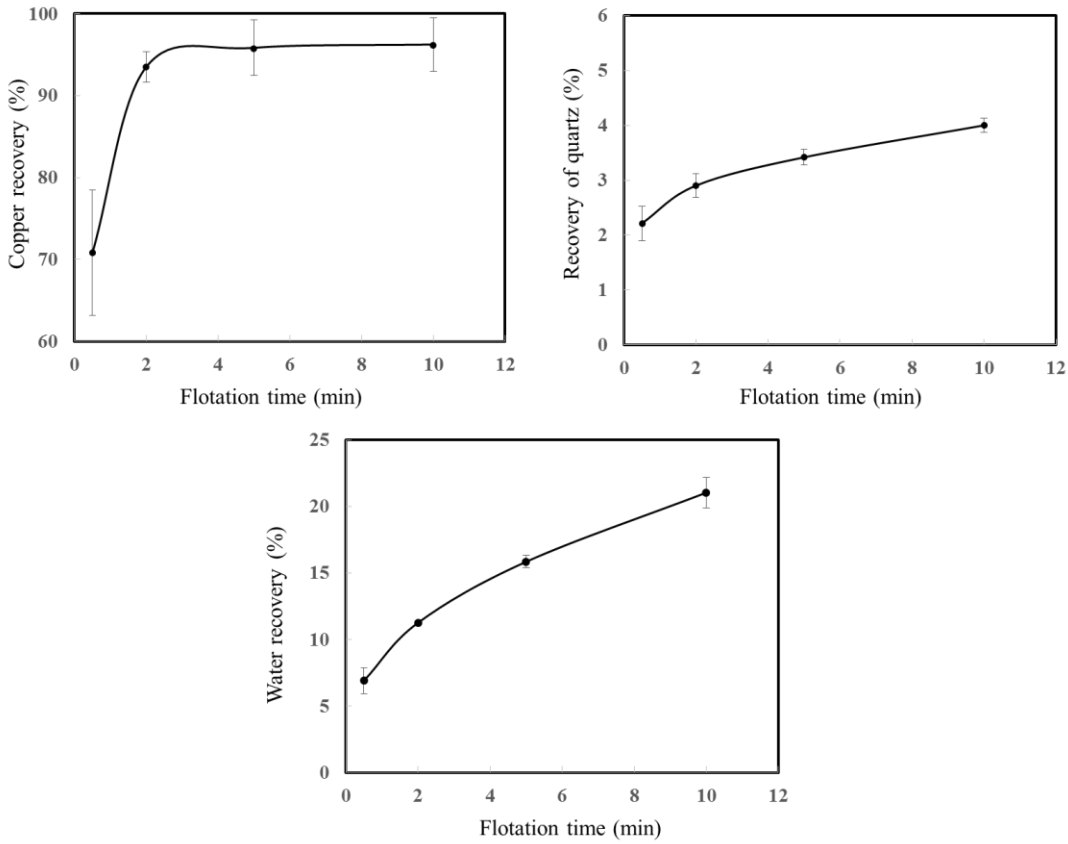

Figure 3-5 Copper recovery, entrainment recovery of quartz and water recovery as a function of flotation time measured in four replicate flotation experiments.

3.3.3 Factorial experimental design and analysis

The design of a good experimental study should be governed by three basic principles: control, randomization, and replication. Control, randomization, and replication are important aspects of experimental design. Control groups and placebo groups are used to avoid confounding the effect of the treatment with other variables. Randomization is necessary in order to make cause-and-effect conclusions. Randomly assigning experimental units to each treatment group can reduce bias that occurs when using other methods, such as allowing participants to choose their own groups or using pre-existing groups that are convenient. Finally, replication is important to be able to infer that a difference observed in your sample is due to a true difference in the population, and not just due to random sampling error.

In this thesis, a two-level factorial experimental design was used to identify the significant factors affecting the degree of entrainment and water recovery in **Chapters 4** and **8**, respectively. In statistics, a factorial experiment generally consists of two or more factors, each with discrete possible values or "levels", and factorial experimental units take on all possible combinations of these levels across all such factors. All of the tests were conducted under relatively homogeneous conditions, in a random order. Such an experiment allows the investigator to study the effect of each factor on the response variable, as well as the effects of interactions between factors on the response variable.

In this two-level factorial experiment designed for the thesis, each variable for testing had a maximum, minimum and centroid value. The difference between the maximum and minimum values represents the range over which each variable was studied. Four central point repeat experiments were used to calculate the experimental error and determine whether the effects observed in experiments were significant or not.

The effects and interactions of the variables on the degree of entrainment at a 95 % confidence level were analysed using Minitab 17 Statistical Software. The software enables a regression model to be fitted to the experimental data to identify important effects. The regression model includes the main effects and the 2-way interactions. Effects of variables are statistically significant when the P values of their coefficient obtained from ANOVA (analysis of variance) are less than the selected significance level of 0.05 (α). P value is defined as the probability of obtaining a result equal to or “more extreme” than what was actually observed, assuming that the null hypothesis (i.e. the tested variable is not significant in this study) is true.

3.3.4 Auxiliary flotation measurements

In **Chapter 2**, it has been hypothesized that forces at the interface determine the motion of solid particles relative to the water. Liquid drag is one of these forces and it will be affected by the velocity of upward liquid flow as well as the fluid rheological properties. Measurements were performed in some of the flotation tests to determine these parameters to assess how the liquid drag force varies as conditions used in the tests are changed.

3.3.4.1 Interface liquid velocity measurement

To estimate the liquid velocity at the interface, liquid content at the interface and the superficial liquid velocity were measured (see Equation 2-3). Superficial liquid velocity was approximated by dividing the water flow measured in the concentrate by the cross-sectional area of a flotation cell. Some methodologies have been developed for measuring gas/liquid holdup in column froths using conductivity, however, little work has been done to estimate the gas or liquid holdup in the froth phase of a mechanical flotation cell (Marchese et al., 1992; Tavera et al., 2001). In this thesis, a froth image analysis approach was introduced to approximate the gas/liquid holdup at the pulp/froth interface. This involved using a video camera with a light source, mounted to a bracket to capture images of the pulp/froth interface through the Perspex wall of the flotation cell. Figure 3-6 shows the schematic of equipment used. Images were captured for the 30 seconds over which each entrainment test was performed. Three clear froth images for each entrainment test were selected randomly and analysed using ImageJ software. The diameters of randomly selected bubbles in the froth layer immediately above the pulp (i.e. 2 mm above the interface) were measured, and the average was used to represent the bubble diameter ($2r$). The length between adjacent air bubbles was also measured, and half of the average length was assumed to be the bubble water thickness (d). As there is a great amount of water at the interface where air bubbles begin to congest, it is reasonable to assume that air bubbles remain spherical in shape. Assuming that the air bubbles (including bubble water) at the interface exhibit hexagonal packing, the volume of air and liquid at the interface can be approximated (see Figure 3-7). The detailed calculation is described in **Appendix A**.

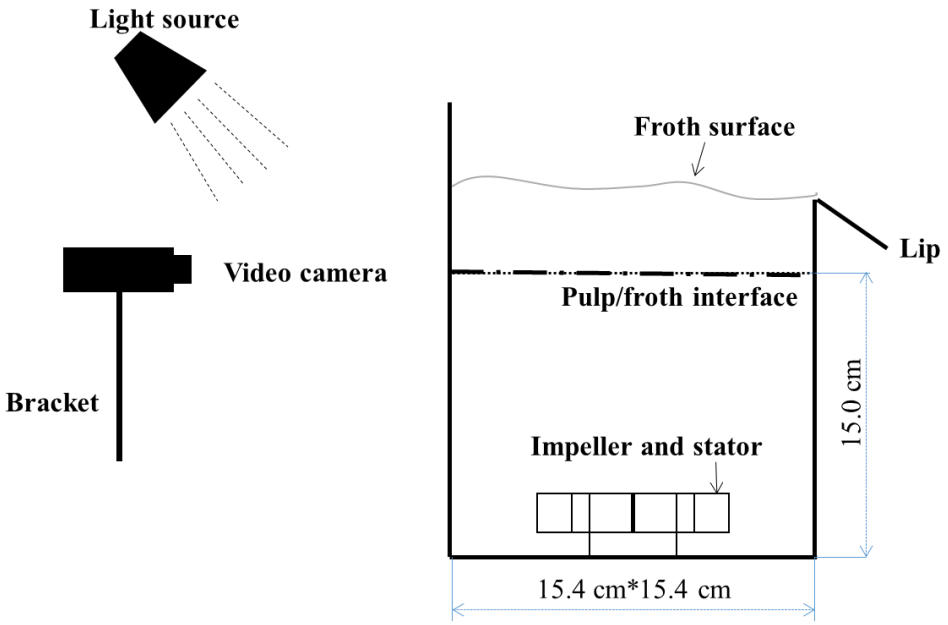

Figure 3-6 Schematic of the equipment used to measure gas/liquid holdup at the pulp/froth interface.

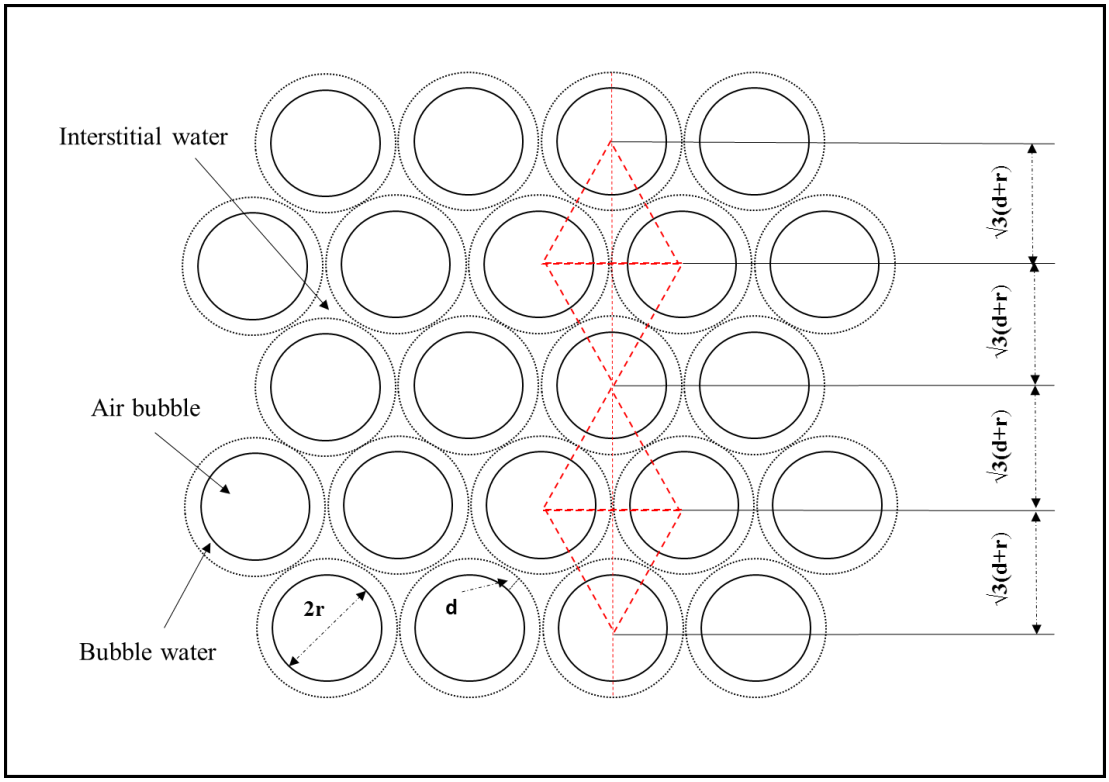

Figure 3-7 An idealised representation of the single-size bubble distribution at the pulp/froth interface used to calculate liquid content.

3.3.4.2 Rheology measurement

Rheology measurements in this study were conducted on a slurry sample prepared in the same manner as the feed to the flotation cell in the batch entrainment tests. Measurements were also performed at a range of different frother concentrations in solution in the absence of solid particles. A Discovery HR-1 hybrid rheometer with a Couette geometry (a rotating bob and a stationary cup with a gap width of 6 mm) from TA instruments was used for the measurements performed at the ambient temperature (25 ℃) (see Figure 3-8). The rheometer was operated in a constant rate-controlled mode with shear rates ramped from 2 s^{-1} to 120 s^{-1} linearly over a period of 2 min. The measured torque was converted by the instrument software into a shear stress from which the viscosity could be obtained from Newton's relationship. The selected shear rate range was based on the finding of Ralston et al. (2007) that the shear rate of 90 s^{-1} is representative of the average shear rate value in a flotation cell. The shear rate at the pulp/froth interface will be smaller.

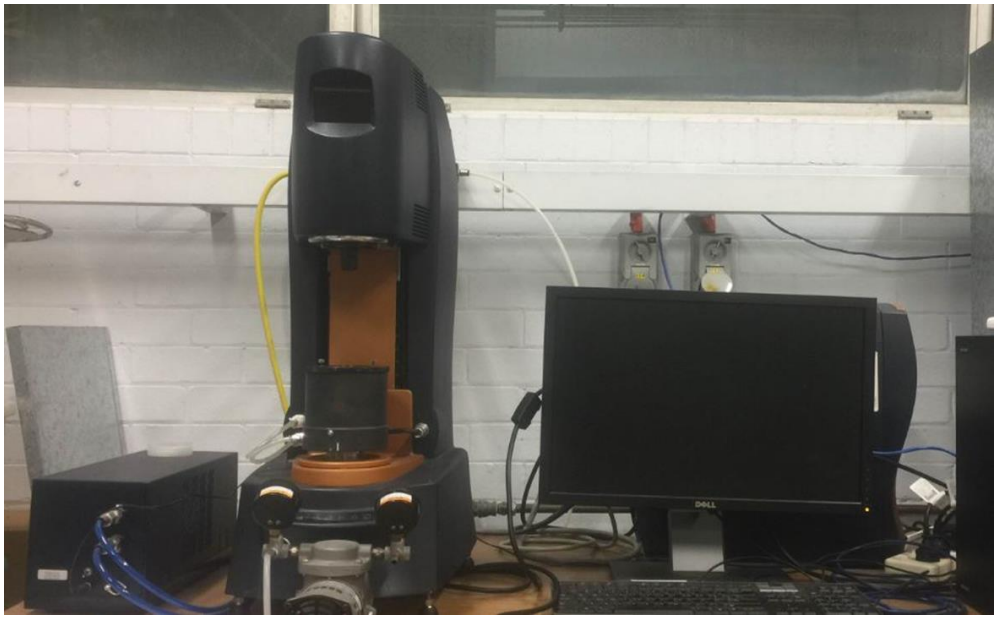

Figure 3-8 The instrument used for rheology measurement.

Chapter 4 The flotation variables that significantly affect the degree of entrainment

This chapter describes an experimental programme used for identifying the significant variables affecting the degree of entrainment and presents the main and interactive effects which were significant with a 95% degree of confidence.

A 2-level factorial batch flotation experiment was designed and the effects of the flotation operating conditions and particle attributes (including impeller speed, gas flowrate, froth height and the specific gravity of the gangue mineral) on the degree of entrainment were investigated.

4.1 Introduction

Literature suggests that there are many flotation variables which affect the degree of entrainment in flotation. It is not clear, however, as to which of these have a significant effect that they should be included in the models used to predict entrainment. A good model should describe the basic phenomena and account for the factors affecting the response significantly (Bisshop, 1974). As outlined in **Chapter 2**, as yet, there has been no comprehensive study of entrainment that statistically investigates and compares the impact of the influencing factors on entrainment recovery.

In this chapter, impeller speed, gas flowrate, froth height and the specific gravity of gangue minerals were the flotation variables investigated in flotation tests performed under similar conditions to that experienced in a typical rougher copper flotation plant. The primary aim was to investigate how these variables affect the degree of entrainment. With a view to providing additional insights into entrainment control, the effect of these variables was also investigated with respect to water recovery (**Chapter 8**).

In the literature outlined in **Chapter 2**, many studies have shown that the degree of entrainment is affected by particle attributes. The importance of particle size has been long recognized as the degree of entrainment is always size dependent with coarse particles exhibiting a low degree of entrainment ($ENT_i \rightarrow 0$) and the fine particles exhibiting a high degree of entrainment ($ENT_i \rightarrow 1$). However, the extent to which particle density affects the degree of entrainment is unresolved and it was therefore chosen as a key variable to be tested. It should be noted that the effect of particle density is often influenced by the effect of particle size, and requires further investigation where the influence of particle size is specifically excluded. Therefore, it was considered better to not vary particle size in the feed but evaluate this effect by sizing the flotation products. These types of sized entrainment results are presented in **Chapter 5**.

The degree of entrainment also changes with operating conditions. Operational variables are always of great importance because they are variables which one can be easily adjusted and manipulated to get the desired flotation performance. Gas flowrate, froth height and impeller speed have all been reported to affect entrainment, but the extent to which these variables affect the degree of entrainment and water recovery and the mechanisms involved are still a matter of conjecture. It was therefore decided to vary these parameters in the experimental program.

Reagent type and dosage, pulp density and copper feed grade will all potentially affect the degree of entrainment, because they will affect conditions in the pulp and the froth structure. It was not feasible to include all of these variables in the test program. However, in batch flotation tests, froth structure and pulp properties change with flotation time due to depletion of the grade of particles in the cell, frother depletion and a reduction in pulp density as the test continues. An analysis of the change in degree of entrainment with time in a batch flotation test should therefore provide a qualitative analysis of the impact of these types of variables on the degree of entrainment. It is however, suggested that in future work, the specific effects of these particular variables be studied.

Initially it was planned to include rheology as one of the variables to be tested in the experimental program. In the preliminary test work performed for this thesis, attempts were made to manipulate pulp viscosity using viscosity modifiers. To achieve even a slight change in the pulp viscosity, a significant dosage of these reagents was required, which resulted in excessive water recovery and froth structure changes to that normally experienced in rougher flotation. Therefore, rheology was removed from the two-level factorial experimental program. It is recommended that the effects of pulp viscosity on entrainment be investigated in future work.

Batch laboratory flotation tests were performed in a 3.5 L mechanical flotation cell according to a 2-level factorial experimental design. All other variables were kept the same to ensure the test results were comparable. The total numbers of tests performed was 20 (including 4 repeats at the central condition). The feed to each test was liberated chalcopyrite mixed with either liberated quartz or hematite (depending on the gangue density specified in the factorial design). The feed copper grade is around 1 wt% and the solids percentage in the pulp is about 37%. 50 g/t of SEX and 10 ppm of Dowfroth 250 were the reagents used for flotation as these are reagents and dosages typically used in chalcopyrite industrial plants. In each flotation test, four flotation concentrates were collected from 0 to 0.5 min, 0.5 to 2 min, 2 to 5 min and 5 to 10 min.

4.2 Results

4.2.1 Determination of variable levels for the factorial experiments

The levels at which each of the variables was tested in the factorial experiments is shown in Table 4-1. These levels of froth height, impeller speed and gas flowrate were those determined experimentally to result in reasonable changes in flotation recovery but not poor flotation behaviour. In flotation, superficial gas velocity (J_g) is the parameter usually used to represent the air flowrate

as it is considered relatively independent of cell size. In industrial applications, J_g usually ranges from 0.6 to 1.5 cm/s (Neethling and Cilliers, 2009). In this study, the minimum, centroid and maximum J_g values chosen were 0.6, 0.8 and 1.0 cm/s, respectively. The corresponding air flowrates were 8.5, 11.5 and 14.5 L/min, respectively. The impeller speeds selected were based on the need to keep the majority of particles in suspension without an overly turbulent pulp/froth interface. The maximum impeller speed chosen was 1200 rpm at which some turbulent effects were visually observed to affect the froth phase. The minimum impeller speed was chosen to be 800 rpm at which there was a small amount of solids observed to be settling in the flotation cell. All froth depths selected were reasonably shallow (i.e. 1.0, 1.5 and 2.0 cm) because deeper froth heights could not be maintained throughout the whole period of the flotation test.

Table 4-1 Levels of each factor used in the factorial experimental design.

Factors	Levels		
	Minimum	Central	Maximum
Impeller speed, rpm	800	1000	1200
Froth height, cm	1.0	1.5	2.0
Air flowrate, L/min	8.5	11.5	14.5
Particle density	2.65	2.65	5.26

It should be noted that, initially, only two gangue minerals were available for use in the experiments and thus particle density was only tested at a high and low level.

4.2.2 Flotation results

The experimental results for the factorial experiments are shown in Table 4-2. Copper recovery was high in the tests. There were large variations in the gangue recovery, which resulted in large variations in concentrate grade. It should be noted that the gangue minerals used in these tests were fully liberated, and they would be recovered into the concentrate only by entrainment. Therefore, the variations in concentrate grade was caused by changes in entrainment recovery.

Gangue recovery in the different tests is a strong function of water recovery which varied significantly (refer to Figure 4-1). However, it also changes as a consequence of variation in the degree of entrainment which can be approximated by the slope of the gangue versus water recovery relationships shown in Figure 4-1.

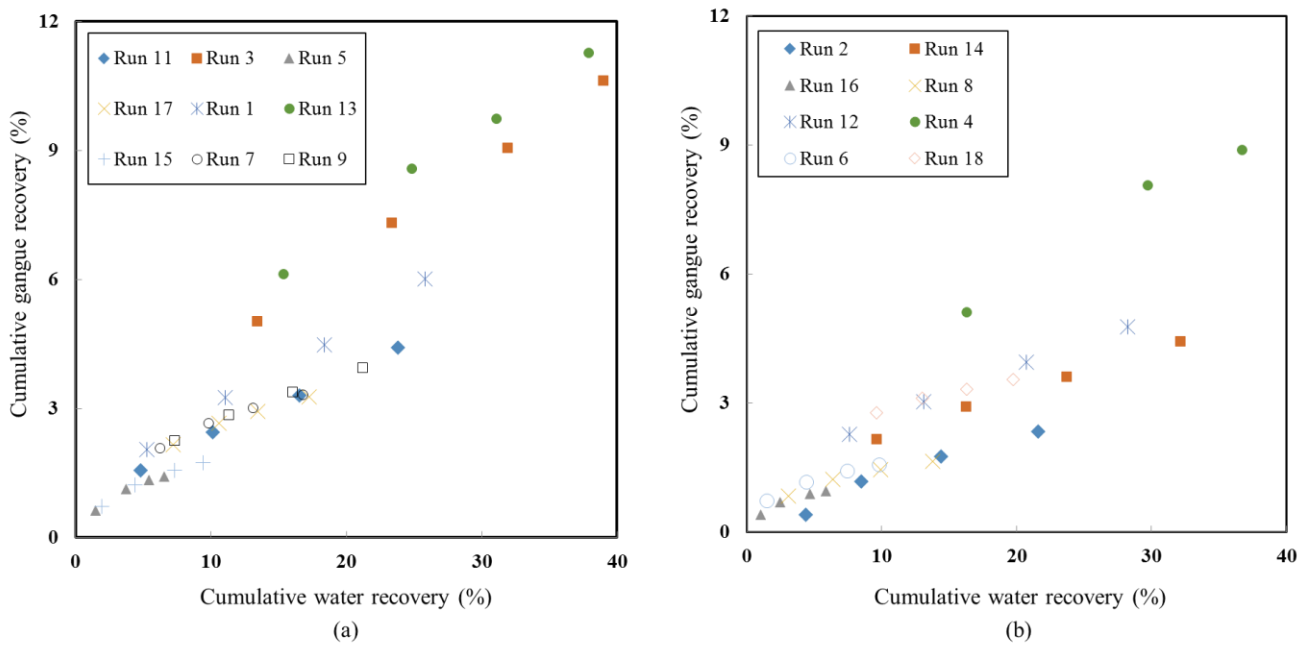

Figure 4-1 The cumulative gangue recovery versus cumulative water recovery: (a) quartz and (b) hematite for concentrates collected after 0.5, 2, 5 and 10 minutes for tests outlined in Table 4-2.

ENT values were calculated for each experimental condition as outlined in **Chapter 3**. The overall *ENT* values ranged from approximately 0.1 to 0.2. In addition, the degree of entrainment was found to vary with flotation time. In each of the twenty batch flotation tests, the *ENT* values dramatically decreased as the experiment progressed. In the following data analysis it will be the overall degree of entrainment that will be the focus as this corresponds to the entrainment over the full 10 minutes of flotation of chalcopyrite, including both roughing and scavenging conditions. A single stage of flotation does not represent the whole process. It should be noted that only the overall degree of entrainment and *ENT1* can be directly compared in the following statistical analysis as these are the only parameters that all have the same feed conditions. An analysis of *ENT1* was not performed in this thesis as over this short time period where a majority of the chalcopyrite is being recovered that other mechanisms may be involved. Potentially the chalcopyrite results in a degree of entrapment that affects the degree of entrainment comparisons between the tests (Ata et al., 2004). Alternatively, the chalcopyrite may not be 100% pure, containing some gangue components. If this gangue is not fully liberated, it will exhibit some floatability or even be a floatable species, which would change the trend. This is considered outside the scope of this thesis, and therefore should be investigated in future work. The results from this thesis pertain to the average over the entire flotation process.

Figure 4-2 shows the overall *ENT* values measured in all the tests (i.e. involving both quartz and hematite) plotted separately against water recovery. The coefficient of correlation (r) of this relationship is -0.010, suggesting that there was no correlation between the measured degree of entrainment and water recovery. This indicates that the variables that affected water recovery were not the same as those that affected the degree of entrainment.

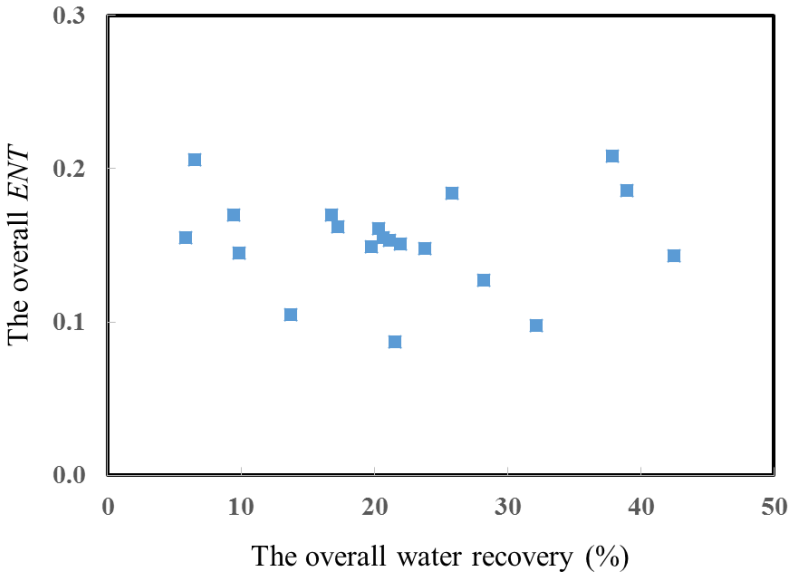

Figure 4-2 The overall *ENT* values versus overall water recovery.

Table 4-2 Summary of two-level factorial experimental results.

Run	*SG*	*IS*, rpm	*AR*, L/min	*FH*, cm	*CR* %	*GR* %	$WR_{overall}$ %	$ENT_{overall}$	*Con Grade*, %	*ENT1*	*ENT2*	*ENT3*	*ENT4*
11	2.65	800	8.5	1	96	4.4	24	0.15	12	0.32	0.15	0.10	0.10
3	2.65	800	14.5	1	94	11	39	0.19	5.9	0.34	0.16	0.11	0.099
5	2.65	800	8.5	2	95	1.4	6.5	0.21	22	0.42	0.20	0.12	0.077
17	2.65	800	14.5	2	95	3.3	17	0.16	15	0.28	0.12	0.079	0.066
1	2.65	1200	8.5	1	98	6.0	26	0.18	10	0.38	0.18	0.13	0.13
13	2.65	1200	14.5	1	99	11	38	0.21	6.4	0.35	0.18	0.10	0.10
15	2.65	1200	8.5	2	97	1.7	9.4	0.17	20	0.37	0.19	0.11	0.073
7	2.65	1200	14.5	2	97	3.3	17	0.17	15	0.32	0.14	0.086	0.070
9	2.65	1000	11.5	1.5	95	4.0	21	0.15	13	0.29	0.13	0.083	0.075
10	2.65	1000	11.5	1.5	96	3.9	21	0.16	13	0.29	0.13	0.091	0.083
19	2.65	1000	11.5	1.5	89	4.0	22	0.15	13	0.32	0.14	0.087	0.074
20	2.65	1000	11.5	1.5	98	3.9	20	0.16	13	0.31	0.13	0.076	0.068
2	5.26	800	8.5	1	97	2.3	22	0.087	18	0.087	0.16	0.078	0.056
14	5.26	800	14.5	1	96	4.4	32	0.098	12	0.21	0.089	0.060	0.052
16	5.26	800	8.5	2	95	1.0	5.9	0.16	25	0.37	0.19	0.083	0.059
8	5.26	800	14.5	2	92	1.6	14	0.11	20	0.26	0.11	0.055	0.041
12	5.26	1200	8.5	1	92	4.8	28	0.13	11	0.28	0.11	0.084	0.065
4	5.26	1200	14.5	1	94	9.6	42	0.14	6.9	0.28	0.13	0.058	0.046
6	5.26	1200	8.5	2	95	1.6	9.8	0.15	21	0.48	0.14	0.077	0.051
18	5.26	1200	14.5	2	94	3.6	20	0.15	14	0.27	0.083	0.048	0.043

Note: *SG* is the specific gravity, *IS* is the impeller speed, *AR* is the air flowrate, *FH* is the froth height, *CR* and *GR* are the assayed copper recovery and gangue recovery, respectively, $WR_{overall}$ is the overall water recovery representing a fraction of water in the cell recovered into the concentrate, *Con grade* is the concentrate grade, $ENT_{overall}$, *ENT 1*, *2*, *3* and *4* are the overall degree of entrainment (total mass gangue in total mass water in concentrate divided by that in tailings) and the degree of entrainment for the four concentrates in a batch flotation test (mass gangue in mass water in each concentrate divided by that in the tailings.

4.2.3 Sensitivity of concentrate grade to the degree of entrainment

As mentioned previously, the overall *ENT* values ranged from approximately 0.1 to 0.2 under the various experimental conditions. It is questionable whether such a change in the *ENT* value will have a significant impact on the concentrate grade. To test this, concentrate grades at different *ENT* values and water recoveries were estimated. In these calculations, copper recovery was set to a constant value. Copper recoveries in the rougher flotation tests performed in this study were generally high, so a copper recovery of 90% was selected. Gangue recovery for each water recovery and degree of entrainment value was simulated using Equation 2-7. Using these copper and gangue recoveries, weighted according to the copper feed grade of 1% used in the test, concentrate grade was then calculated. Figure 4-3 shows the concentrate grade as a function of water recovery and the degree of entrainment determined from these simulations.

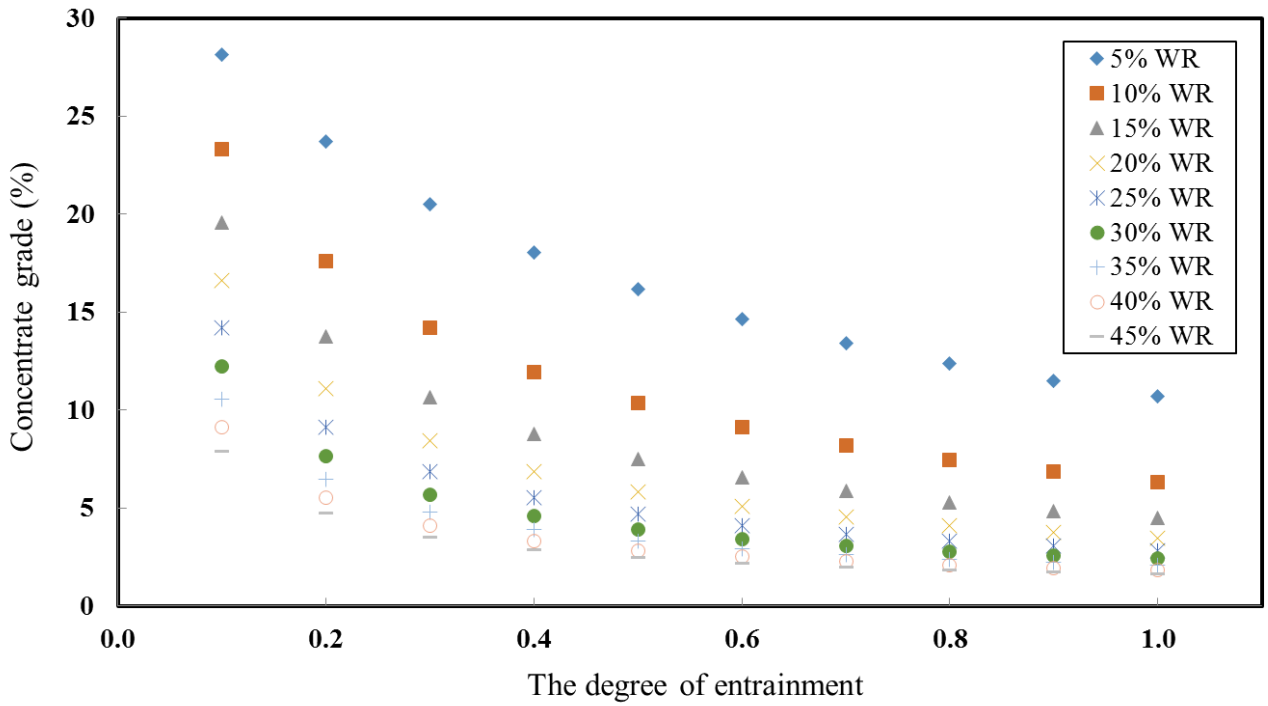

Figure 4-3 The simulated concentrate grade at different water recoveries and *ENT* values.

At each water recovery, the concentrate grade decreases with an increase in the degree of entrainment. The reduction is fairly obvious at relatively low water recovery, but even at high water recoveries, there is a discernible change in concentrate grade when the degree of entrainment increases from 0.1 to 0.3. Indeed, examining the experimental results of the present study, a dramatic decrease in concentrate grade was seen when the overall degree of entrainment increased

from approximately 0.1 to 0.2 (see Table 4-2). Therefore, the degree of entrainment can be critical in terms of influencing the concentrate grade and should not be ignored.

4.2.4 Statistical significance of variables and interactions

The effects and interactions of the four variables on the overall degree of entrainment at a 95% confidence level were analysed using Minitab 17 Statistical Software. The results of the regression analysis are shown in Table 4-3.

Table 4-3 Results of a statistical evaluation of the effect of four variables and their interactions on the degree of entrainment.

Term	P value	Effect on response
Impeller speed	**0.054**	
Particle density	**0.000**	**Negative (exhibits maximum)**
Gas flowrate	**0.988**	
Froth height	**0.259**	
Impeller speed* Particle density	**0.218**	
Impeller speed* Gas flowrate	**0.218**	
Impeller speed* Froth height	**0.072**	
Particle density* Gas flowrate	**0.595**	
Particle density *Froth height	**0.116**	
Gas flowrate* Froth height	**0.026**	**Negative (exhibits minimum)**
Curvature	**0.007**	

As can be seen from Table 4-3, the overall degree of entrainment was significantly affected by particle density and the interaction between gas flowrate and froth height. The effect of impeller speed almost reaches the conventional level of significance (P = 0.05) and so one must conclude that there is some evidence that impeller speed may also affect entrainment. The gas flowrate*froth height interaction means that both factors affect entrainment but in association; thus, the nature of the dependency of entrainment on froth height depends on the gas flowrate. It should be noted that the “non-significant” factors and interactions do not imply no effect or interaction, but rather that any such effect cannot be determined with confidence based on the prevailing error from the data obtained. In addition, the P value of 0.007 for “curvature” in Table 4-3 indicates that the relationship between one or more of the significant factors and the degree of entrainment is non-linear.

Alternatively, the effects of variables on the response in a factorial design can also be evaluated using a normal probability plot (Figure 4-4). The fitted line of this plot indicates where the points

would be expected to fall if the effects were zero, and the variation observed was random and normally distributed. The greater the distance from the line, the more significant the effect of a tested variable. Points that fall to the left of the line have a negative effect while points to the right have a positive effect. As can be seen, two effects, particle density and the interaction between gas flowrate and froth height stand out and have a much more significant effect than the other variables. Both have a negative correlation with the overall degree of entrainment. The effect of particle density is more significant than the interaction between gas flowrate and froth height.

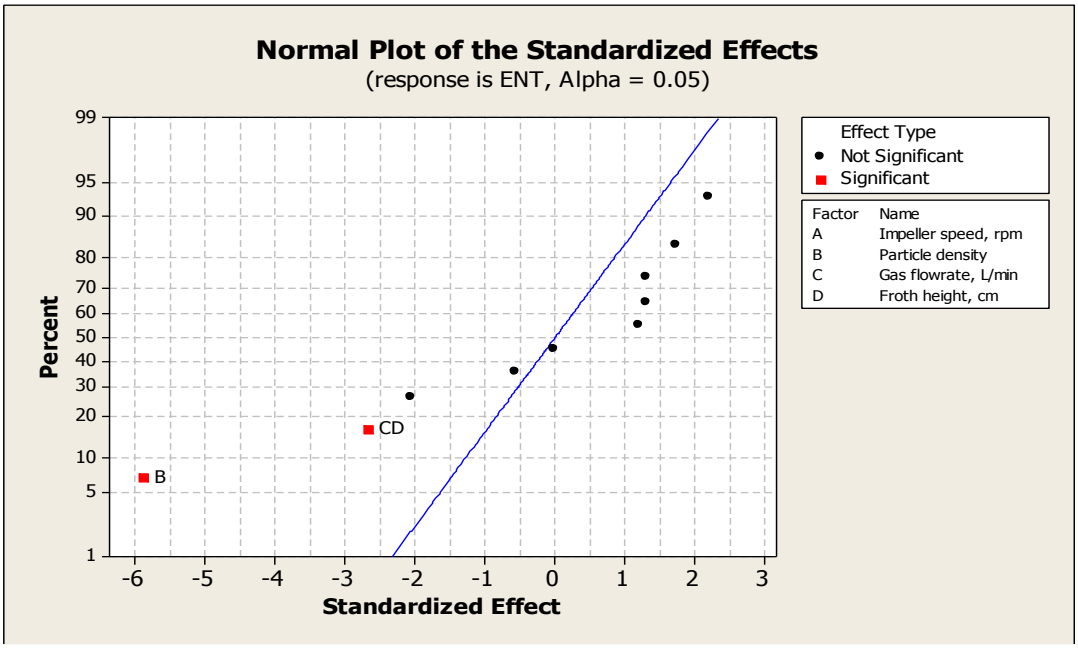

Figure 4-4 Normal probability plot created by Minitab when analysing the effect of the variables of the experiment and their interactions on the degree of entrainment.

4.3 Discussion

4.3.1 Particle density

The results have shown that particle density has a significant effect on the degree of entrainment. Quartz with a specific gravity of 2.65 results in a higher degree of entrainment than hematite with a specific gravity of 5.26. Figure 4-5 shows the average *ENT* value measured in different concentrates of the batch flotation test for those tests performed using quartz and hematite. The *ENT* values of the quartz were always larger than those of hematite. This difference can be interpreted as the effect of particle density (as the particle size distributions were similar).

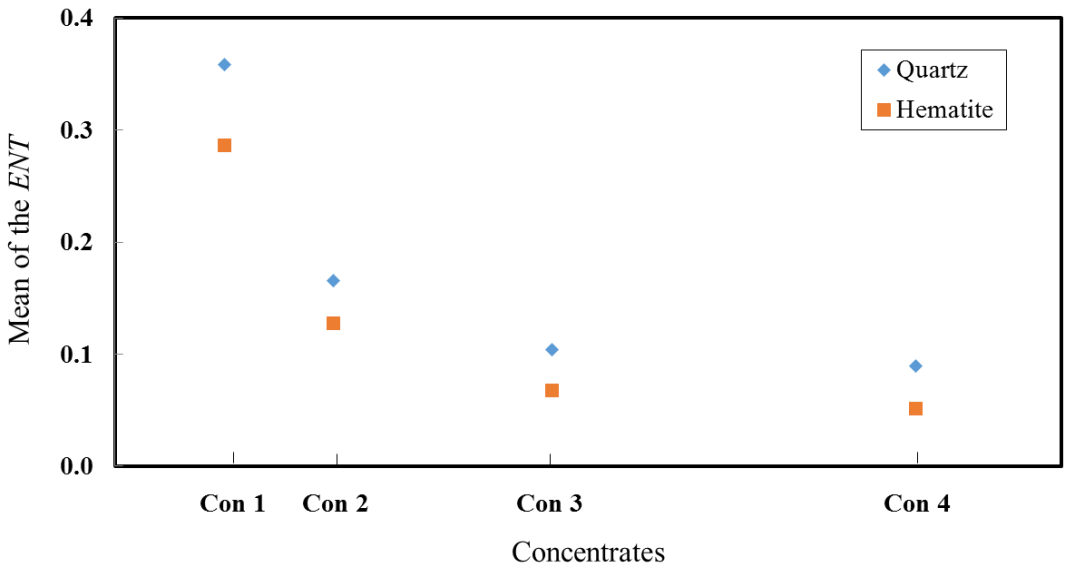

Figure 4-5 The mean of the *ENT* of four concentrates (Con 1: 0-0.5 min; Con 2: 0.5-2 min; Con 3: 2-5 min; Con 4: 5-10 min) as a function of particle density.

Figure 4-6 shows the average overall degree of entrainment plotted against particle density for all the tests performed under the different operating conditions. The correlation is negative and the average changes from 0.18 for quartz to 0.13 for hematite. As shown in Figure 4-3, a decrease in the degree of entrainment from 0.2 to 0.1 can lead to a large change in concentrate grade, regardless of the water recovery, which indicates that the effect of particle density on the degree of entrainment is significant and should not be ignored when the degree of entrainment is modelled. No conclusion can be drawn as to whether this correlation is linear as only two particle densities were used in the experimental design.

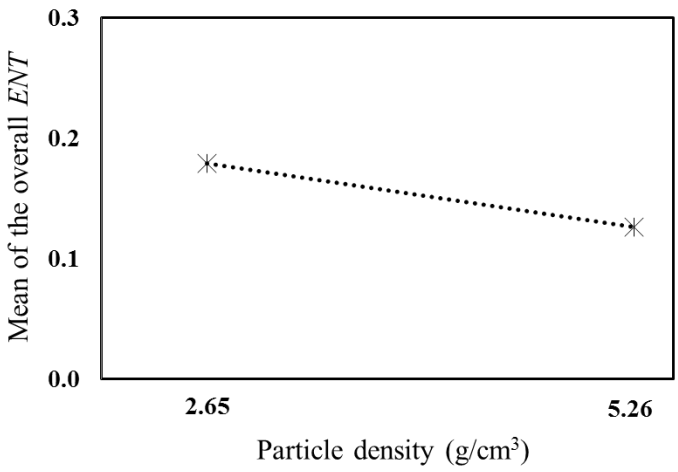

Figure 4-6 The effect of particle density on the mean degree of entrainment.

The decrease in the degree of entrainment with increasing particle density is in line with that observed in many other studies (Johnson, 1972; Bisshop, 1974; Thorne et al., 1976; Maachar and Dobby, 1992; Drzymala, 1994). The higher entrainment of low-density particles is presumably associated with the fact that they tend to move with water due to their lower sedimentation velocities. As a result, the degree of solids suspension in the pulp phase may be poorer for higher density minerals, resulting in less carry over from the pulp into the froth – although this is unlikely to be the case in these experiments because the pulp in small scale laboratory tests is usually perfectly mixed. In addition, high density minerals have a greater chance of settling and draining through the plateau borders within the froth phase than the lower density minerals. Therefore, the degree of entrainment decreases when the particle density is increased.

Traditionally entrainment recovery has been empirically modelled based on particle size alone. These results suggest that it would be better to model entrainment based on particle mass which is a consequence of both particle size and density. To achieve that, a better understanding of the mechanism underpinning the effect of particle size and density on the degree of entrainment is required. This will be the objective of the work that will be presented in **Chapter 5**.

4.3.2 Impeller speed

The P-value for the significance of the impeller speed (0.054) would indicate that the effect is likely to be statistically significant. However, the analysis shows that the magnitude of the effect, i.e. how much changing impeller speed affects *ENT*, is minor, suggesting that there is no significant effect on the degree of entrainment. In general, it is often possible to get a statistically significant effect whose magnitude is small and therefore is of little or no practical effect.

Figure 4-7 shows the average degree of entrainment for quartz and hematite measured in different batch flotation concentrates at different impeller speeds. There was little variation in *ENT* values except for hematite in the first concentrates. This difference might be due to a change in froth structure caused by a variation in the flotation of copper in this first concentrate.

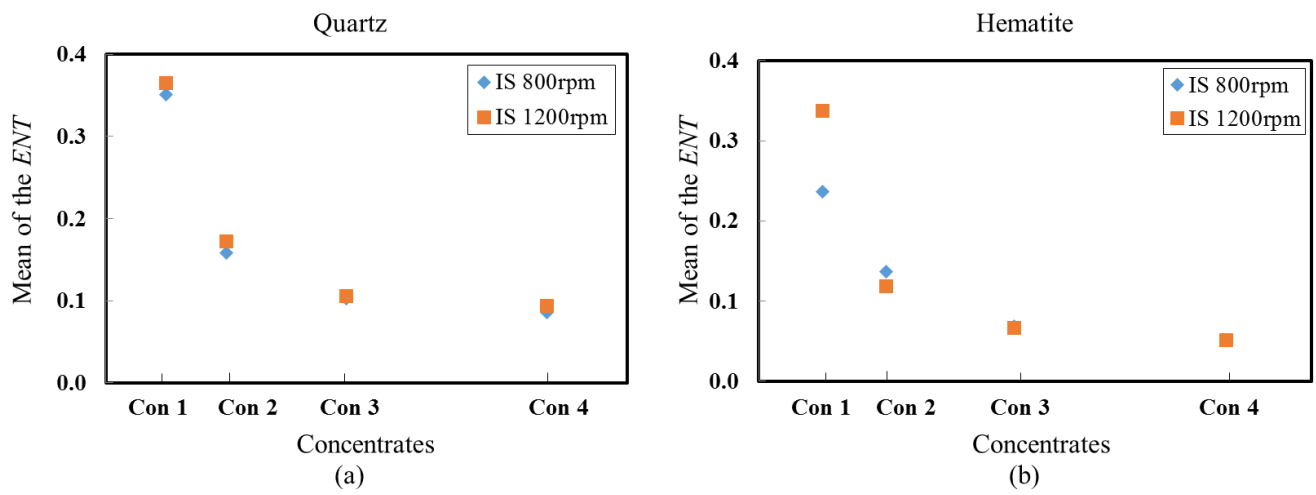

Figure 4-7 The effect of impeller speed on the degree of entrainment of four concentrates (Con 1: 0-0.5 min; Con 2: 0.5-2 min; Con 3: 2-5 min; Con 4: 5-10 min): (a) quartz and (b) hematite.

The effect of impeller speed on the average degree of entrainment is shown in Figure 4-8 for all the tests as well as those only performed using quartz and hematite. A very slight increase in the mean *ENT* value was found with increasing impeller speed but it is not significant and would not be expected to result in an appreciable change in concentrate grade.

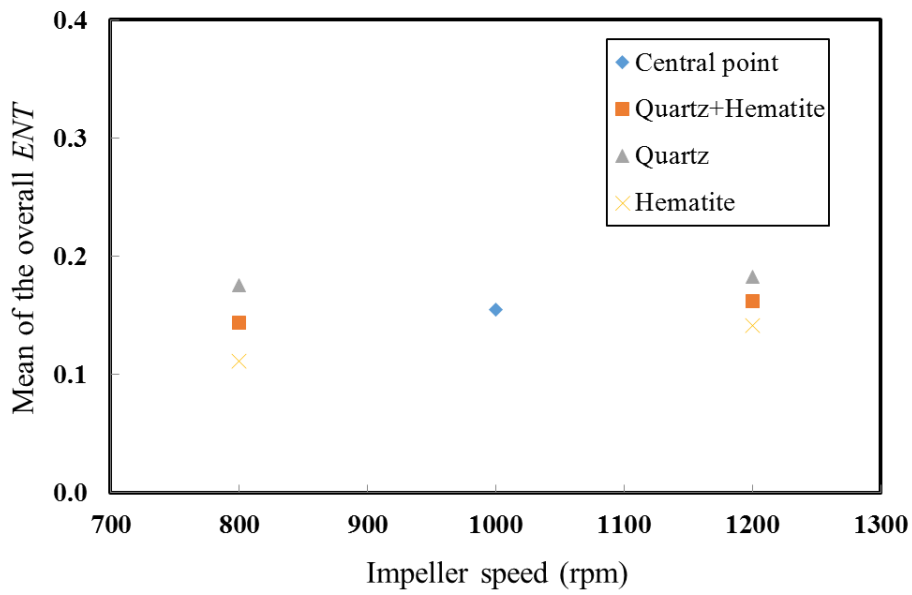

Figure 4-8 The effect of impeller speed on the degree of entrainment.

It is interesting to note, however, that there were observable increases in gangue recovery (and thus decreases in concentrate grade) with an increase in impeller speed in the test results. This was a

consequence of an increase in water recovery, as will be discussed in **Chapter 8**. This increase in gangue recovery by entrainment with impeller speed is in agreement with observations made in other studies in the literature (Cilek, 2009; Akdemir and Sönmez, 2003).

4.3.3 The interaction between gas flowrate and froth height

Table 4-3 shows that the degree of entrainment was significantly affected by the interaction between gas flowrate and froth height. This is illustrated in Figures 4-9 and 4-10 which show the average overall entrainment value as a function of gas flowrate at different froth heights in all the tests and subsets of tests. In all these graphs, at a shallow froth depth, the degree of entrainment increases as gas flowrate increases. Conversely, at the deeper froth depths, the degree of entrainment decreases as gas flowrate increases.

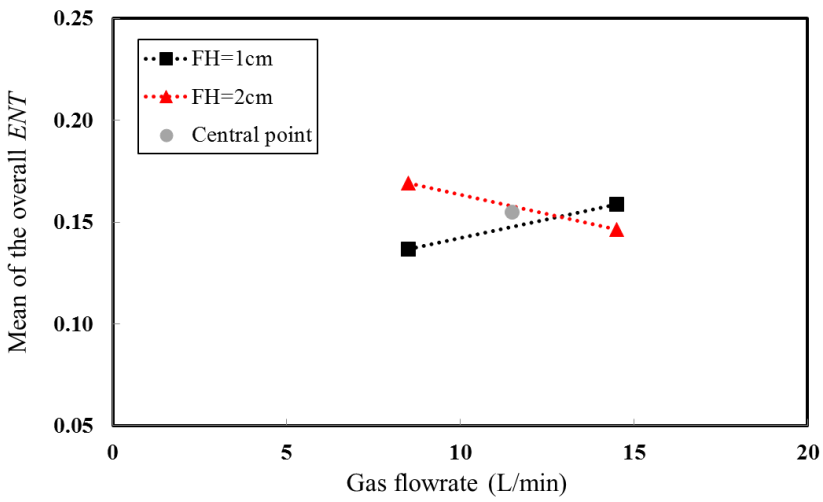

Figure 4-9 The interaction of gas flowrate and froth height on the degree of entrainment.

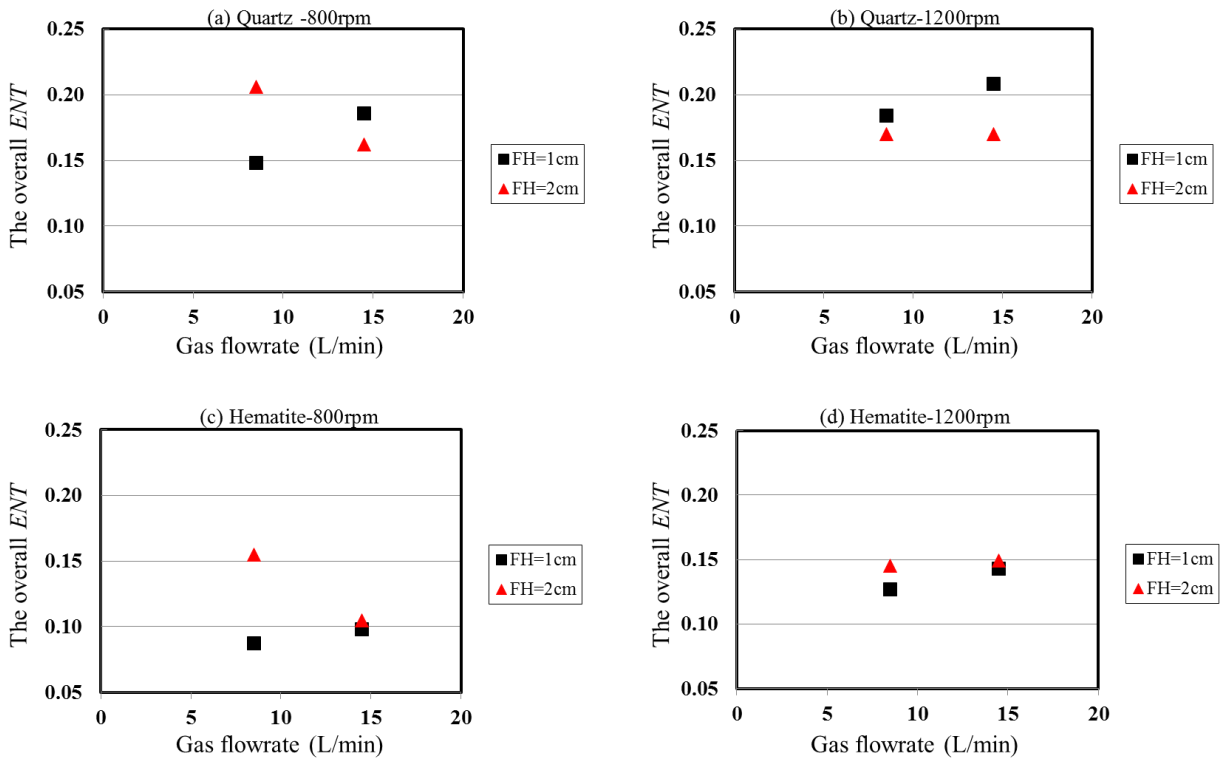

Figure 4-10 The degree of entrainment of quartz and hematite under various operating conditions.

This is a relatively unexpected result as intuitively one would expect that either an increase in froth height or a decrease in gas flowrate would result in more time for drainage and therefore a decrease in both the degree of entrainment and water recovery and subsequently the gangue recovery. In this study there are examples where an increase in gas flowrate can decrease the degree of entrainment (deep froths) and examples where increasing froth height (at low gas flowrates) increases the degree of entrainment. Increasing degrees of entrainment with froth height (at a low gas flowrate) have previously been observed by Zheng et al. (2006b) who collected data from an Outokumpu 3 m^3 pilot tank cell operating under various operating conditions (four different gas flowrates and four different froth heights) using ore diverted from the feed of a rougher bank of the Xstrata Mt. Isa Mines copper concentrator. Figure 4-11 shows the degree of entrainment of quartz for different particle sizes at two froth heights at the lowest gas flowrate of 1085 L/min (J_g = 0.91 cm/s) measured in this study. The degree of entrainment of the finer particle sizes is greater at the froth height of 23.2 cm than that at the froth height of 19.0 cm.

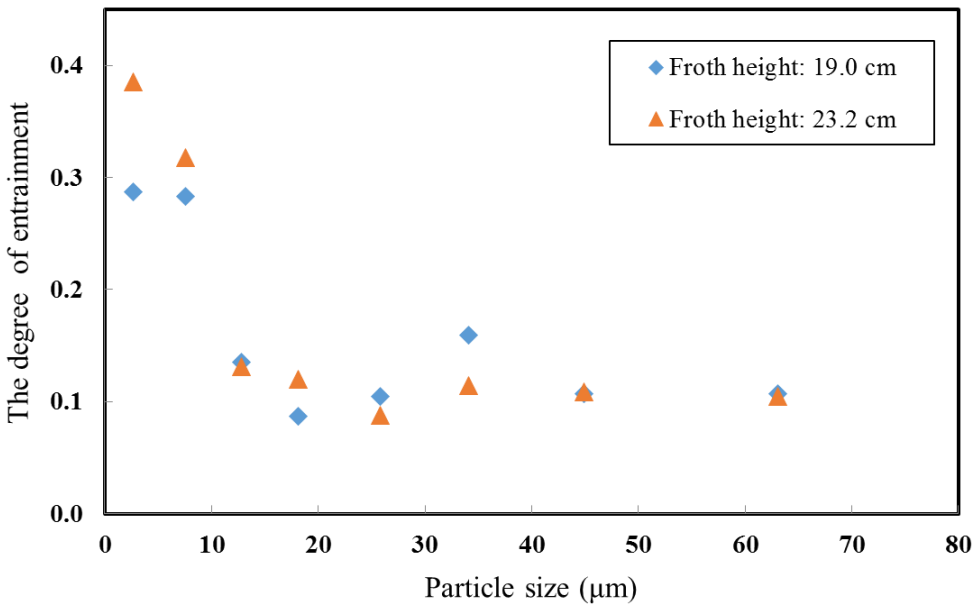

Figure 4-11 The degree of entrainment of quartz as a function of particle size at the froth height of 19.0 and 23.2 cm at the gas flowrate of 1085 L/min (after Zheng et al., 2006b).

More research will be carried out in **Chapter 7** to determine the reason for the observed results.

4.3.4 Froth residence time

Intuitively, froth residence time may affect entrainment (water recovery and the degree of entrainment) as longer time would result in more water drainage and potentially more solid particles to drain preferentially to the water (Zheng et al., 2006b).

In the literature, froth residence time is commonly calculated based on the air (refer to Equation 2-9) or the slurry in the froth (see Equation 4-1) (Gorain et al., 1998; Lynch et al., 1981; Schwarz and Grano, 2005; Szatkowski, 1987):

$$\lambda_{slurry} = \frac{(1-\varepsilon_g)\cdot V_f}{Q_s} \qquad 4\text{-}1$$

where λ_{slurry} is the froth residence time of slurry, ε_g is the mean air hold-up in the froth, Q_s is the volumetric flowrate of the concentrate, incorporating both solids and liquid flowrates and V_f is the volume of the froth.

Figure 4-12 shows the effect of froth residence time of gas (0.98 s, 1.86 s and 3.35 s) (calculated by Equation 2-9) on the mean of the degree of entrainment measured in this experimental study. Little change was observed in the degree of entrainment when the froth residence time increased from 0.98 s to 3.35 s.

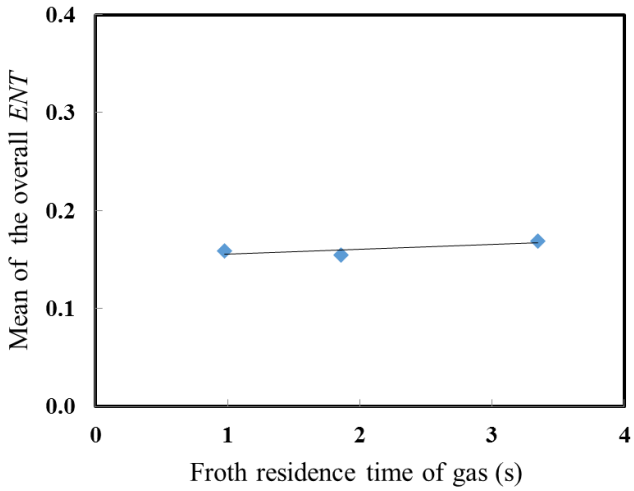

Figure 4-12 The degree of entrainment as a function of froth residence time of gas.

The froth residence time of the slurry may be better correlated with entrainment than the residence time of the air. In this study, froth residence time of slurry was calculated by dividing the froth volume by the volumetric flow of water and solids in the concentrate, corrected for an estimate of the air holdup in the froth (see Equation 4-1). The mean air holdup in the froth phase will be high (>95%), and in this work was estimated by dividing the air volumetric flowrate by the combined volumetric flowrate of concentrate and air as performed by Zheng et al. (2004). Froth residence time calculated in this alternative way also did not exhibit a relationship with the measured degree of entrainment (see Figure 4-13).

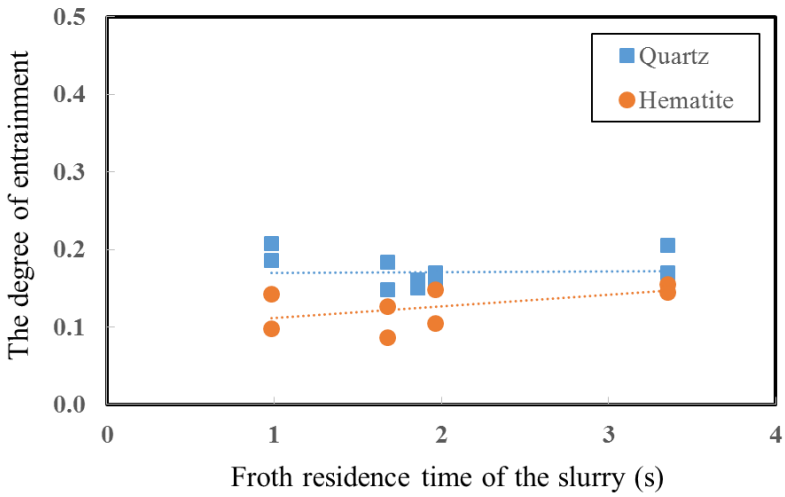

Figure 4-13 The degree of entrainment as a function of froth residence time of slurry.

This is contrast to the water recovery results. As will be discussed in **Chapter 8**, water recovery varied significantly in the experiments of this test and was found to be strongly correlated with froth residence time. An increase in froth residence time resulted in less water recovery and therefore less gangue recovery which thus increased concentrate grade.

Therefore, it can be concluded that froth residence time affects gangue recovery by entrainment through an effect on water recovery, but does not affect the relative drainage of solids to water significantly (i.e. the degree of entrainment).

4.3.5 Change of *ENT* with flotation time

Interestingly, the degree of entrainment was found to decrease throughout the batch flotation experiment with the first concentrate of all 20 tests having a much higher degree of entrainment value than that measured in the later concentrates (see Table 4-2). The change is significant with entrainment averaging about 0.3 to 0.4 in the first concentrate and 0.1 in the last (see Figure 4-5).

Potentially this variation was caused by a change in particle size. The particle size feeding all experiments is the same but the particle size feeding the later stages of the test could be coarser (therefore resulting in lower *ENT* values) because of the preferential removal of fines by entrainment. A change in solids percentage in the pulp is also expected to change the degree of entrainment.

Calculations performed, however, show that other variables must also be involved in changing the degree of entrainment with flotation time. This is corroborated by Tests 5 and 16 where gangue recovery is very low but the degree of entrainment still decreases significantly throughout the test. Under these conditions, the particle size and the gangue solids percentage in the pulp will remain relatively constant cannot therefore be the variables responsible for the appreciable variations in the degree of entrainment during Test 5 and Test 16.

During the experiment, the copper feed grade (and therefore the copper loading) will be decreasing. To assess whether the change in copper feed grade is the primary driver for the result, flotation tests were conducted using quartz without the addition of copper or collector to the feed. These tests were performed using a frother dosage of 10 ppm, an impeller speed of 1000 rpm, a gas flowrate of 11.5 L/min and a froth height of 1.5 cm. Figure 4-14 shows the degree of entrainment of quartz in the concentrates produced during these experiments. It was observed that under this operating condition, the addition of chalcopyrite slightly increased the degree of entrainment in the first concentrate, else it had little impact on the degree of entrainment. The *ENT* values for quartz decrease dramatically, with a similar trend to that observed in the factorial test results regardless of whether chalcopyrite is present in the flotation feed.

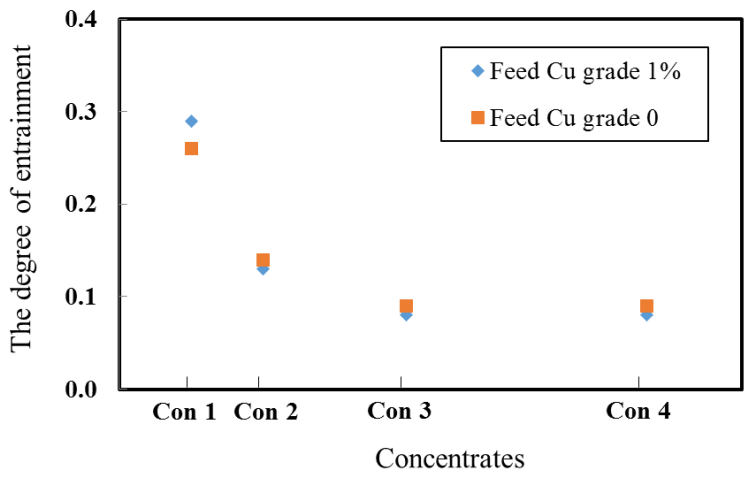

Figure 4-14 The degree of entrainment of quartz at the froth concentration of 10 ppm (Con 1:0-0.5 min; Con 2: 0.5-2 min; Con 3: 2-5 min; Con 4: 5-10 min).

Potentially the change in the degree of entrainment may also be caused by frother which is an important variable which will be affecting froth structure. Frother can potentially deplete during the course of a flotation test as it preferentially reports to the concentrate (Melo and Laskowski, 2007). Flotation tests were conducted using quartz at different concentrations of Dowfroth 250. The

flotation tests were conducted for 30 seconds, at a gas flowrate of 11.5 L/min and a froth height of 1 cm. Figure 4-15 shows the variation in the degree of entrainment when frother concentration was increased.

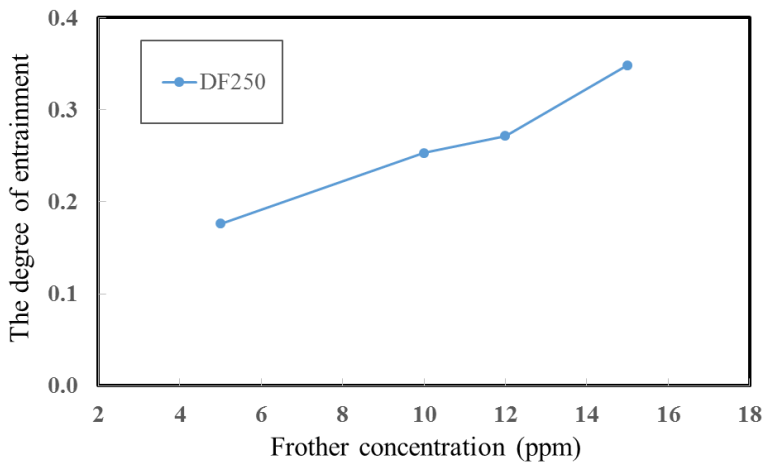

Figure 4-15 The degree of entrainment of quartz at different froth concentrations.

Dowfroth 250 concentration would need to decrease from 10 ppm to 5 ppm in order to obtain an entrainment level comparable to ENT_2 in the factorial experiments. Considering that the initial Dowfroth 250 concentration in the factorial experiment was 10 ppm, it is difficult to believe that the concentration decreased to less than 5 ppm within the first 30 seconds of flotation, as to produce the measured change in the degree of entrainment ($ENT_2 << ENT_1$).

The reason for the change of the degree of entrainment with flotation time is therefore uncertain. This might be a result of all the variables discussed above combining with the operational conditions (i.e. gas flowrate, froth height, impeller speed) to affect froth structure and characteristics and thus the degree of entrainment. It is recommended that this be studied in future work.

Though the reason for the degree of entrainment changing with flotation time was not identified, the testwork indicates that frother concentration would result in appreciable changes in the degree of entrainment. In **Chapter 6**, therefore, flotation tests were conducted to investigate the mechanism by which frother affects the degree of entrainment.

4.4 Summary

A factorial batch flotation experimental program was performed using liberated chalcopyrite, the floatable mineral, mixed with quartz and hematite which were non-floatable and recovered primarily by entrainment. Data analysis was performed on the experimental results from the factorial experiment to identify the primary factors affecting the degree of entrainment. The following conclusions were drawn:

- There was no correlation observed between the measured water recovery and the degree of entrainment, which indicates that the key drivers of water recovery were not the same as those that affected the degree of entrainment.

- Particle density and the interaction between gas flowrate and froth height had a significant effect on the degree of entrainment. The degree of entrainment also changed significantly with flotation time throughout each experiment.

- Froth residence time showed no relationship with the degree of entrainment. It implies that the drainage of gangue solid particles relative to the water in the froth is not time dependent. Froth residence time did, however, have a significant effect on entrainment because of its effect on water recovery.

This chapter has identified that particle density (not just particle size) should be considered when modelling the degree of entrainment. The effect of particle size and density on the degree of entrainment might be because both factors will affect the sedimentation velocity of gangue mineral particles (Smith and Warren, 1989). Particle settling velocity has also been identified as an important component in the fundamental model of Neethling and Cilliers (2009), but the assumptions made in this model have yet to be verified using experimental evidence. In **Chapter 5**, additional experimental testwork will be performed and analysed to investigate the mechanism underpinning the effect of particle size and density on the degree of entrainment. This will involve an assessment of whether particle sedimentation velocity is the key mechanism. The possibility of developing new empirical models for the degree of entrainment using the identified mechanism will also be explored.

This chapter also suggests that frother can significantly affect degree of entrainment. In flotation, gangue recovery by entrainment is often observed to change with frother. This is often attributed to

a change in water recovery, but the results of experiments also show that this effect can be caused by frother significantly changing the degree of entrainment. However, the mechanism by which this occurs is still poorly understood and requires further investigation. **Chapter 6** investigates the mechanism by which frother type and concentration affect the degree of entrainment.

When it comes to the effect of gas flowrate and froth height on entrainment, researchers have presumed that the effect is due to these variables affecting froth residence time which provides the time for entrained water and solid particles to drain out of the froth to the pulp. This chapter, however, shows that froth residence time does not significantly affect the degree of entrainment. This suggests more work is still required to understand the underpinning mechanism by which these variables affect the degree of entrainment. **Chapter 7** presents the results of a study performed to investigate the reason why gas flowrate and froth height affect the degree of entrainment.

To reduce the effects of entrainment on the concentrate grade, manipulating operational variables to only minimise the degree of entrainment will not be sufficient as water recovery also plays a major role. **Chapter 8** analyses the results from the 2-level factorial batch flotation tests performed in this chapter with respect to their effect on water recovery. The aim was to take the opportunity to determine the effect of the investigated variables on water recovery as well as to provide insights into how best to model water recovery.

Chapter 5 The effect of particle size and density on the degree of entrainment

This chapter presents testwork performed to evaluate the mechanism underpinning the effects of particle size and density on the degree of entrainment and proposes an empirical model for the degree of entrainment incorporating the identified mechanism. It also provides a validation of the proposed model using both laboratory scale and industrial scale data.

5.1 Introduction

The degree of entrainment of an entrainment tracer is often used to predict the degree of entrainment of other minerals in the flotation system, regardless of their density. There is a lack of consensus in the literature as to whether density should be incorporated when modelling entrainment.

However, **Chapter 4** demonstrated that particle density was one of the most significant factors affecting the degree of entrainment which should not be ignored. The objective of the work performed in this chapter was to determine if the mechanism by which particle size and particle density affect the degree of entrainment can be determined by an analysis of experimental information. An additional objective was to determine if a new empirical model for the degree of entrainment could be developed based upon the identified mechanism.

In this study, entrainment tests were performed, separately, with three fully liberated non-floating gangue minerals with different specific gravity, i.e. quartz (SG=2.65), ilmenite (SG=4.7) and hematite (SG=5.26), under the same operating conditions, adding only frother (Dowfroth 250 at 10 ppm) to the test. Flotation would usually be performed using a valuable mineral in the system. In these experiments, however, the valuable mineral was eliminated to remove any complication resulting from variation in the valuable mineral recovery due to changes in the test conditions.

The entrainment tests were conducted in the batch laboratory flotation cell described in **Chapter 3**, operated at an impeller speed of 1000 rpm, a froth height of 1 cm and a gas flowrate of 11.5 L/min. For each entrainment test, only one single concentrate was collected from 0 to 10 min. At the completion of the test, the concentrate and tailing were filtered, dried, weighed, sampled and sized. The sizing was performed by wet-screening using sieves in the range of 150 to 20 μm. It was decided to screen rather than cyclosize the -38 micron fraction as gangue density affects the mean size of the cyclosizer fractions which would make direct comparison of test results difficult.

5.2 Results and discussion

5.2.1 *ENT* values of quartz, ilmenite and hematite

The calculated degrees of entrainment obtained from the entrainment experiments performed using the laboratory flotation cell are summarised in Table 5-1 and Figure 5-1. As outlined in **Chapter 3**,

the degree of entrainment was calculated by comparing the concentration of gangue in the concentrate and the final tailing samples. It should also be noted that the particle size used for the experimental data points in Figure 5-1 is the arithmetic mean of the neighbouring screen sizes. The calculation of the degree of entrainment in this chapter is described in more detail in **Appendix B**.

Table 5-1 Calculated degrees of entrainment of quartz, ilmenite and hematite in the laboratory entrainment tests.

Size (µm)	-20	+20-38	+38-53	+53-75	+75-106	+106-150	+150
***ENT*(quartz)**	**0.45**	**0.13**	**0.052**	**0.018**	**0.0068**	**0.0033**	**0.0021**
***ENT*(ilmenite)**	**0.35**	**0.046**	**0.012**	**0.0024**	**0.00050**	**0.00080**	**0**
***ENT*(hematite)**	**0.27**	**0.051**	**0.0095**	**0.0029**	**0.00090**	**0.00030**	**0**

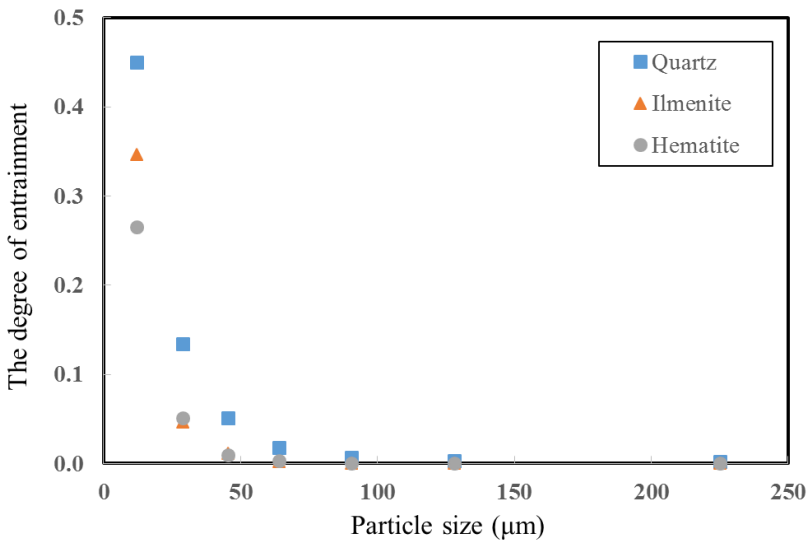

Figure 5-1 The degree of entrainment of quartz, ilmenite and hematite in the laboratory entrainment tests.

Figure 5-1 clearly shows that under the prevailing laboratory cell operating conditions, the *ENT* values decreased with an increase in particle size in a non-linear manner. This is consistent with findings in the literature (Engelbrecht and Woodburn, 1975; Laplante et al., 1989). In addition, the *ENT* values of quartz (SG=2.65) were consistently greater than those of ilmenite (SG=4.7) and hematite (SG=5.26) at each size fraction. This variation becomes increasingly appreciable with decreasing particle size. The *ENT* values for ilmenite largely fall between those of quartz and hematite (with some anomalies in the coarser sizes which is attributed to the experimental errors associated with the low degree of entrainment values). These results prove that the common

practice of using the degree of entrainment of a single tracer to represent the degree of entrainment of all other species in the system, without taking into account their mineral densities, is inaccurate.

The empirical model proposed by Savassi et al. (1998), represented by Equations 1-1 and 1-2 in **Chapter 1**, has been widely used in industrial applications. This empirical model was evaluated using the experimental data obtained from the entrainment tests performed in the 3.5 L laboratory mechanical flotation cell. Figure 5-2 shows the results of fitting the *ENT* model of Savassi et al. (1998) to the experimental data for quartz, ilmenite and hematite. As the model takes into account only the particle size, experimental data for the three gangue minerals are used in the model fitting separately. It can be seen that the model fits the experimental data well for quartz, ilmenite and hematite. The fitted values of the constant ξ and δ for the three gangue minerals are found to be:

$\xi = 24.56$ µm and $\delta = 0.57$ for quartz

$\xi = 19.25$ µm and $\delta = 0.31$ for ilmenite

$\xi = 17.11$ µm and $\delta = 0.68$ for hematite

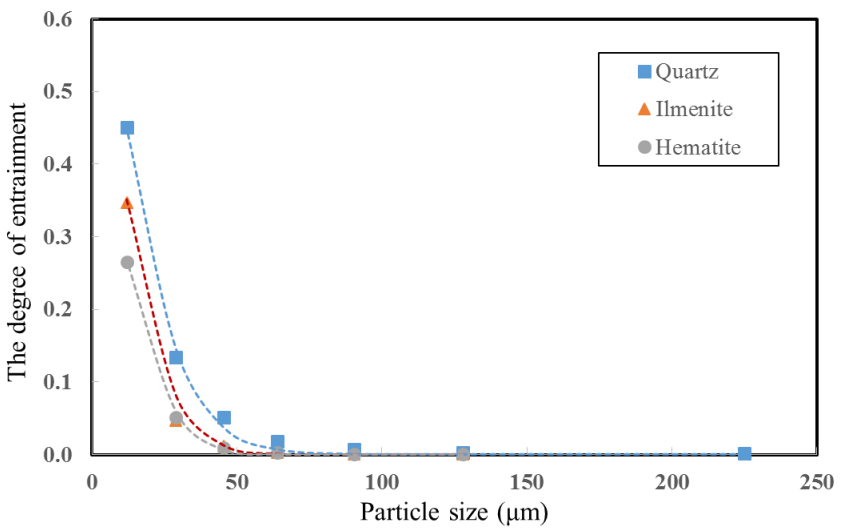

Figure 5-2 Results of fitting the *ENT* model of Savassi et al. (1998) to the experimental data for quartz, ilmenite and hematite.

The significant difference in the model constants ξ and δ suggests that the model derived from one mineral cannot be used for other minerals under the same operational conditions. This difference results from the effect of particle density. As shown in Figure 5-1, an increase in the density of a

particle results in a decrease in its degree of entrainment and this changes both the entrainment parameter ξ and the drainage parameter δ in the model. To achieve the same constants ξ and δ for the three different gangue minerals, the effect of particle density must be considered.

5.2.2 The mechanism underpinning the effect of particle size and density

The degree of entrainment represents the degree of drainage of solid particles relative to water. Differences in the *ENT* value of particles with different size and density are likely to be a consequence of differences in their particle settling rate within the water just below or within the froth phase where flow can be considered laminar. Under these conditions, the particle settling rate of entrained particles in the fluid can be approximated using Stokes' Law (Smith and Warren, 1989). According to this law, the settling velocity results from a balance between the particle weight and the upward drag force (Elger et al., 2013). The weight of a solid particle in fluid (i.e. apparent immersed weight) is given by:

$$G_0 = V_s \cdot (\rho_s - \rho_w) \cdot g \qquad 5\text{-}1$$

where G_0 is the apparent immersed weight of the particles in the water (the weight of the particles minus the buoyancy force), V_s is the particle volume, ρ_s is the specific gravity of the solid particles, ρ_w is the density of the water and g is the gravitational acceleration. Note that ρ_w may be elevated by the presence of fine suspended particles in the water.

Drag force under laminar conditions will be proportional to the particle velocity relative to the water velocity (v), in the direction of the water motion (refer to Equation 2-2). The particle settling velocity will be that which results in sufficient drag that it exactly balances the apparent immersed weight of the solid particle. The settling velocity (v_{set}) will then be proportional to the difference in densities between the entrained particles and water, and to the square of the particle diameter, as shown by Equation 5-2. The squared relationship between settling velocity and particle size is a consequence of the apparent immersed weight increasing as a function of the particle diameter cubed which has to be balanced against the Stokes' drag which is increasing proportionally to particle size.

$$v_{set} \propto (\rho_s - \rho_w) \cdot d_i^2 \qquad 5\text{-}2$$

Figure 5-3 shows the degree of entrainment for the three gangue minerals as shown in Table 5-1 plotted against $(\rho_s - \rho_w)\cdot d_i^2$ where water density of 1 g/cm^3 and arithmetic mean particle size (μm) were used. As can be seen, the degree of entrainment is correlated via a power function with $(\rho_s - \rho_w)\cdot d_i^2$, suggesting that particle settling rate that results from the balance between drag force and apparent immersed weight is the key factor determining the degree of entrainment observed in the flotation experiments. Particles with a greater size and density settle faster and experience greater drainage relative to water than particles with a smaller size and density.

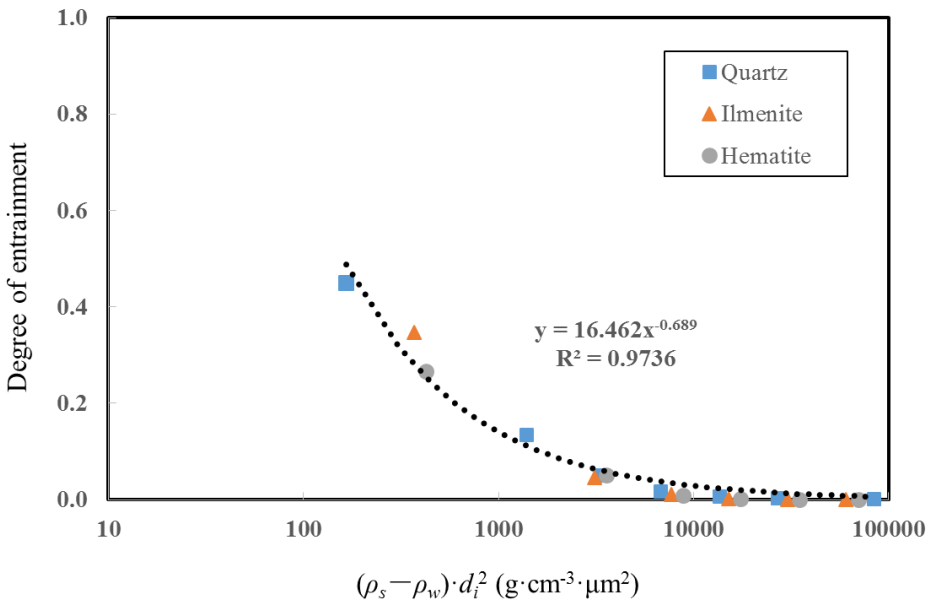

Figure 5-3 The degree of entrainment for the three gangue minerals as a function of $(\rho_s - \rho_w)\cdot d_i^2$.

In summary, variations in mineral particle size and density result in different drag force and apparent immersed weight which affect the degree of entrainment. Thus, for the purpose of prediction, particle setting velocity incorporating both particle size and density is a more appropriate parameter which can be used to model the degree of entrainment.

5.2.3 An empirical model based on particle size and density

An empirical model for the degree of entrainment incorporating $(\rho_s - \rho_w)\cdot d_i^2$ was then proposed, as described by Equation 5-3. This is a rational function which was chosen as it visually aligns with the shape of the relationship (refer to Figure 5-3) and fits the boundary conditions of the system.

$$ENT_i = \frac{1}{1 + p \cdot \left((\rho_s - \rho_w) \cdot d_i^2 \right)^q} \qquad 5\text{-}3$$

where p (in $(g \cdot cm^{-3} \cdot \mu m^2)^{-q}$) and q are constants.

Equation 5-3 satisfies the following boundary conditions:

$$d_i \to \infty, \rho_s \to \infty \Rightarrow ENT \to 0$$
$$d_i \to 0, \rho_s \to 0 \Rightarrow ENT \to 1$$

Figure 5-4 shows the results of fitting Equation 5-3 to the experimental data obtained from the laboratory flotation tests reported in Table 5-1. As can be seen, the mathematical relationship fits the data reasonably well for the quartz, ilmenite and hematite. The corresponding parameters p and q are found to be:

$p = 0.061\ (g \cdot cm^{-3} \cdot \mu m^2)^{-0.689}$ and $q = 0.689$

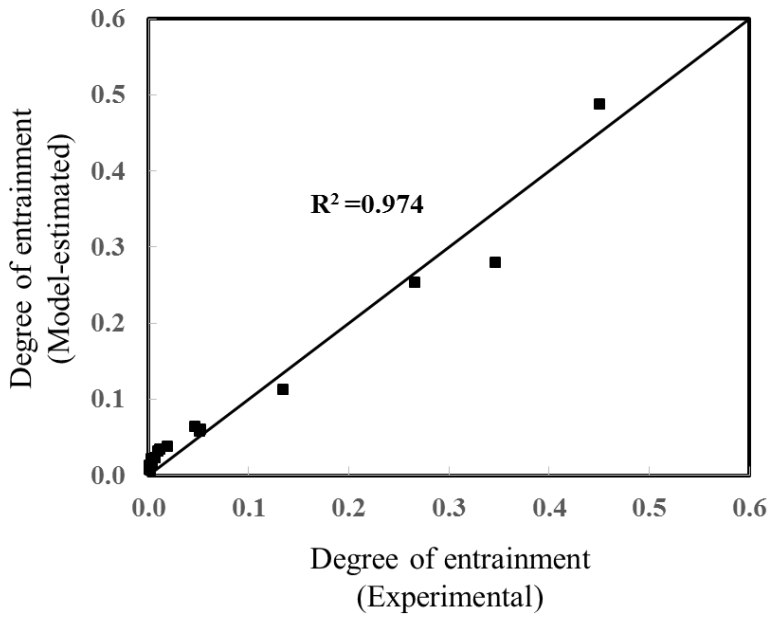

Figure 5-4 Experimental *ENT* values measured in the quartz, ilmenite and hematite laboratory flotation tests reported in Table 5-1 versus the results obtained by fitting all data using Equation 5-3.

It should be noted that in the experiments performed in this current study only particle density was varied while the other operating conditions, which have been shown to affect the degree of

entrainment, were kept constant. It is also expected that the other variables which affect particle settling rate, like fluid viscosity and particle shape, will also have a significant effect on the degree of entrainment. It is therefore expected that the constants *p* and *q* will vary with a significant change in the flotation environment.

The model will provide, however, a method to accurately predict the degree of entrainment (and thus the entrainment recovery) for different minerals in the flotation feed based on a measured degree of entrainment of a tracer mineral. The model will also provide an estimate of the change in entrainment recovery with a change in particle size or mineral composition feeding a flotation cell when the other flotation parameters are not changed significantly. For these calculations to be accurate, one needs to take into account the degree of liberation of the mineral in the system as this will affect its average particle density.

It is also worth noting that the form of the rational function proposed in Equation 5-3 when modified to only consider the effect of particle size (Equation 5-4) gives an improved degree of fit to the entrainment data collected in this study to that of the empirical model of Savassi et al. (1998) (refer to Figure 5-2). This is demonstrated in Figure 5-5 which shows the results of fitting Equation 5-4, separately, to the degree of entrainment experimental data obtained for each of the quartz, ilmenite and hematite laboratory flotation tests reported in Table 5-1.

$$ENT_i = \frac{1}{1 + m \cdot d_i^{\,n}} \qquad 5\text{-}4$$

where *m* (in μm^{-n}) and *n* are constants.

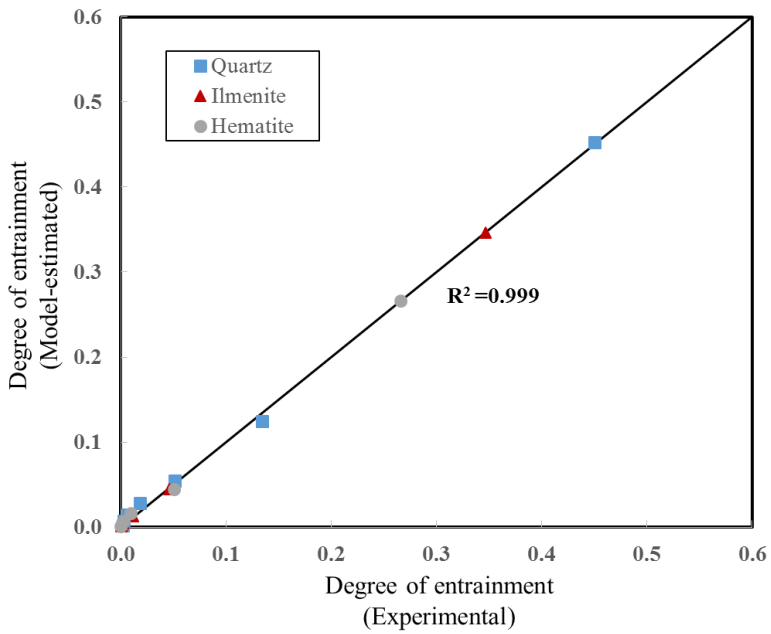

Figure 5-5 Experimental *ENT* values measured in the quartz, ilmenite and hematite laboratory flotation tests reported in Table 5-1 versus results obtained by fitting the data for each mineral separately using Equation 5-4

5.2.4 Industrial application of the new model

The new empirical model (Equation 5-3) was further examined using industrial data from a survey of Glencore Ernest Henry Mine, in 2007. The data were obtained from the first rougher cell (127 m^3) of the copper concentrator. The rougher feed contained 37% solids by mass with a P80 value of approximately 220 µm, and a feed grade of 0.7% copper. The flotation cell was operated at a froth height of 10 cm and a superficial gas velocity of 1.11 cm/s, with a frother (PPG type) dosage of 5 g/t and a collector (a mix of PAX and SIBX) dosage of 40-60 g/t. Samples were taken from the feed, concentrate and tailing. All samples were then analysed for solid%, particle size distribution, chemical assays and mineral liberation. The degrees of entrainment were calculated for individual size fractions based on the overall mass of fully liberated particles in the concentrate and the tailing. It should be noted that the mineral liberation data were based on the free surface of polished sections. In this study, two gangue minerals with distinct particle density differences, quartz (SG=2.65) and magnetite (SG=5.28), were analysed.

The calculated degrees of entrainment in the industrial flotation cell are summarised in Table 5-2. Similar to the laboratory experimental data shown in Table 5-1, the *ENT* values of quartz obtained from the industrial flotation cell decrease with an increase in particle size in a non-linear manner. The *ENT* values of magnetite, the gangue mineral with the higher specific gravity, are smaller than

those of quartz, indicating significantly less recovery by entrainment. The trend in the magnetite results, however, is not as expected with the degree of entrainment increasing in the coarser sizes. This may be due to experimental error or, more probably, the fact that some of the gangue minerals identified as liberated are locked with floatable sulphide mineral species and mis-categorised due to stereological error. They were reporting to the concentrate due to true flotation and therefore resulted in erroneously higher entrainment values. This is a problem observed often when analysing gangue data for a real ore and why it is better to study entrainment using gangue which is known to be fully liberated.

Table 5-2 Calculated degrees of entrainment in the first rougher cell at the Ernest Henry copper mine.

Size (µm)	-20	+20-38	+38-53	+53-75
***ENT*(quartz)**	**0.25**	**0.050**	**0.042**	**0.027**
***ENT*(magnetite)**	**0.099**	**0.018**	**0.026**	**0.026**

Figure 5-6 shows the results of fitting Equation 5-3 to all the industrial experimental data shown in Table 5-2. As can be seen, the empirical model appears to fit the data reasonably well for quartz and magnetite, and the corresponding parameters for the empirical model are found to be:

$p = 0.050\ (\mathrm{g{\cdot}cm^{-3}{\cdot}\mu m^{2}})^{-0.816}$ and $q = 0.816$

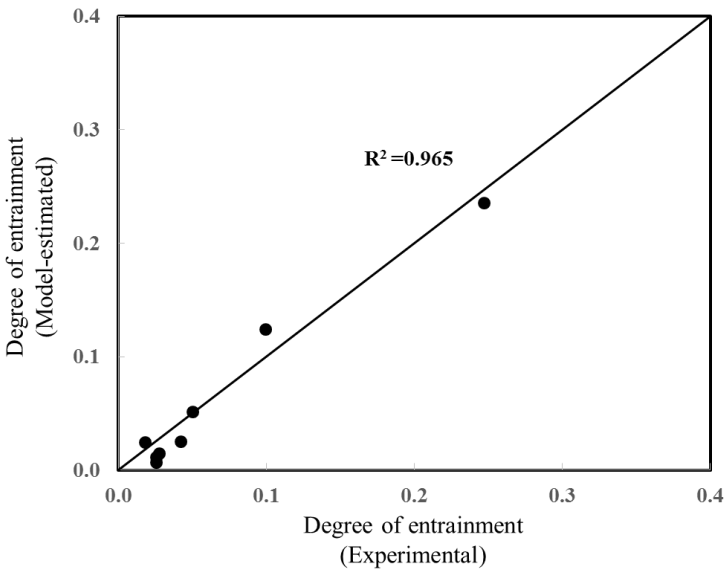

Figure 5-6 Experimental *ENT* values for quartz and magnetite measured in the first rougher cell of the Ernest Henry Mines copper concentrate reported in Table 5-2 versus the results obtained by fitting all data using Equation 5-3.

5.3 Summary

A power law relationship was observed between the degree of entrainment and a function related to the particle settling rate, providing experimental validation that it is the balance between the drag force and the particle apparent immersed weight in the water that determines the extent to which particles overcome the effects of gravity and report to the concentrate due to entrainment. An increase in particle size and density both result in an increase in the particle settling rate because they increase the apparent immersed weight of the particle to a greater extent than any increase in the drag force as a consequence of the particle having a greater surface area as it gets bigger in size. This explains why the degree of entrainment decreases with increases in the particle size and/or particle density.

An empirical mathematical model for the degree of entrainment was proposed to relate the degree of entrainment to particle size and density. The empirical model provided a reasonable fit to both the laboratory and industrial experimental data set, which indicates that it can be used for predicting the degree of entrainment for minerals with different particle density.

Chapter 6 The impact of frother on the degree of entrainment

This chapter presents the results of experiments performed to study the mechanism by which frother (i.e. frother type and concentration) affects the degree of entrainment.

6.1 Introduction

In **Chapter 4** it was found that frother concentration could result in an appreciable change in the degree of entrainment. As outlined in **Chapter 2**, the recovery of gangue minerals by entrainment in flotation has been observed to change with a change in frother dosage. This effect is often believed to be due to the frother affecting the water recovery. The results from **Chapter 4**, as well as other studies in the literature, indicate that frother also changes entrainment recovery by changing the degree of entrainment. However, the mechanism by which this occurs is still poorly understood.

As discussed in **Chapter 2**, researchers have observed that the gangue concentration within the froth above the pulp/froth interface remains relatively constant. This suggests that the entrainment classification may predominantly occur at the interface. Whether gangue particles move upward into the froth with the water or remain in the pulp will be a consequence of a balance of forces. There will be a downward force due to the weight of the particle and an upward drag force due to the water flowing upwards into the froth (as demonstrated by the results in **Chapter 5**). Drag force will be affected by a number of factors including the fluid velocity and viscosity. Frother changes the water flow into the concentrate, and potentially has an impact on the liquid velocity at the interface and thus the drag force, resulting in different degree of entrainment values. Frother concentration and type may also affect the fluid viscosity which in turn affects the drag force. As the polar groups of frother are orientated to the liquid phase, they interact with water molecules through hydrogen bonds, which in turn increases the liquid viscosity (Laskowski, 1993; Cruz et al., 2013).

This chapter will present the results obtained by performing a series of gangue only entrainment flotation tests in the 3.5 L laboratory flotation cell using two different frothers, Dowfroth 250 and MIBC, at a range of frother dosage rates. The feed to these tests was fully liberated quartz and frother was the only reagent added. The tests were performed without valuable minerals present to remove any complication that might arise due to variation in the valuable mineral recovery with changes in the test conditions. During the tests, liquid velocity at the interface and fluid viscosity were measured. The characteristics of the materials used, the experimental set up and the measurement methodology have been described in **Chapter 3.** The objective of these tests was to more comprehensively study the effects of frother on entrainment and to determine whether these effects can be correlated with the liquid velocity/viscosity at the interface. If this correlation exists,

it would provide experimental evidence that it is changes in drag force at the interface that is the mechanism by which frother type and concentration affect the degree of entrainment.

All the tests were performed using an impeller speed of 1000 rpm, a froth height of 1 cm and a gas flowrate of 11.5 L/min. For each entrainment test, only one single concentrate was collected from 0 to 30 s. In **Chapter 4**, it was discussed that a short flotation time may not be suitable for *ENT* comparisons, because of a possible entrapment in the presence of chalcopyrite or the impurity chalcopyrite which would overestimate the degree of entrainment values. In this chapter, however, the valuable mineral was removed in the tests, which eliminated this interference. Moreover, given that this chapter aims to investigate the mechanism underpinning the effect of frother type and concentration, a longer flotation time would result in a decrease in frother concentration, as in flotation applications there is a depletion of frother concentration in the pulp due to frother preferentially reporting to the concentrate in the recovered water (Melo and Laskowski, 2007). It was assumed that such a short time would not change the frother concentration in the system significantly. Tests were performed using either Dowfroth 250 or MIBC at dosages of 5, 10, 12 and 15 ppm. The pure frothers were diluted with Brisbane tap water to produce a 1.5 mg/ml solution before being added to the flotation cell and conditioned for 1 min. At the completion of the test, the concentrate and tailing were filtered, dried and weighed.

6.2 Results and discussion

6.2.1 Entrainment test results

Figure 6-1 shows the change in the gangue recovery by entrainment and water recovery as the frother concentration of Dowfroth 250 and MIBC was increased in the entrainment tests. Error bars (i.e. standard deviation) in Figure 6-1 were estimated based on the results from four replicates of a flotation test which was described in **Chapter 3**. As can be seen, quartz recovery increases with increasing frother concentration when using either Dowfroth 250 or MIBC. This is in agreement with other studies (Melo and Laskowski, 2007; Rahal et al., 2001). The entrainment recovery of quartz increased from 0.15% to 1.6% as the MIBC concentration was increased from 5 ppm to 15 ppm. Dowfroth 250 had a more pronounced effect on the quartz recovery than MIBC with quartz recovery increasing from 0.41% to 6.5% as the Dowfroth 250 concentration was increased from 5 ppm to 15 ppm.

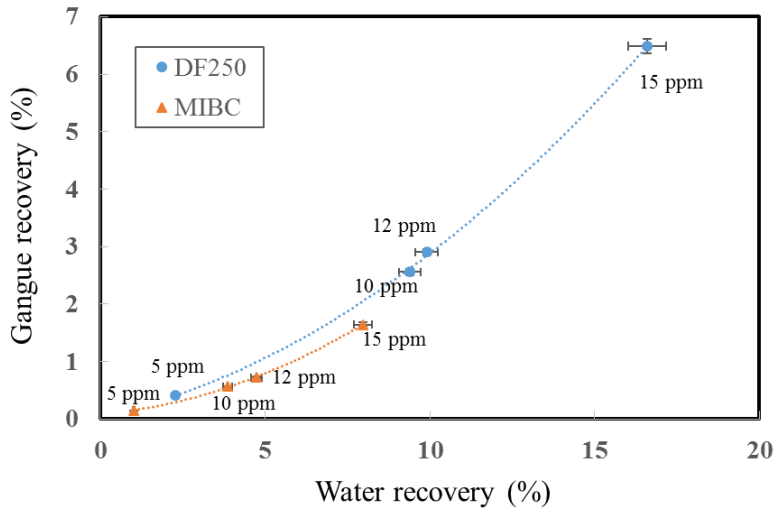

Figure 6-1 The effect of frother concentration of Dowfroth 250 and MIBC on the entrainment recovery of quartz and water recovery.

As expected, quartz recovery in these tests exhibited a strong correlation with water recovery which varied significantly in the experiments. As shown in Figure 6-1, water recovery increased from 1.0% to 8.0% as the frother concentration of MIBC was increased from 5 ppm to 15 ppm, and increased from 2.3% to 17% when the frother concentration of Dowfroth 250 was increased over the same range. These variations in water recovery are a significant reason why quartz recoveries vary significantly in these tests.

Quartz recovery in the different tests also changes as a consequence of variation in the degree of entrainment, which according to Equation 2-8, can be approximated by the slope of the quartz versus water recovery relationships shown in Figure 6-1. The upwardly curving relationships in Figure 6-1 is an indication that the degree of entrainment is increasing with frother concentration. This observation is in line with other studies (Rahal et al., 2001; Kracht et al., 2016). The calculated degree of entrainment as a function of frother type and concentration is shown in Figure 6-2. As can be seen, the degree of entrainment increased with increasing concentration of both Dowfroth 250 and MIBC. When increasing the concentration of Dowfroth 250 from 5 ppm to 15 ppm, the degree of entrainment on an un-sized basis increased from 0.18 to 0.35. For the same change in the concentration of MIBC, the degree of entrainment only increased from 0.14 to 0.19. It should be noted that this magnitude of change in the degree of entrainment can result in a significant change in the concentrate grade as demonstrated in simulations performed in **Chapter 4** (see Figure 4-3).

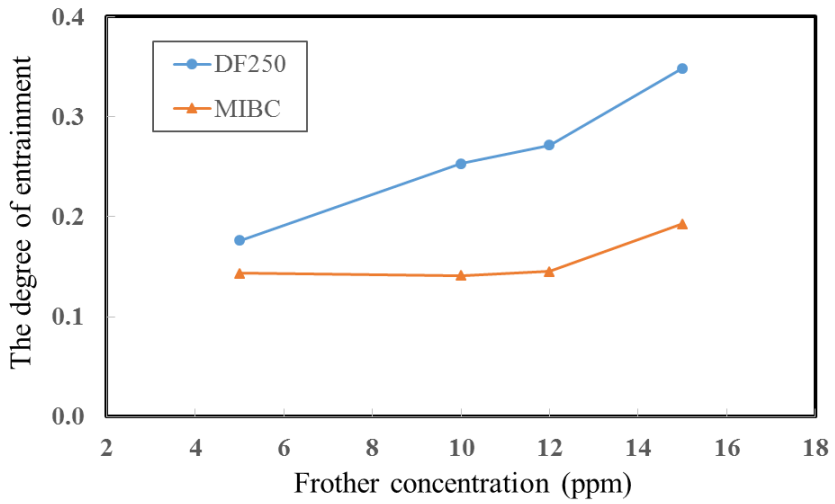

Figure 6-2 The effect of frother concentration of Dowfroth 250 and MIBC on the degree of entrainment.

The variation in the degree of entrainment for quartz due to frother (i.e. frother type and concentration) means that there has been a change in hydrophilic gangue concentration in the water collected into the concentrate. This could either be a consequence of incomplete solid suspension at the top of the pulp or preferential drainage of solid particles to water within the froth phase. In this study, the pulp phase is assumed to be perfectly mixed, as it is reasonable to make such an assumption in a 3.5 laboratory flotation cell. Thus it is concluded that the rejection of entrained material largely occurs in the froth phase. It is hypothesised that, as outlined in the introduction to this chapter, this rejection occurs at the pulp/froth interface.

6.2.2 Liquid velocity at the interface

Frother concentration may be changing the degree of entrainment because it is affecting the drag force at the interface. This drag force will be related to the velocity of the water being pushed up into the froth phase by the rising bubbles. As outlined in **Chapter 2**, the actual liquid velocity at the interface can be estimated using the overall superficial liquid velocity and the liquid content at the interface. In this study, superficial liquid velocity was approximated by dividing the water flow measured in the concentrate by the cross-sectional area of the flotation cell. As shown in Figure 6-1, water recovery increased with increasing concentrations of both Dowfroth 250 and MIBC. As the water added to the cell was relatively similar in the batch entrainment tests, this means that the increase in the concentration of Dowfroth 250 and MIBC resulted in increasing water flow to concentrate and thus increasing superficial liquid velocity in the tests. Moreover, Dowfroth 250

caused a greater change in the superficial liquid velocity than MIBC over the same frother concentration range.

Figure 6-3 shows the variations in the liquid content at the interface measured by analysing images of the pulp/froth interface region collected during the batch entrainment tests performed using Dowfroth 250 and MIBC at different frother concentrations. Liquid content at the interface decreased from 85% to 70% when the frother concentration of Dowfroth 250 was increased from 5 ppm to 15 ppm. The general trend when using MIBC is that the liquid content at the interface decreased with increasing frother concentration, though there was a fluctuation at 12 ppm which is likely due to experimental error. The error bars in Figure 6-3 were calculated by measuring the degree of variation in the liquid content calculated using the bubble diameters and bubble water thicknesses measured in the three images analysed for each test.

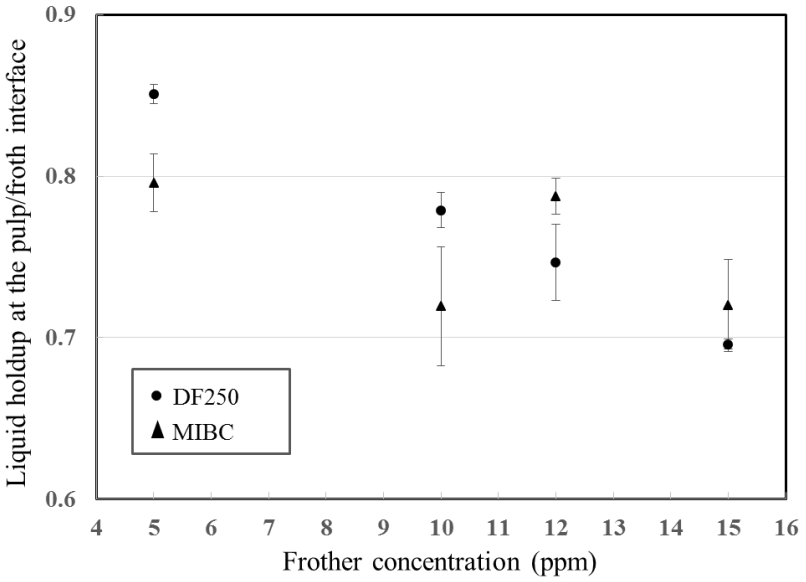

Figure 6-3 The effect of frother concentration on the liquid holdup at the pulp/froth interface.

Generally, liquid content at the interface decreases as frother concentration increases and this is due to the significant change in the bubble size distribution at the interface when the dosage of both frothers increase. Figure 6-4 shows examples of the froth images of the pulp/froth interface captured during each entrainment test. These images indicate that bubbles at the interface become smaller and denser with increasing frother concentration for both frothers. The decrease in the bubble size at the high frother concentration would suggest that increased concentration of frother decreases the rate of bubble coalescence at the interface region.

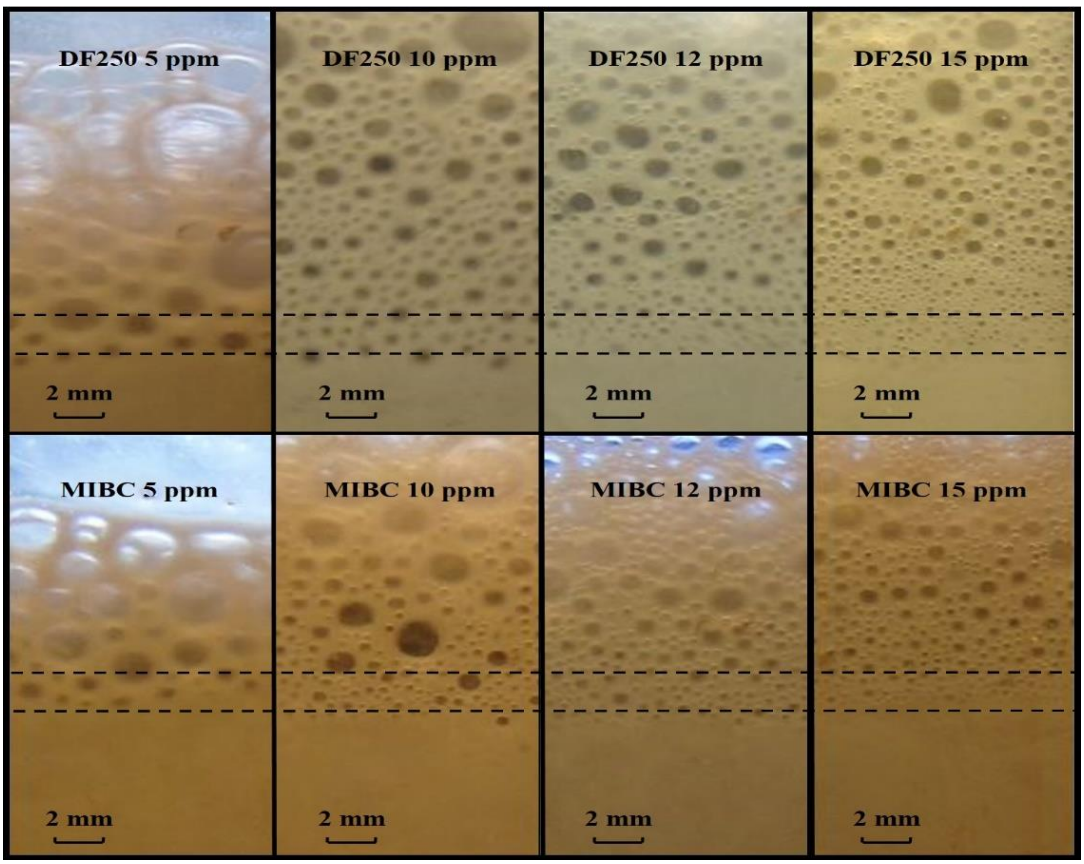

Figure 6-4 Images of the froth captured during the batch entrainment tests performed using different frother types at different frother concentrations.

At the interface region, this large population of fine congested bubbles observed at the higher frother concentration results in an increase in the gas holdup (i.e. the decrease in the liquid holdup). This change appears to be consistent with the observation by Yianatos et al. (2009) that gas holdup increases with decreasing bubble size in the region below the interface. As shown in Figure 6-3, the variation in liquid holdup caused by Dowfroth 250 was greater than that by MIBC. This is because different frothers exhibit different degrees of foamability with increasing frother concentration. When different frothers are used, bubbles of different size distributions are commonly produced even at the same frother concentration (Gupta et al., 2007).

An increase in frother concentration leads to both an increase in superficial liquid velocity and a decrease in the liquid holdup at the pulp/froth interface which, according to Equation 2-3, results in an increase in the actual liquid velocity at the interface for both frothers. The effect of frother on the liquid velocity at the interface is shown in Figure 6-5. As can be seen, the effective liquid velocity at the interface for both frothers increased when the frother concentration was increased, and the variation caused by Dowfroth 250 was greater than that by MIBC. Moreover, it was interesting to

note that there was a similarity in the shape of the trends displayed in Figure 6-5 and Figure 6-2, implying that there may be a direct correlation between the degree of entrainment and the liquid velocity at the pulp/froth interface.

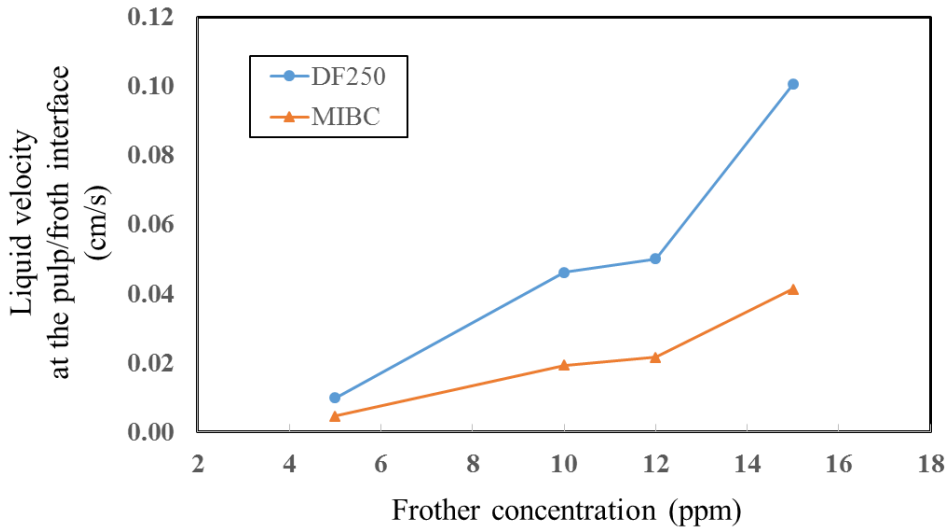

Figure 6-5 The effect of frother concentration of Dowfroth 250 and MIBC on the liquid velocity at the interface.

Figure 6-6 shows the relationship between the degree of entrainment and the liquid velocity at the interface measured in the tests performed using different frother concentrations and types of frother. The degree of entrainment is linearly correlated with the liquid velocity at the interface. The relationship is independent of frother type with the data associated with both frothers falling onto the one relationship. This experimental result suggests that the mechanism by which the frothers affects the degree of entrainment is because they change the upward liquid velocity at the interface, which ultimately affects the drag force applied to the particles in this zone. A greater drag force means that particles of a greater size and density are dragged into the froth to follow the water phase. A greater concentration of solid particles in the water results in a higher degree of entrainment. The linear relationship between the degree of entrainment and velocity suggests the flow is laminar at the interface as drag force is proportional to velocity for laminar flow but proportional to the square of velocity for turbulent flow.

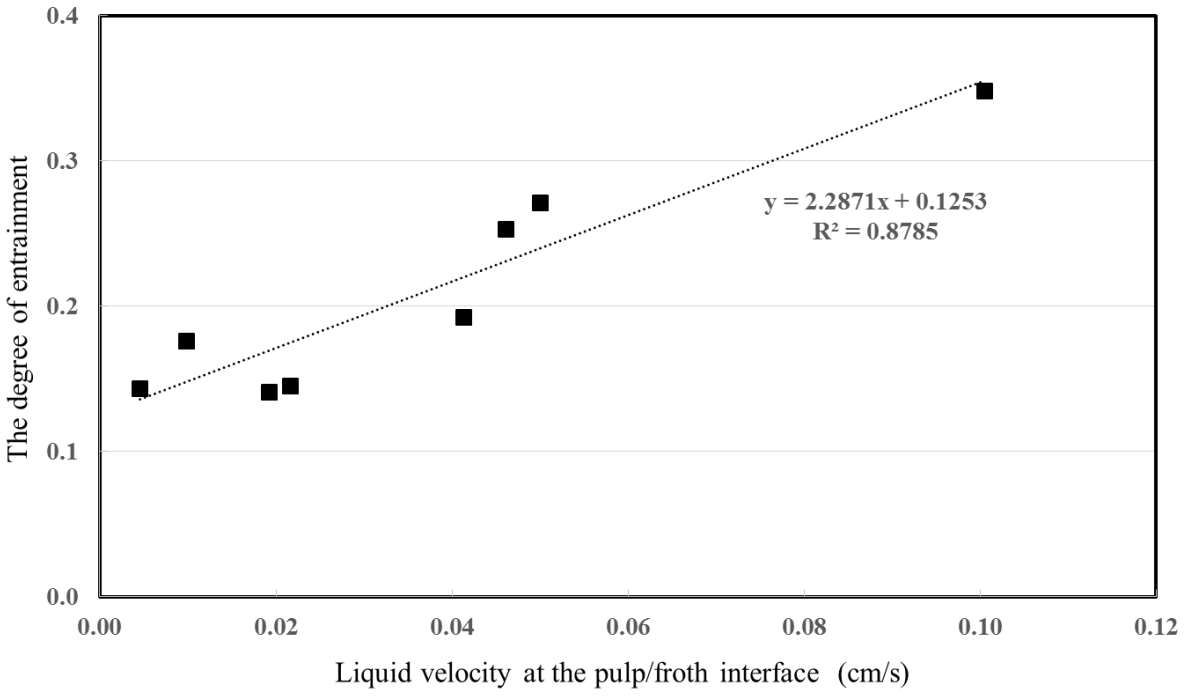

Figure 6-6 The correlation between the liquid velocity at the pulp/froth interface and the degree of entrainment at different frother types and concentrations.

In summary, when there is an increase in frother concentration, bubble size and the water surrounding the bubbles decrease at the interface, which leads to an increase in gas holdup (i.e. a decrease in liquid holdup). The increase in frother concentration also increases the net water flowrate to the final concentrate, resulting in an increase in the net water flowrate at the interface. The result is that at a higher frother concentration, there is a higher actual liquid velocity at the interface, resulting in a greater drag force acting upwards on the solid particles, i.e. a greater amount of solid particles are being transferred to the concentrate per unit mass of water.

Similarly, when there is a change in frother type, there is also a change in bubble size distribution, leading to a change in the net liquid flowrate at the interface, and hence the drag force acting on the solid particles. Thus, different degree of entrainment values are obtained.

A number of studies have emphasized the significance of the interface region for separation of gangue minerals from valuable minerals (Cutting et al., 1984; Szatkowski, 1987; Yianatos et al., 1987; Tao et al., 2000; Smith and Warren, 1989; Ata, 2001; Schwarz, 2004; Neethling and Cilliers, 2009; Zheng, 2009). The experimental results in this study suggest that the drainage of solid particles relative to water occurs at the interface region. This is supported by observations made in **Chapter 4** where it was observed that the degree of entrainment was not correlated to froth

residence time. Further, the results also support the Bubble Swarm Theory as a dominant mechanism for the transfer of water and gangue solid particles across the interface region (Smith and Warren, 1989). That is, bubble swarms congest at the interface and push water upwards, resulting in a drag force for solid particles to move upwards with water. If the drag force on the particle is sufficient for it to rise at the interface region, it will continue within the water phase and only return to the pulp phase due to water drainage.

6.2.3 Rheological effects

Rheology anecdotally affects entrainment significantly, although to date the effects are ill understood. In the pulp, a change in the viscosity influences the turbulence, which potentially affects the state of solids suspension below the pulp/froth interface which will affect the amount of solids entering the froth. It also affects the bubble size distribution and gas holdup which will affect the froth characteristics and therefore the degree of drainage in the froth (Shabalala et al., 2011). At the pulp/froth interface, if the flow is laminar, slurry viscosity will also increase the drag force on the solid particles as drag force is proportional to the fluid viscosity (Elger et al., 2013).

Frother could therefore be affecting the degree of entrainment because it is resulting in a change in the pulp viscosity at the interface. To test this hypothesis, viscosity measurements were performed of the pulp with different concentrations of frother added. It was assumed that the pulp phase is perfectly mixed, so the viscosity of the slurry will be reasonably representative of viscosity at the pulp/froth interface region. Viscosity measurements were also performed of a water solution in which frothers had been added at different concentrations.

Figure 6-7 shows the measured viscosities of solution containing different concentrations of Dowfroth 250 and MIBC. As can be seen, the higher concentration of Dowfroth 250 and MIBC did not appear to significantly affect the solution viscosity.

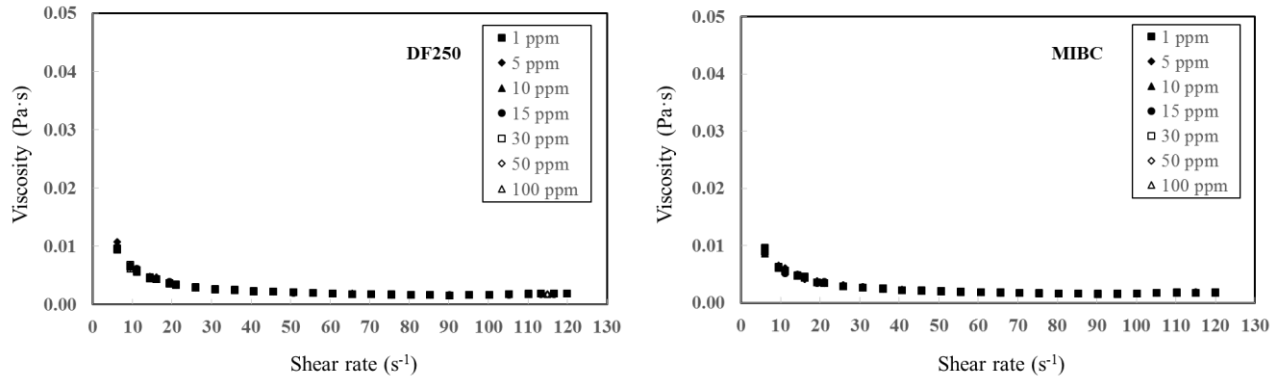

Figure 6-7 Viscosity of solution measured in the presence of Dowfroth 250 and MIBC at the pH of 9.5.

The viscosities of the quartz slurry containing different frother concentrations (i.e. 5 ppm, 10 ppm and 15 ppm) were also measured. As shown in Figure 6-8, the presence of quartz in the system slightly increased the viscosity, as quartz particles started to interact with each other in the flow, creating resistance to flow. However, an increase in the concentration of both Dowfroth 250 and MIBC did not appear to change the slurry viscosity. From the study by Cruz et al. (2013) where the effect of frother on the slurry viscosity of clay minerals was investigated, increasing Interfroth 6500 concentration to 30 ppm did result in an increase in the viscosity of 20 wt.% kaolinite suspensions and 8 wt.% bentonite suspensions. This implies that the effect of frother on the fluid viscosity is mineral type dependent. In this study, the reason for the unchanged frother solution viscosity and quartz slurry viscosity with increasing concentration of DF250 and MIBC is unknown, but it may be associated with saturated hydration between molecules and water or no interaction between the quartz mineral and frother molecules (Schott, 1968; Laskowski, 1993; Yalçın et al., 2002).

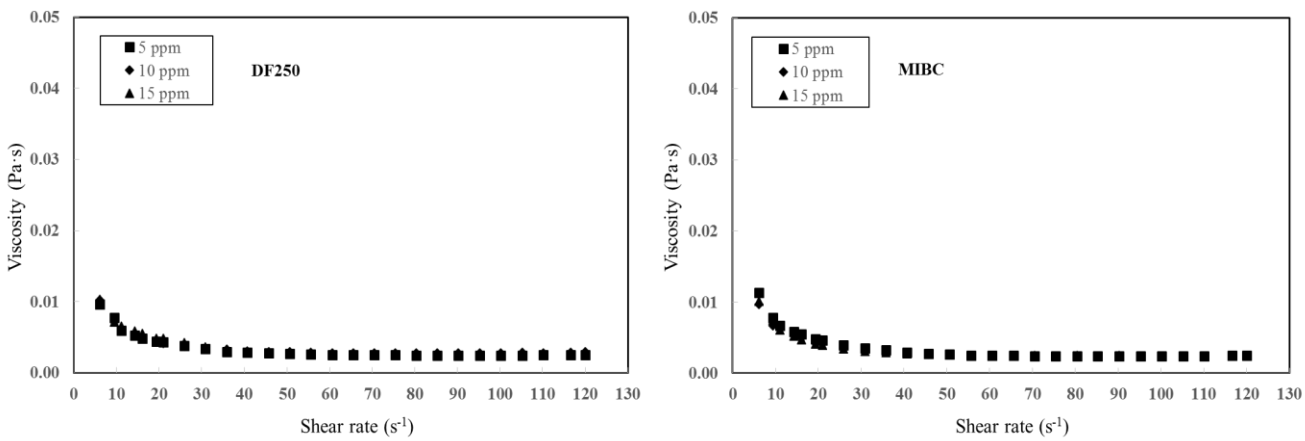

Figure 6-8 Viscosity of 37 wt.% quartz suspension measured in the presence of Dowfroth 250 and MIBC at the pH of 9.5.

It can be therefore concluded that changes in viscosity was not the main reason for the increases in the degree of entrainment for quartz observed in this study with increasing frother concentration of Dowfroth 250 or MIBC. It does not discount, however, the scenario that frother will have an effect on slurry viscosity in other mineral systems, resulting in different degrees of entrainment through the mechanisms outlined above.

Other parameters (e.g. % solids, particle size, particle interactions) are known to affect slurry viscosity and therefore will potentially also affect the degree of entrainment by changing the drag force. This should be studied in future work.

6.3 Summary

The recovery of quartz by entrainment was shown to be affected by DF250 and MIBC, as a consequence of frother changing both the water recovery and the degree of entrainment.

When frother concentration was increased, bubble size was shown to decrease significantly at the pulp/froth interface, which resulted in an increase in the gas holdup (i.e. a decrease in the liquid holdup) and thus less volume for water to pass through. The water superficial flowrate was also found to increase. These changes produced a higher liquid velocity through the Plateau borders at the interface and therefore increased the drag force acting on the solid particles. Therefore, the amount of particles that can overcome the downward gravitational force to be recovered by the entrainment mechanism to the final concentrate increased, and the degree of entrainment increased. This was supported by the observed linear relationship between the degree of entrainment and liquid velocity at the pulp/froth interface obtained from the tests performed using different frother types and concentrations. These results suggest that the liquid velocity at the interface is a key factor which affects the degree of entrainment.

Chapter 7 The effect of gas flowrate and froth height on the degree of entrainment

This chapter investigates the mechanism by which gas flowrate and froth height affect the degree of entrainment.

7.1 Introduction

In **Chapter 4**, an interactive effect between gas flowrate and froth height was found to have a statistically significant influence on the degree of entrainment. It was initially assumed that this effect was due to the gas flowrate and froth height affecting the froth residence time and therefore the time available for particles to drain in the water contained within the froth. However, the degree of entrainment in these tests did not correlate with froth residence time and therefore some other mechanism must be involved.

Chapter 6 demonstrated that the degree of entrainment was correlated with the liquid velocity at the interface presumably because it changes the drag force on the particles. It was hypothesised that gas flowrate and froth height might affect the degree of entrainment via the same mechanism. Gas flowrate and froth height will be affecting the velocity of the water flow at the pulp/froth interface and therefore, potentially, the drag force in this region. To test this hypothesis, entrainment tests were performed using fully liberated quartz in a 3.5 L laboratory mechanical flotation cell operated at different froth heights and gas flowrates and the liquid velocity at the pulp/froth interface was measured at each condition using the approach outlined in **Chapter 3.** The objective was to determine the underpinning mechanism.

A total of four tests were performed and Table 7-1 summarises the froth height and gas flowrate used in each experiment. It should be noted that these chosen operating conditions match those used in the factorial experimental program in **Chapter 4**. Dowfroth 250 was the frother used in the experiments which was added to give a frother concentration of 10 ppm. All other experimental conditions were the same as those used in the entrainment tests outlined in **Chapter 6** (including the fact that the test was only performed over 30 seconds). Again, valuable mineral was not added to the experiment to remove any complication that might arise due to variation in the valuable mineral recovery with changes in the test conditions. The aim was to study the change in the degree of entrainment due to gas flowrate and froth height only.

Table 7-1 Details of the entrainment tests performed at different froth heights and gas flowrates.

Test	Froth height, cm	Gas flowrate, L/min	*Jg*, cm/s
1	**1**	**8.5**	**0.6**
2	**1**	**14.5**	**1**
3	**2**	**8.5**	**0.6**
4	**2**	**14.5**	**1**

7.2 Results and discussion

A summary of the experimental results is shown in Table 7-2. Unfortunately, no experimental data were obtained from Test 3, as the froth height of 2 cm could not be maintained throughout the test at the low gas flowrate of 8.5 L/min.

Table 7-2 Summary of the experimental results from entrainment tests performed at different from heights and gas flowrates.

Test	*GR* %	*WR*%	*ENT*	Q_l/A_c, cm/s	$\varepsilon_{interface}$ %	v_l,cm/s
1	1.5	6.1	0.24	0.023	67	0.035
2	4.3	14	0.29	0.053	71	0.075
3	-	-	-	-	-	-
4	1.3	6.2	0.19	0.023	79	0.029

Note: *GR* is the gangue recovery, *WR* is the water recovery representing the fraction of water in the cell recovered into the concentrate, *ENT* is the un-sized degree of entrainment, Q_l/A_c is the superficial liquid velocity, $\varepsilon_{interface}$ is the liquid holdup at the pulp/froth interface and v_l is the liquid velocity at the interface.

Quartz is fully liberated in these tests. Therefore, it is reasonable to assume that all the quartz particles recovered into the concentrate were due to entrainment. As can be seen in Table 7-2, there are large variations in the quartz recovery by entrainment as gas flowrate and froth height are varied. This is despite the total flotation time being relatively short (30 seconds). Quartz recovery increased from 1.5% to 4.3% when the froth height was maintained at 1 cm and the gas flowrate was increased from 8.5 L/min from 14.5 L/min. At a gas flowrate of 14.5 L/min, quartz recovery decreased from 4.3% to 1.3% as the froth height was increased from 1 cm to 2 cm. The recovery of quartz in the different tests will likely be due to changes in both the water recovery as well as variations in the degree of entrainment.

Figure 7-1 shows the changes in gangue recovery, water recovery and the degree of entrainment with a change in froth height and gas flowrate. There was an increase in gangue recovery, water recovery and the degree of entrainment as gas flowrate increased from 8.5 L/min to 14.5 L/min at the shallow froth depth of 1 cm. It was also observed that gangue recovery, water recovery and the degree of entrainment increased when froth height was decreased from 2 cm to 1 cm at the high gas flowrate of 14.5 L/min. These results are in line with the types of trends observed in previous studies (Englebrecht and Woodburn, 1975; Cutting et al. 1981; Langberg and Jameson, 1989; Zheng et al. 2006b).

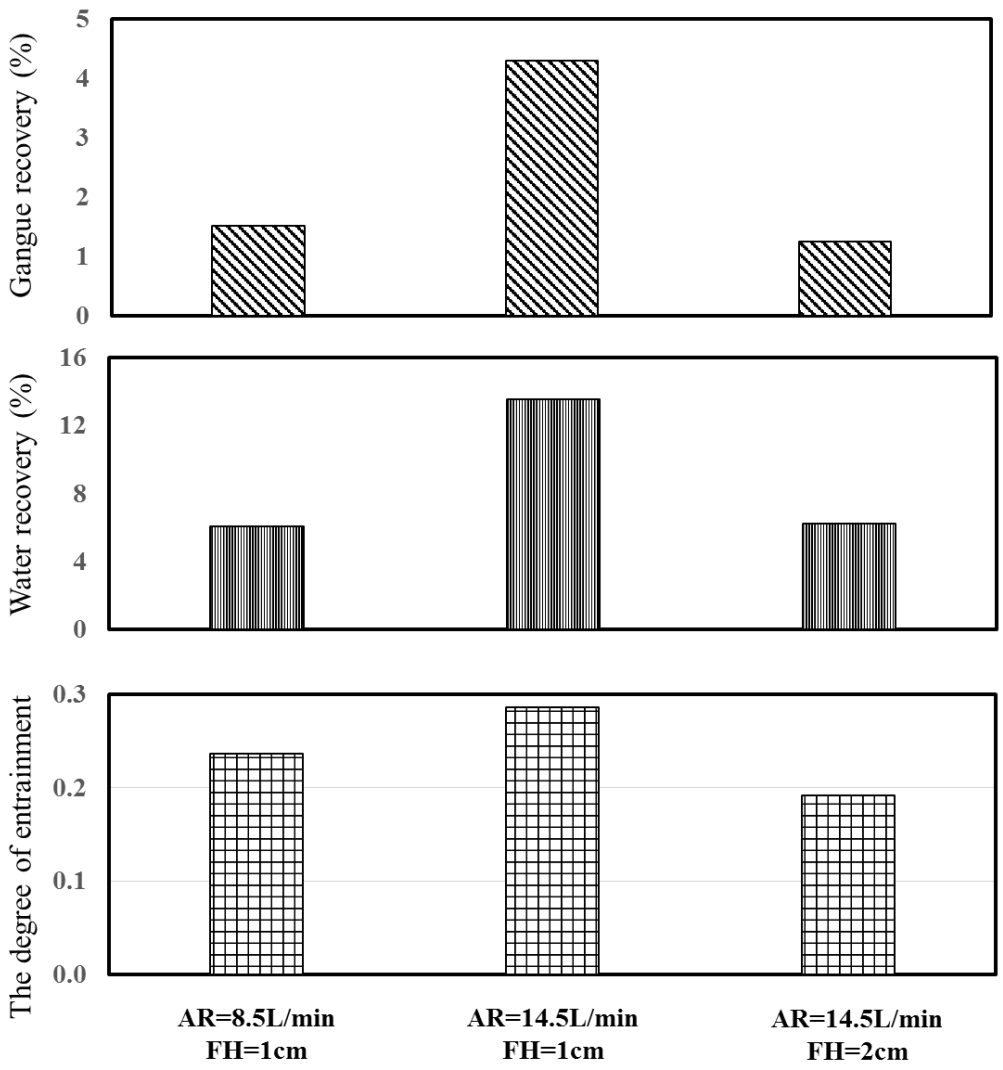

Figure 7-1 Gangue recovery, water recovery and the degree of entrainment measured at different air rates and froth heights.

The water flow to concentrate in these tests increased when the gas flowrate increased or the froth height decreased (Table 7-2). The highest superficial liquid velocity was therefore achieved in Test 2 performed using a shallow froth depth and a high gas flowrate.

Figure 7-2 shows the measured liquid holdup at the pulp/froth interface for different combinations of froth height and gas flowrate. The results indicate that either an increase in froth height or an increase in gas flowrate results in an increase in the water content at the interface. Presumably the reason why an increase in froth height (at a particular gas rate) results in an increase in water content at the interface is because this results in less water being recovered to concentrate due to the fact that it provides more time for drainage through the froth phase. This results in a greater volume of water in the lower regions of the froth. The increase in water content at the interface at a higher gas flowrate (for a particular froth depth) is probably because the congestion of a larger number of

bubbles at the interface, enables a greater amount of water to enter the froth. This is counteracted somewhat by the higher gas flowrate resulting in shorter froth residence times and thus less drainage of water from the upper regions of the froth phase. Figure 7-3 shows examples of the froth images of the pulp/froth interface captured during each entrainment test. Bubble size in this region was calculated to be similar at the different gas flowrates and froth heights, implying that the change in water content at the interface is because of the greater ratio of water to bubbles in this zone.

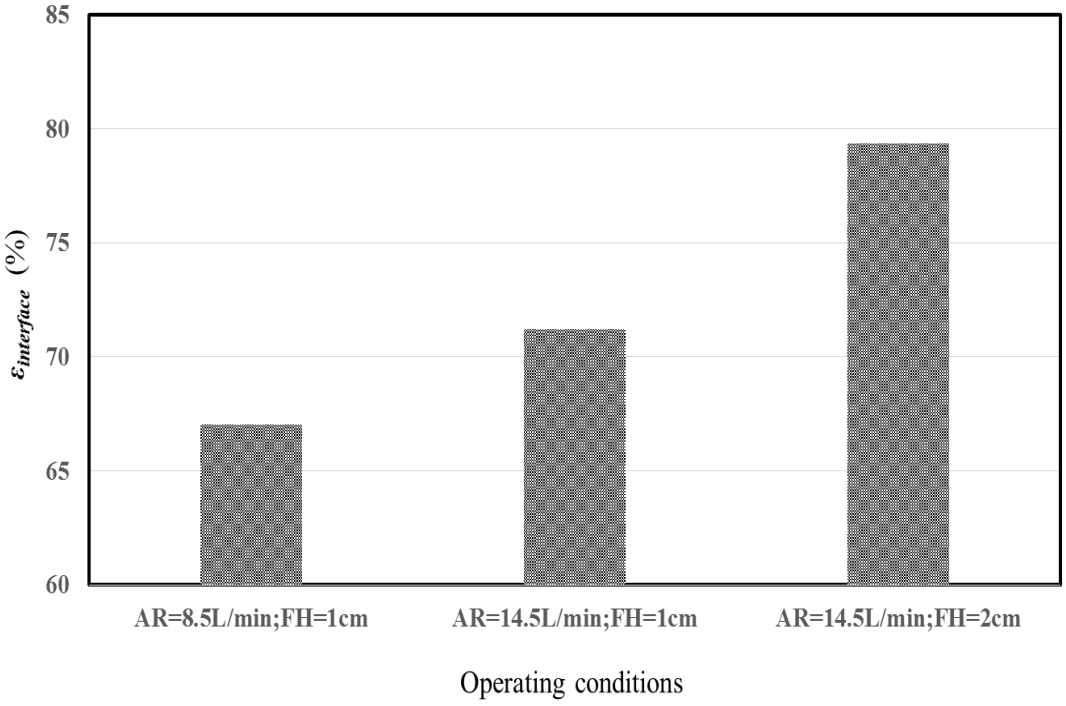

Figure 7-2 Liquid holdup at the pulp/froth interface measured in the batch entrainment tests performed at different gas flowrates and froth heights.

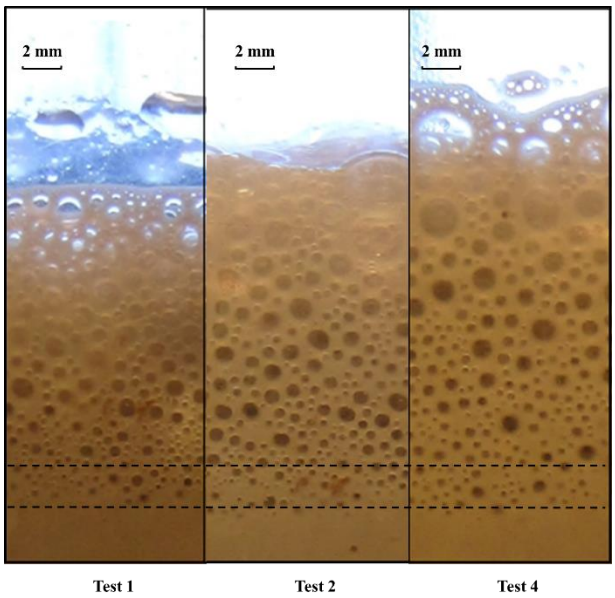

Figure 7-3 Images of the froth captured during the batch entrainment tests performed at different gas flowrates and froth heights.

Using the measured superficial liquid velocity and water content at the pulp/froth interface, the actual liquid velocity at the interface was calculated using Equation 2-3. Figure 7-4 shows the liquid velocity at the interface calculated for the tests performed at different gas flowrates and froth heights. As can be seen, at a high gas flowrate and shallow froth depth, there was a relatively large liquid velocity at the pulp/froth interface. When gas flowrate was decreased from 14.5 L/min to 8.5 L/min at the froth height of 1 cm, or when froth height was increased from 1 cm to 2 cm at the gas flowrate of 14.5 L/min, the liquid velocity at the interface decreased.

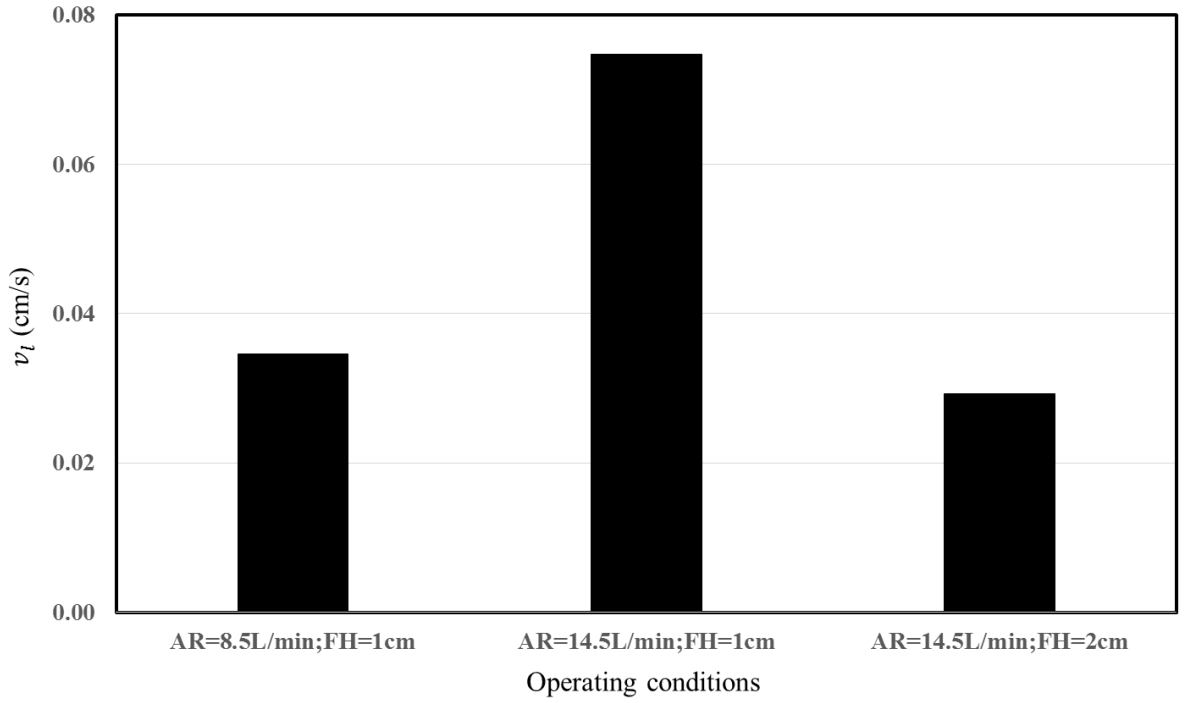

Figure 7-4 Effective liquid velocities at the pulp/froth interface at different air rates and froth heights.

Figure 7-5 shows the correlation between the degree of entrainment and the liquid velocity at the pulp/froth interface for the tests performed at different froth heights and gas flowrates. As observed in **Chapter 6**, the degree of entrainment is again a strong function of the liquid velocity at the interface, and the relationship is linear. This implies that the changes in the degree of entrainment obtained with changes in gas flowrate and froth height are largely due to these variables changing the liquid velocity at the pulp/froth interface which changes the drag force acting on the particles. This changes the concentration of gangue in the water that reports to the froth phase.

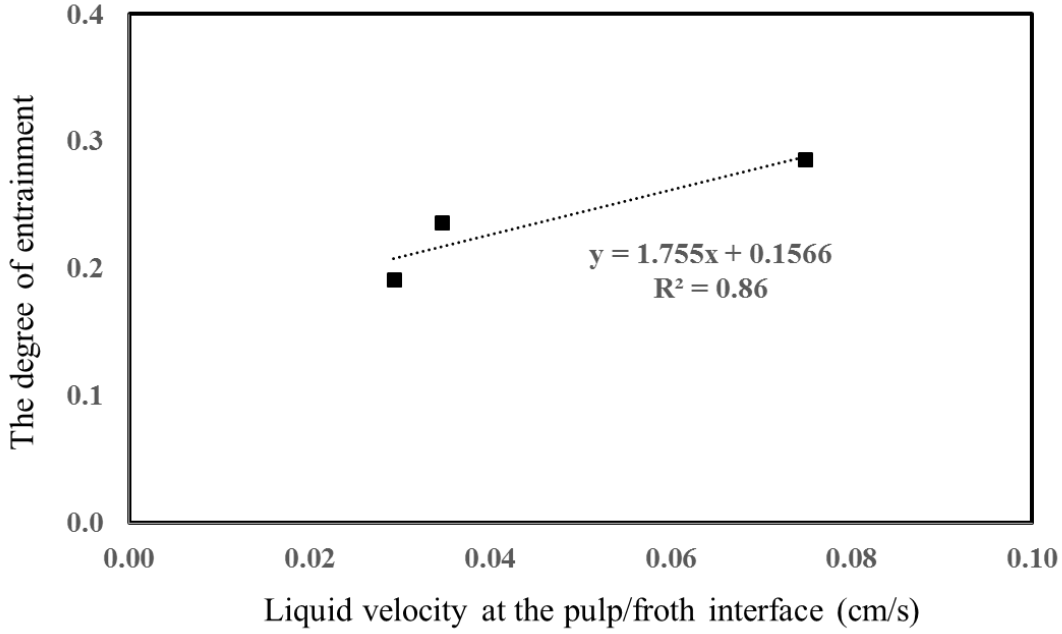

Figure 7-5 The correlation between the liquid velocity at the pulp/froth interface and the degree of entrainment of liberated quartz particles at different froth heights and gas flowrates.

In **Chapter 6**, frother type and concentration were also found to change the degree of entrainment because they resulted in a change in the liquid velocity and therefore the drag force at the pulp/froth interface (refer to Figure 6-8). If the same mechanism is underpinning the effect of frother, gas flowrate and froth height on the degree of entrainment, there should be one single degree of entrainment value that corresponds to one liquid velocity irrespective of these operational parameters. Figure 7-6 shows the correlation between the degree of entrainment and the liquid velocity at the pulp/froth interface measured in the tests performed at different gas flowrates and froth heights and that obtained in the tests outlined in **Chapter 6** performed using different frother types and concentrations. The single linear relationship between the liquid velocity at the pulp/froth interface and the degree of entrainment at different operating conditions strongly supports the argument that the changes in the degree of entrainment with gas flowrate, froth height, frother type and concentration were a consequence of a change in liquid velocity and therefore drag force at the interface. The linear relationship also suggests that the liquid flow through the interface region is laminar as this will result in the drag force being proportional to velocity.

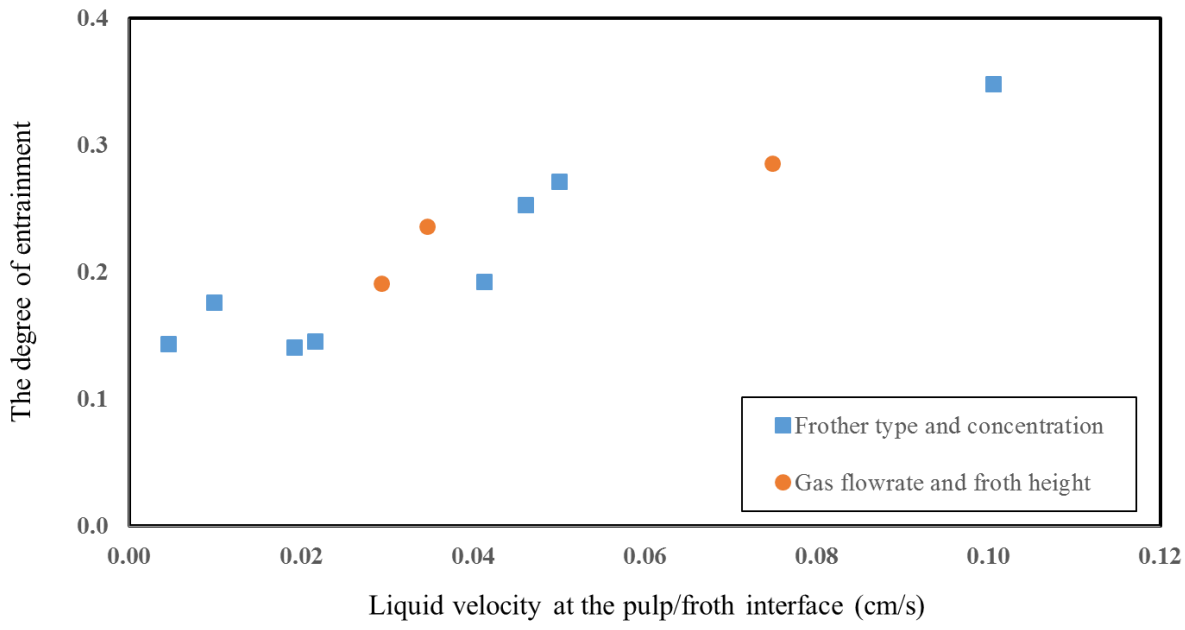

Figure 7-6 The correlation between the liquid velocity at the pulp/froth interface and the degree of entrainment of liberated quartz particles at different operating conditions.

In summary, the gas flowrate and froth height affect the upward water velocity at the interface by affecting both the amount of water being pushed through the zone and the volume through which this water must travel (as a consequence of changes in the air content). These effects are a consequence of froth height and gas flowrate affecting froth residence time and therefore the degree of drainage in the froth. Gas flowrate also affects the amount of water being pushed into the froth phase from the pulp. Consequently, gas flowrate and froth height interactively change the liquid velocity at the interface and thus the drag force acting on solid particles, resulting in different degrees of drainage of solid particles relative to water at the interface.

The observations made in **Chapter 4**, however, cannot all be explained by this reasoning. At low gas flowrates, the degree of entrainment was found to increase as froth depth was deepened (refer to Figures 4-9 and 4-10). This is a trend also observed by Zheng et al (2006b) in entrainment measurements performed in an industrial scale continuously operated flotation cell. At low gas flowrates and deep froths, a very low liquid velocity at the interface is expected which, according to the above reasoning, should result in lower degree of entrainment values. The higher results suggest that other mechanisms may be involved. It has been reported that the drainage of solid particles in a froth can be hindered due to the narrowing plateau borders that occur when water drains from the froth. This mechanism is known as “entrapment” and is more likely to occur when valuable minerals are attached to the bubbles (Ata et al., 2004). As froth height at a low gas flowrate is increased, more water will drain but the solid drainage may largely be hindered by the entrapment

mechanism, resulting in an increase in the concentration of entrained materials in the upper regions of a deep froth. This will result in the solids concentration increasing (rather than staying constant as would be expected due to increased drag forces). Thus, there will be another distinct region in a froth in terms of the solid concentration of gangue minerals.

Thus, the evidence presented in this thesis (combined with observations by others) suggests that the profile of hydrophilic gangue particle concentrations in a flotation cell can be generally described as follows:

- From the bottom of the flotation cell to the region below the pulp/froth interface, there is classification of solids due to the state of solids suspension in the pulp phase.
- Across the pulp/froth interface region, there is a rapid drop in the gangue concentration as a result of the drag force due to the upward flow of water only being sufficient to overcome the weight of the smaller and less dense particles. Only a proportion of particles are therefore dragged into the froth phase with the water.
- After this decrease, the gangue particle concentration remains relatively constant as the upward velocity of water and thus the drag force acting on the particles increases with froth height due to a reduction in the froth liquid content. Only particles in the downward flowing water drain back to the pulp.
- If the froth is sufficiently deep, there comes a point where the solid concentration increases with froth height as the entrained particles become entrapped in the bubbles but the water content decreases due to drainage.

The degree of entrainment will be proportional to the concentration of gangue in the water in the upper regions of the froth which reports to the final concentrate.

Unfortunately there was no experimental data able to be produced to confirm that the degree of entrainment increases when a low gas flowrate and deep froth depth is used, and is no longer proportional to the liquid velocity at the interface. The froth proved unstable at this condition (Test 3) without valuable mineral present in the system. This should be investigated in future work, as the effect of deep froth and low gas flowrate on entrainment is of great importance especially in cleaner circuit in a flotation plant. The study does, however, provide the experimental evidence of how froth height and gas flowrate affect the degree of entrainment outside of these conditions.

7.3 Summary

Entrainment tests using fully liberated quartz were performed at different froth heights and gas flowrates in a 3.5 L flotation cell. The degree of entrainment in these tests varied significantly and was found to be linearly correlated with liquid velocity at the interface, suggesting that the observed changes in the degree of entrainment was due to changes in the drag force acting upon the particles at the interface region. This is the same mechanism by which frother was found to affect the degree of entrainment in **Chapter 6**. This underlying mechanism was further substantiated by the fact that the relationship between liquid velocity and the degree of entrainment was independent of whether gas rate, froth height, type of frother or frother concentration was varied in the experiments.

When gas flowrate and froth height was varied, bubble size was shown to be similar at the pulp/froth interface but water content between adjacent bubbles changed significantly. This resulted in a significant variation in liquid holdup and thus the volume for water to pass through. The water superficial flowrate was also found to change. These changes produced variation in liquid velocity through the Plateau borders at the interface, and therefore altered the drag force acting on the solids. As a result, the amount of particles that overcame the downward gravitational force to be recovered by the entrainment mechanism to the final concentrate changed, leading to differences in the degree of entrainment.

This mechanism does not explain the high degree of entrainment observed when deep froths and low gas flowrates were used in a test presented in **Chapter 4**. It is speculated that this change is a result of entrapment occurring when the froth becomes “dry”. This suggests that in some cases there is a region in the upper portion of froth where the solid concentration increases with froth height due to the solid particles be entrapped in the bubbles whilst water continues to drain.

Chapter 8 The effect of flotation operating conditions and particle properties on water recovery

This chapter analyses the results from the two-level factorial batch flotation experiment described in ***Chapter 4*** *with respect to water recovery. In this chapter, the data are analysed to identify which of the examined flotation conditions and particle properties (including interactions) significantly affect water recovery. The potential mechanism underpinning the identified effects is explored and an approach to modelling water recovery is presented.*

8.1 Introduction

In **Chapter 4**, a two-level batch flotation experimental program was conducted and analysed to identify the operational variables and particle attributes which affect the degree of entrainment significantly. In these tests impeller speed, froth height, gas flowrate and particle density were the factors varied. It should be noted, however, that the recovery of gangue mineral particles by entrainment is not only dependent on the degree of entrainment but also water recovery. To minimise the overall entrainment recovery, therefore, the effect of these variables on water recovery should also be considered.

In this chapter, the water recovery data obtained from the two-level factorial experiments outlined in **Chapter 4** was studied. The aim was to identify which of the investigated parameters have a significant effect on the water recovery and if possible, determine the mechanism in order to provide insights into how best to model water recovery. As outlined in **Chapter 2**, an empirical model to predict water recovery as a function of the feed and operating characteristics of a flotation cell is yet to be developed.

8.2 Results

8.2.1 Flotation results

A summary of the water and gangue recovery and concentrate grade measured in the two-level factorial batch flotation experiments is shown in Table 4-2. As stated in **Chapter 4**, copper recoveries in the rougher flotation tests performed in this study were generally high and large variations in concentrate grades and water recoveries were observed. As the gangue minerals used in these tests were fully liberated, it is reasonable to assume they were recovered into the concentrate only by entrainment. Therefore, the variation in the concentrate grade was caused primarily by changes in the entrainment recovery of the gangue minerals. Entrainment is primarily driven by water recovery and it is therefore expected that both gangue recovery and concentrate grade will be correlated with water flow to concentrate. Figure 8-1 shows the concentrate grade obtained as a function of water recovery in the factorial experiments. As expected, concentrate grade was strongly correlated with water recovery, decreasing significantly with an increase in water recovery. Higher water recoveries resulted in greater gangue recovery, as shown in Figure 4-1 in **Chapter 4**.

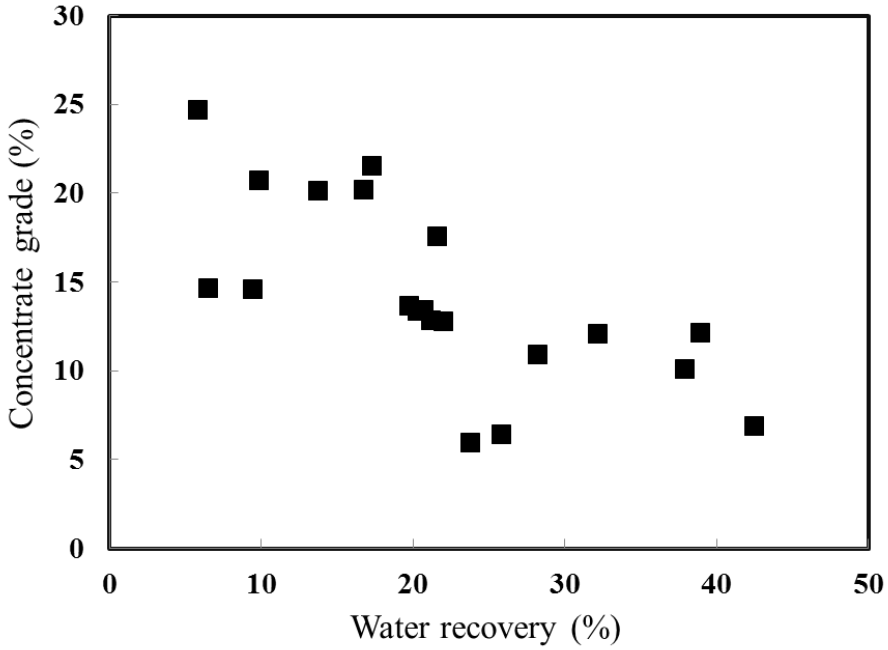

Figure 8-1 The relationship between water recovery and concentrate grade.

8.2.2 Statistical significance of variables

The effects and interactions of the four variables on the water recovery at a 95% confidence level were analysed using Minitab 17 statistical software. The results of the regression analysis are shown in Table 8-1.

Table 8-1 Results of a statistical evaluation of the effect of four variables and their interactions on the water recovery.

Term	P value	Effect on response
Impeller speed	0.001	Positive
Particle density	0.980	
Gas flowrate	0.000	Positive
Froth height	0.000	Negative (exhibits maximum)
Impeller speed* Particle density	0.003	Positive
Impeller speed* Gas flowrate	0.894	
Impeller speed* Froth height	0.363	
Particle density* Gas flowrate	0.655	
Particle density *Froth height	0.836	
Gas flowrate* Froth height	0.021	Negative (exhibits minimum)
Curvature	0.269	

The effect of the variables on the response can also be evaluated using a normal probability plot, as shown in Figure 8-2. As can be seen, three main effects and two interactions stand out and have a much more significant effect than the other variables. These significant variables in a decreasing order of importance are: (1) froth height, (2) gas flowrate, (3) impeller speed, (4) the interaction between impeller speed and particle density and (5) the interaction between gas flowrate and froth height.

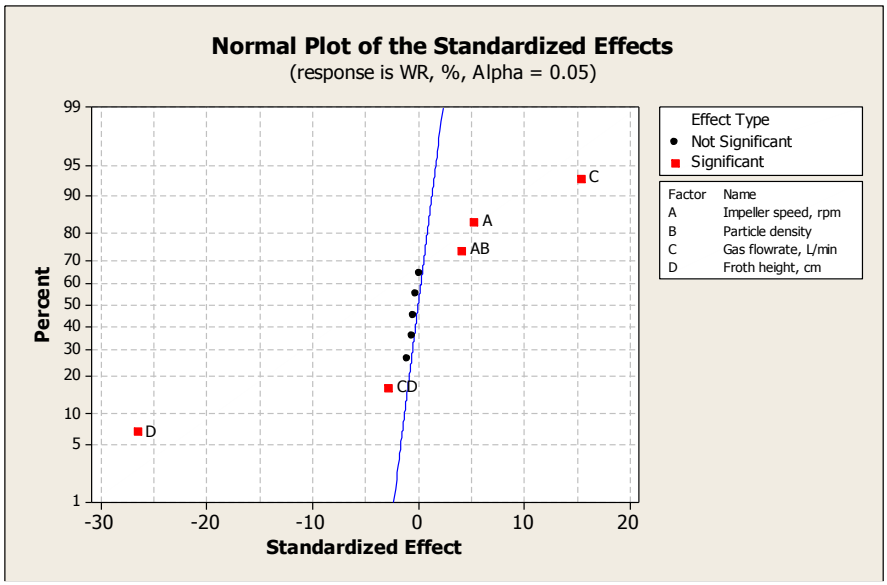

Figure 8-2 Normal probability plot created by Minitab when analysing the effect of the experimental variables and their interactions on the water recovery.

In Table 8-1, the P value related to "curvature" (0.269) is relatively large indicating, statistically, there is no evidence of curvature between the examined factors and the measured water recovery, in the range over which the factors were investigated. A linear mathematical relationship was therefore able to be developed between water recovery and the factors and interactions identified from the factorial experiment (see Equation 8-1):

$$WR = 8.083255 + 0.00877r + 2.83948G - 11.24258H_F + 0.000176(r \cdot \rho) - 0.67094(G \cdot H_F) \qquad 8\text{-}1$$

where WR is water recovery, r is impeller speed, G is gas flowrate, H_F is froth height and ρ is particle density.

The estimated water recovery using Equation 8-1 versus the measured water recovery was plotted as shown in Figure 8-3. It can be concluded that the mathematical relationship provides a good

estimation of water recovery as the differences between the observed values and the estimated values are small. The relationship therefore incorporates most of the significant factors (and their interactions) that are affecting the water recovery produced in the tests.

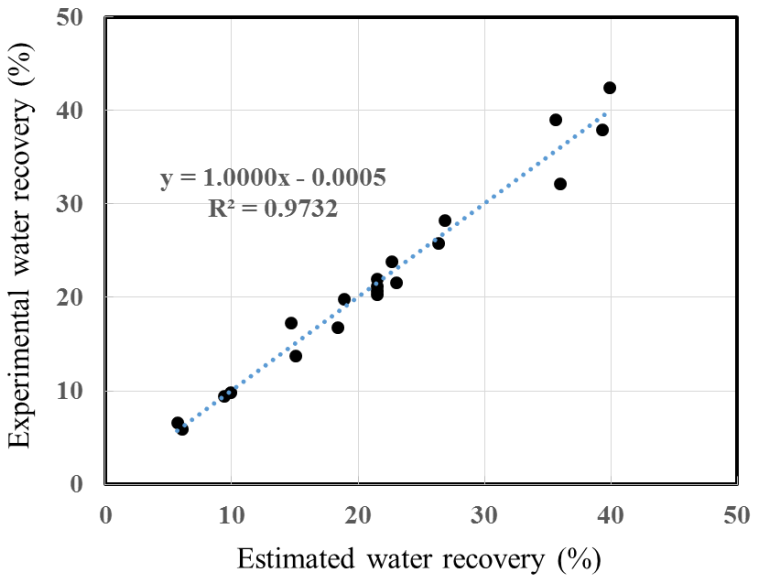

Figure 8-3 The estimated water recovery obtained from the mathematical model compared to that measured experimentally.

8.3 Discussion

8.3.1 The effect of impeller speed

The statistical analysis indicated that impeller speed was one of the variables that had a significant effect on the water recovery. Figure 8-4 shows the average water recovery plotted against impeller speed for all the tests performed under the different operating conditions. The correlation was positive and the average water recovery changed from 20% at 800 rpm to 24% at 1200 rpm. An analysis of Figure 8-1 highlights how this type of change in water recovery results in an appreciable change in concentrate grade due to an increase in entrainment recovery.

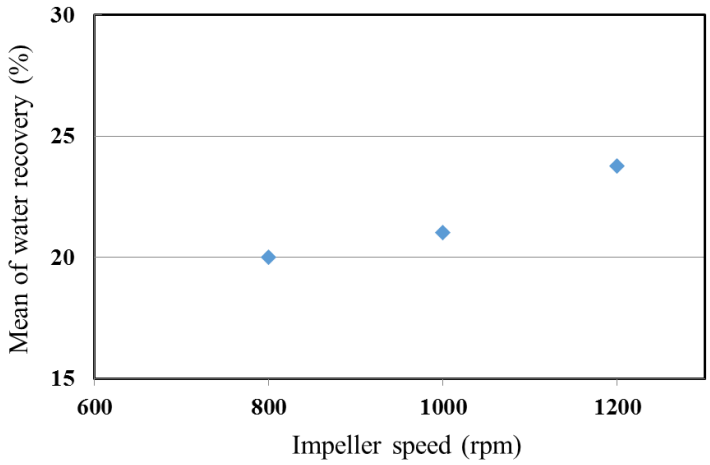

Figure 8-4 The average water recovery as a function of impeller speed in all the tests.

The increase of water recovery with increasing impeller speed may be a consequence of a change in the amount of water entering the froth. An impeller/stator system in a flotation cell agitates the slurry and disperses the air into bubbles to provide opportunities for collision between mineral particles and bubbles. When there is an increase in impeller speed, the flow leaving the rotor will have a higher velocity and there will be a stronger jet of fluid leaving the stator to strike the cell wall and move upwards (Fallenius, 1987).

The kinetic energy fluctuation at the pulp/froth interface was measured in an effort to determine if the fluid properties in this region were changed significantly with a change in impeller speed. Kinetic energy fluctuation was measured using a Piezoelectric Vibration Sensor (PVS) which converts mechanical vibrations to AC voltage when a piezoelectric film in the water flow deflects back and forth (Meng et al., 2014). Figure 8-5 shows the plane where a series of measurements were performed across the cell cross-section. The plane of measurements was located 1 cm away from the wall and 1 cm below the pulp/froth interface. The measurement at each flotation condition was performed at multiple positions and then averaged.

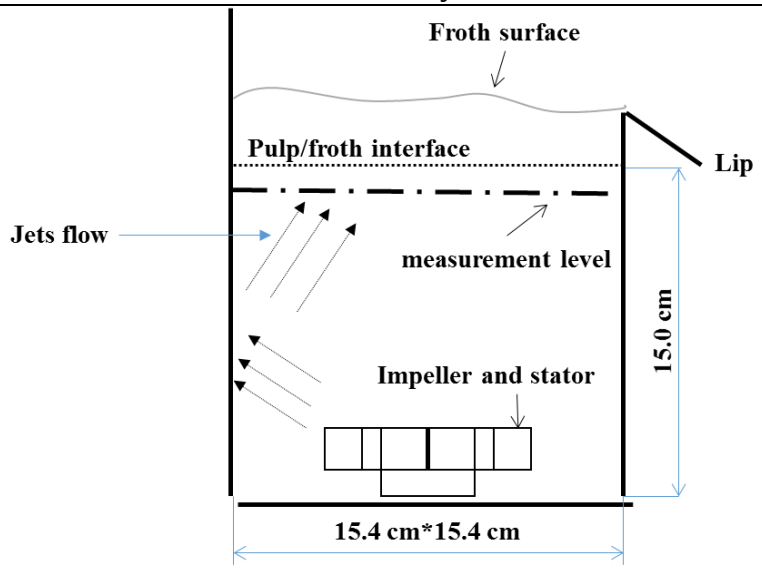

Figure 8-5 Side view outlining the level where turbulence was measured in the 3.5 L flotation cell.

Figure 8-6 shows the average kinetic energy fluctuation at the pulp/froth interface as a function of impeller speed for both quartz and hematite systems (a froth height of 1.5 cm and a gas flowrate of 11.5 L/min were used). The results show a significant increase in the kinetic energy fluctuation of fluid at the interface in both quartz and hematite systems when the impeller speed was increased. It is therefore likely that the velocity of the flow was also increased.

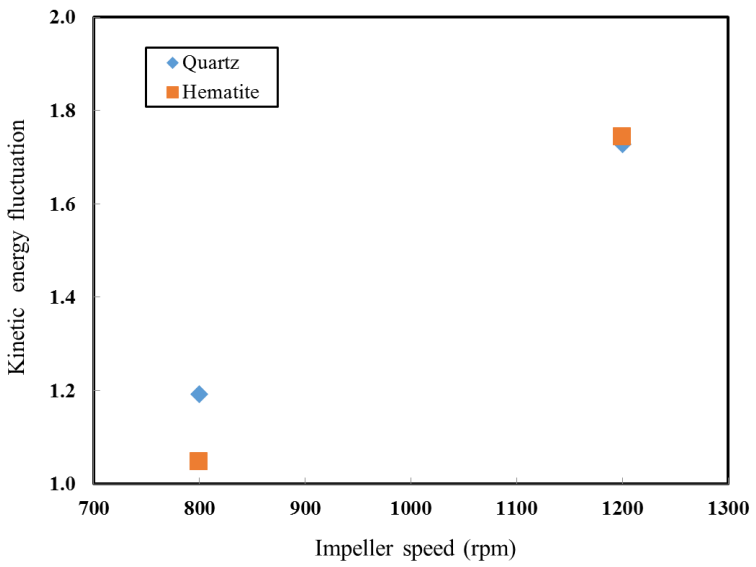

Figure 8-6 Kinetic energy fluctuation (turbulence) at the pulp/froth interface for quartz and hematite systems at different impeller speeds.

The effect of impeller speeds on water recovery may not only be a consequence of an increase in fluid momentum at the interface of the cell. It has been found that, for a particular gas flowrate, a greater number of small bubbles are produced at higher impeller speeds (Grau and Heiskanen, 2005). Small bubbles often result in more water into the froth, and they also change the froth structure in favour of promoting water transfer to the concentrate in the froth. This is because bubble size is one of the key factors determining the Plateau border length. The relationship between the length of the Plateau borders (L) and the bubble radius (r) in the foam has been established as: $L \propto r^{-2}$ (Neethling and Cilliers, 2003). A small bubble radius may therefore result in more resistance to water drainage and an increase in the amount of free water recovered.

8.3.2 The interaction of impeller speed and particle density

The results of the factorial experiments indicated that although particle density did not have a significant effect on water recovery, an interaction between particle density and impeller speed did have a significant effect. This is illustrated in Figures 8-7 and 8-8 which show the average water recovery of quartz (SG=2.65) and hematite (SG=5.26) at different impeller speeds in all the tests and for subsets of the tests. In the quartz system, water recovery increased slightly when impeller speed increased. In comparison, there was always an appreciable increase in water recovery in the hematite system with increasing impeller speed. This implies that the change in water recovery caused by impeller speed was dependent on which gangue mineral species was present in the flotation cell.

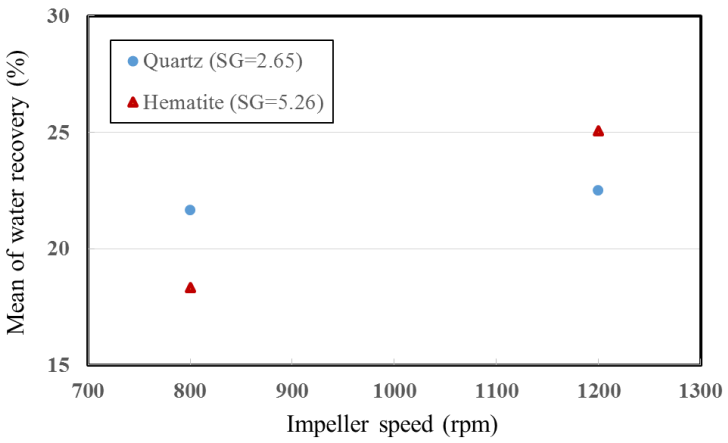

Figure 8-7 The relationship between impeller speed and water recovery for tests performed using gangue of different particle density.

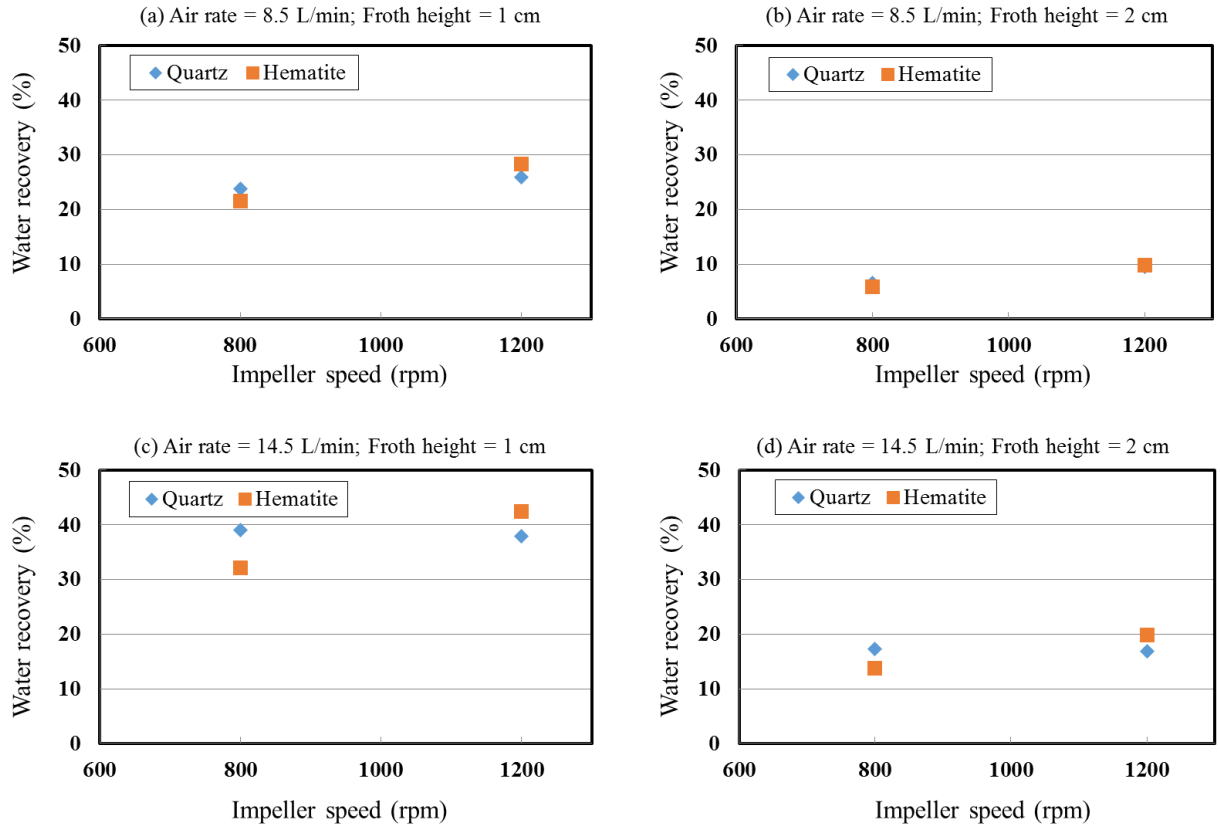

Figure 8-8 Water recovery versus impeller speed for tests performed using quartz and hematite for various combinations of cell operating conditions.

It is interesting to observe that the low-density gangue mineral (quartz) resulted in only a slight increase in water recovery when impeller speed increased, while the high-density gangue mineral (hematite) resulted in a much larger difference in water recovery under the same conditions (see Figure 8-8). It is speculated that the hematite system which had a significantly higher mineral density than the quartz system potentially resulted in a great force propelling water into the froth phase. It could also be a consequence of differences in the change in bubble size in different gangue mineral systems.

8.3.3 The effects of gas flowrate, froth height and their interactions

Water recovery was found to vary significantly with gas flowrate and froth height and there was also an interactive effect of these two variables. The overall effect of gas flowrate on water recovery is illustrated in Figure 8-9, which shows a positive linear relationship between water recovery and gas flowrate. An increase in gas flowrate is believed to result in a larger number of bubbles carrying water into the froth phase from the pulp phase. In addition, gas flowrate also affects the extent of

drainage of water. An increase in gas flowrate increases the froth rising velocity and shortens the froth residence time and, therefore, a lower proportion of water drains back to the pulp from the froth (Zheng et al., 2006b).

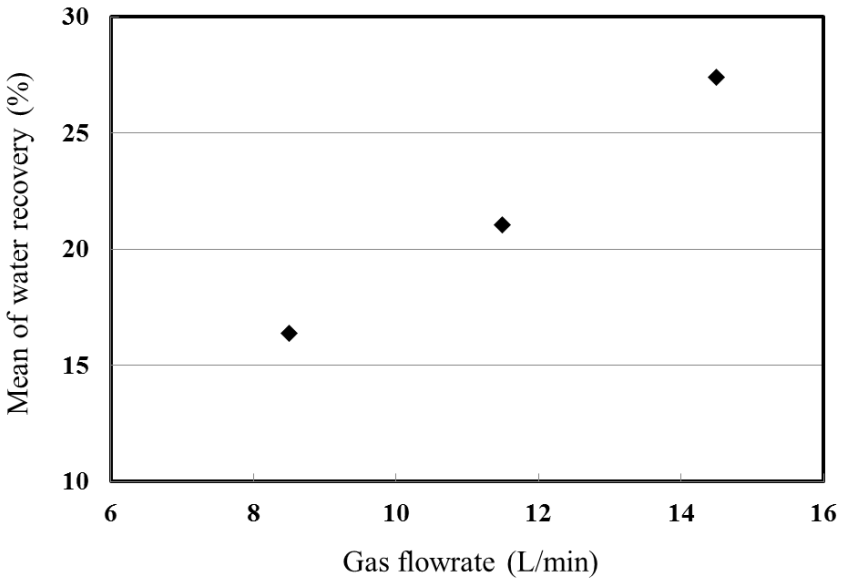

Figure 8-9 The average water recovery as a function of gas flowrate in all the tests.

Figure 8-10 shows the overall effect of froth height on water recovery. A negative correlation was found between water recovery and froth height with a marked decrease in water recovery when the froth height was increased from 1 cm to 2 cm. This is in line with other studies (Englebrecht and Woodburn, 1975; Cutting et al. 1981; Zheng et al. 2006b) and is probably due to two reasons. Firstly, an increase in froth height increases froth residence time providing a great opportunity for water in the froth to drain back to the pulp. The high amount of water drainage results in different water content (liquid hold-up) throughout the froth phase with a significant decrease from the pulp/froth interface to the concentrate launder and then to the top of the froth (Cutting et al., 1981). Hence, there is less water overflowing and collected at the concentrate launder when froth height increases. Secondly, the “dry froth” caused by the increased time for drainage of water (and associated solids contained in the water) could have a significant impact on froth structure and its stability. As observed by Ekmekçi et al (2003), with an increase in froth height, the drainage of water in the froth thins the bubble films leading to bubble coalescence and collapse with a lower froth stability and a brittle froth structure. An unstable froth may accelerate water drainage. Consequently, less water is recovered to the concentrate when the froth height is increased.

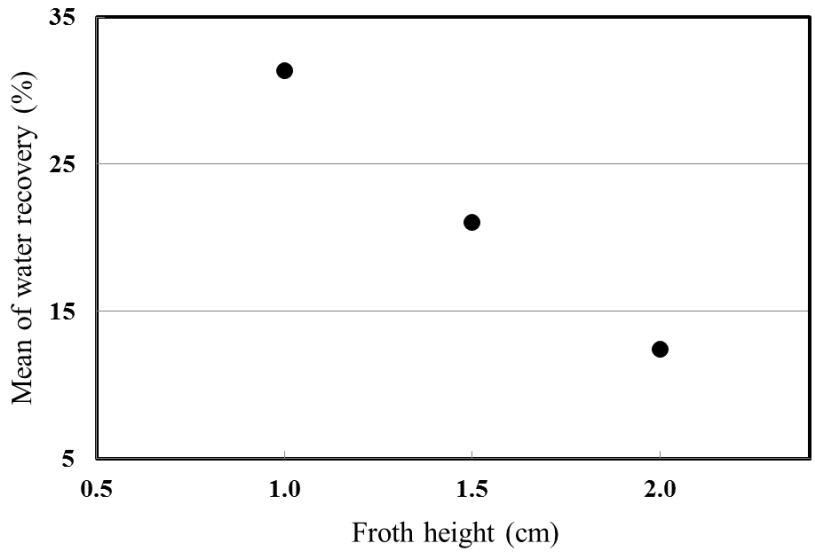

Figure 8-10 The average water recovery as a function of froth height in all the tests.

It was interesting to find that the interaction between gas flowrate and froth height was of significance in influencing water recovery (see Table 8-1). The interactive effect is illustrated in Figures 8-11 and 8-12 which show the average water recovery as a function of gas flowrate at different froth heights in all the tests and for subsets of the tests, respectively. Water recovery was less sensitive to gas flowrate as froth height was increased. There was a larger variation in water recovery at the shallow froth height (1 cm) than at the deep froth height (2 cm) when increasing the gas flowrate. Interestingly, water recovery was also more sensitive to the froth height as gas flowrate was increased, which is in agreement with the finding of Langberg and Jameson (1989). A larger decrease in water recovery was found at the high gas flowrate when increasing the froth height. These findings show that gas flowrate interacts with froth height at different levels to impact on water recovery.

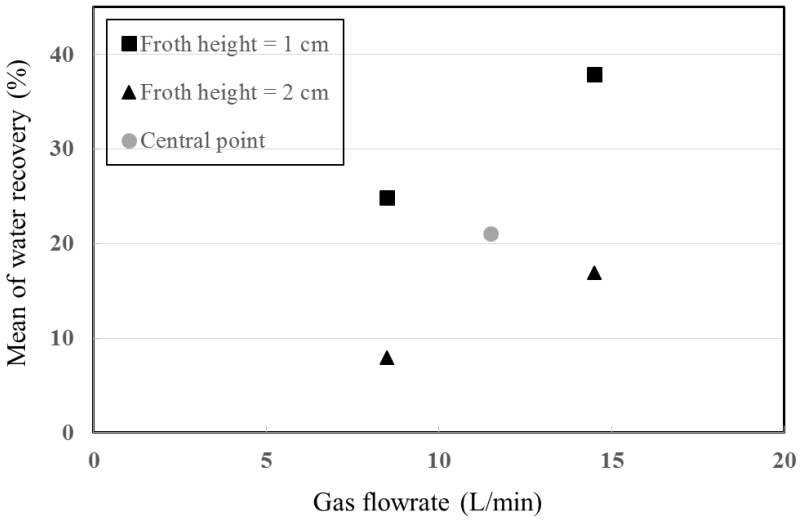

Figure 8-11 Average water recovery as a function of gas flowrate split into tests performed at different froth heights.

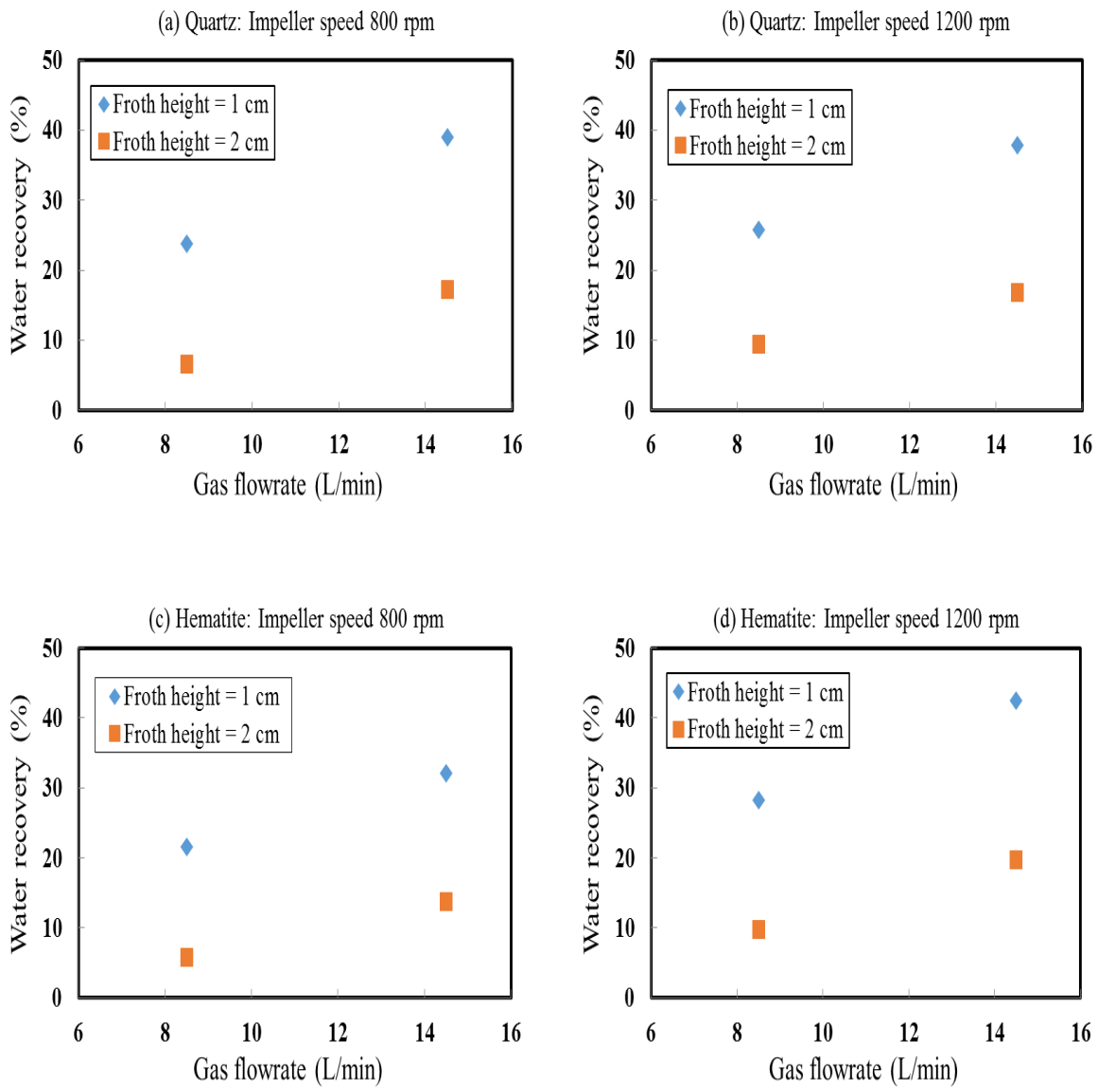

Figure 8-12 Water recovery as a function of gas flowrate split into tests performed at different froth depths for various combinations of other conditions.

As will be discussed in the next section, this interaction is in line with what was observed in the flotation tests when froth residence time increased. At longer froth residence time, which occurred when using deeper froth heights and lower gas flowrates, the rate of decrease in water recovery decreased.

8.3.4 Froth residence time

Gas flowrate and froth height had the most significant effect on the water recovered in this study. It is hypothesised that this is due to their effects on froth residence time which is often considered a

driver of water recovery as it provides the time for water to drain out of froth. Froth residence time is commonly calculated based on the gas froth residence time – the time the gas volumetric flow will take to move through the froth volume, as shown by Equation 2-9.

The effect of froth residence time of gas on water recovery under the various conditions investigated in this study is illustrated in Figure 8-13. As can be seen, different water recoveries were obtained at the same froth residence time. When the effects of impeller speed and particle density were removed, however, water recovery was related to froth residence time as shown in Figure 8-14. This is in line with the empirical water recovery model proposed by Savassi (1998):

$$R_w = c \cdot \lambda_{air}{}^{d} \qquad 8\text{-}2$$

where *c* and *d* are constant.

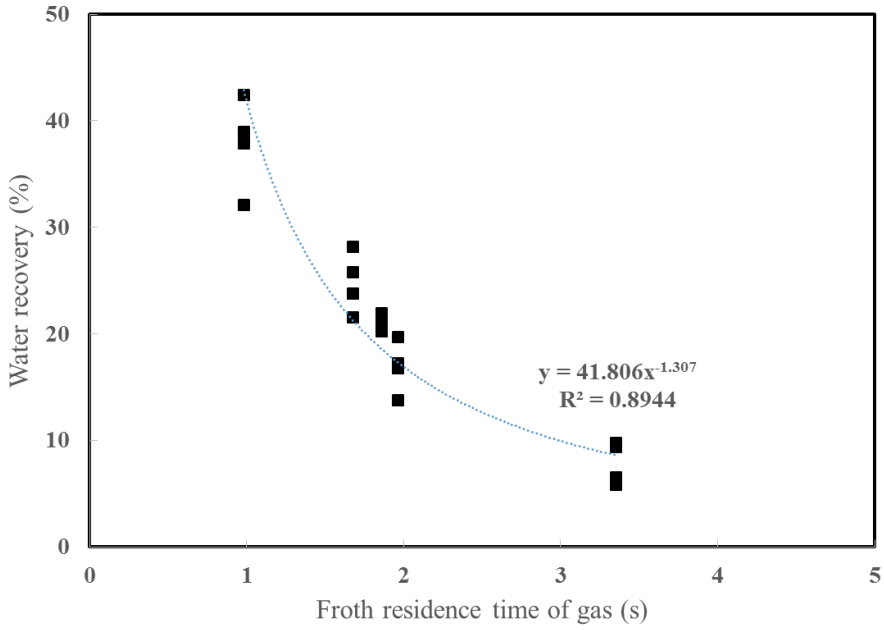

Figure 8-13 Correlation between water recovery and froth residence time of gas using all batch flotation test results.

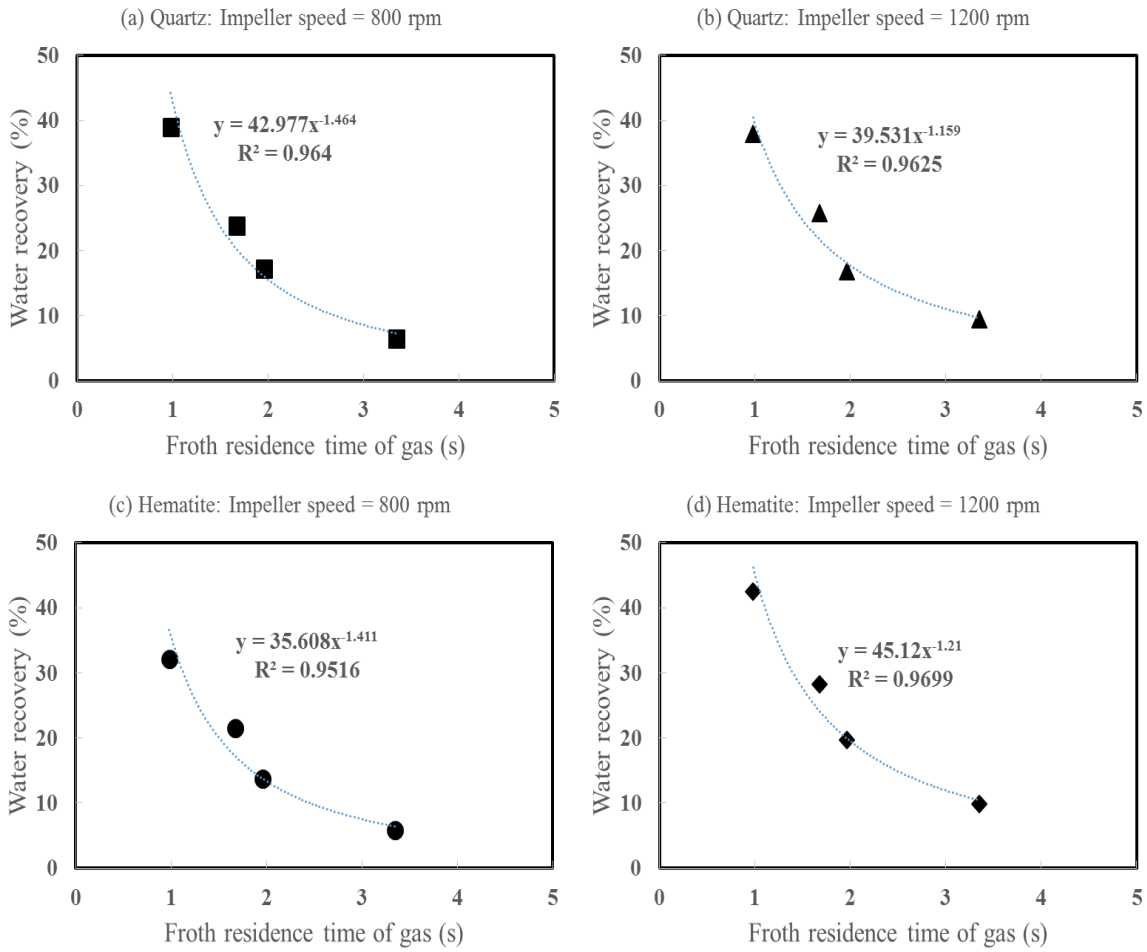

Figure 8-14 Experimental water recovery as a function of froth residence time under various operating conditions modelled using the empirical power relationship proposed by Savassi (1998).

The significant discrepancy between the measured and the predicted water recoveries when all the experimental data were used in the Savassi model (Equation 8-2) indicates that water recovery cannot simply be predicted using the froth residence time alone (see Figure 8-13). Pulp conditions such as impeller speed and particle density also contribute greatly to water recovery, probably because of the change in the water transfer to the froth from the pulp.

The Savassi power model fits the data adequately but it doesn't represent that, fundamentally, water recovery is a consequence of two processes – water flow into the froth and water drainage. If it is assumed that water drains from the froth at a constant rate, *b*, and that as the froth residence time approaches zero, the water recovery approaches the water recovered from the pulp into the froth, then Equation 8-3 can be derived.

Thus water recovery can be represented by an exponential function of froth residence time which is a function of two parameters of the system which have a physical significance: water recovery entering the froth from the pulp (R_w^{pulp}) and the rate of water drainage out of the froth to the pulp (b).

$$R_w = R_w^{pulp} \cdot \exp(-b \cdot \lambda_{air}) \qquad 8\text{-}3$$

This model was fitted to each combination of the impeller speed and gangue mineral data sets, assuming that the drainage parameter of the system would remain constant. Figure 8-15 plots the experimental water recoveries at different cell operating conditions and the results of fitting Equation 8-3 to these experimental data. This exponential model fits the experimental data better than the Savassi model with the regression coefficient being higher in all cases.

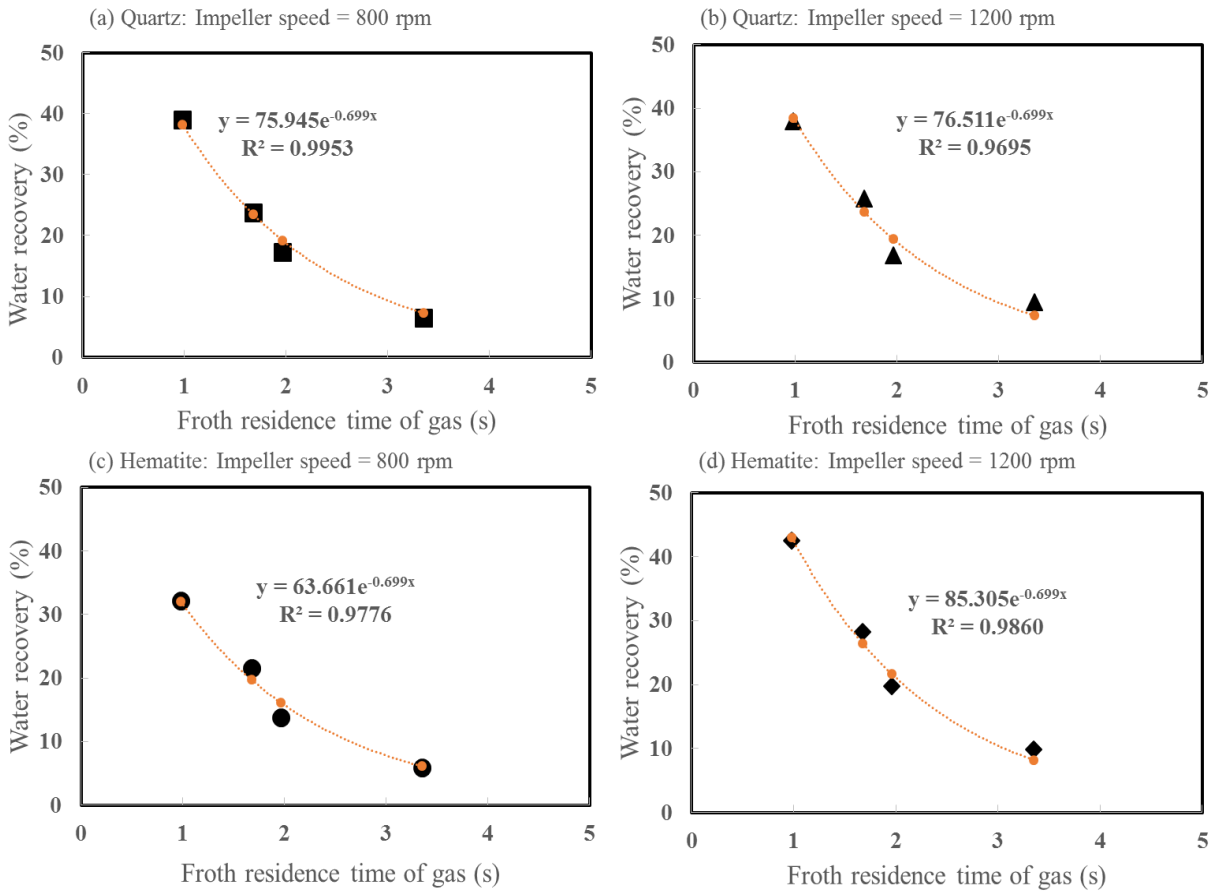

Figure 8-15 Modelling of water recovery as an exponential function of froth residence time under various operating conditions.

Fitting of the data using a different drainage parameter for each combination of variables did not result in an appreciable improvement in the degree of fit of the model. It therefore is concluded in this system that the rate of drainage with froth residence time remains relatively constant.

The results of the model fitting suggest, however, that recovery of water from the pulp to the froth did not remain constant. In the high-density gangue mineral system (hematite), the amount of water recovered from the pulp to the froth was much greater at the higher impeller speed than at the lower speed. However, the water reporting into the froth in the low-density gangue mineral system (quartz) was relatively independent of impeller speed. The mechanism behind is not known definitively but is likely to be due to either a change in momentum at the pulp interface or a change in bubble size, which could result in a change in the amount of water transferred into the froth.

It should be noted that the water flow from the pulp into the froth is also likely to be affected by gas flowrate which was ignored in the above analysis with the gas flowrate only affecting froth residence time. There was insufficient experimental data to uncover these possible underlying relationships in this study. More data would be required at additional froth depths.

In this study, water recovery was empirically modelled, acknowledging that two distinctively different mechanisms simultaneously play a role. The results from this experimental program demonstrate that both drainage and the water flow into the froth change as operating conditions in the flotation cell are varied and these changes impact on the overall water recovery produced during flotation. More research work is required to unravel and determine how cell operating conditions affect these two mechanisms with a view to ultimately developing a robust method of predicting water recovery in flotation simulation models.

8.4 Summary

Water recovery varied significantly as batch flotation test conditions were changed. Statistically, water recovery was significantly affected by (1) froth height, (2) gas flowrate, (3) impeller speed, (4) the interaction between impeller speed and particle density and (5) the interaction between gas flowrate and froth height (in the order of reducing effects). Froth height and gas flowrate were the primary drivers as they affected the froth residence time in the froth and therefore the time available for drainage. There was an increase in the amount of water entering the froth phase when the impeller speed increased, especially when the gangue had a high specific gravity. For conditions

under which the water flow to the froth from the pulp was kept relatively constant (i.e. constant impeller speed and particle density), water recovery could be modelled effectively as an exponential function of froth residence time. The results of this work highlight the need, when modelling water recovery, to consider both the water flow into the froth and water drainage in the froth as separate mechanisms.

Chapter 9 Conclusions and future work

9.1 Introduction

In this thesis, a series of batch flotation tests were performed using a 3.5L flotation cell according to a two-level factorial experimental design. This was followed by gangue only entrainment tests. The aim was to develop a comprehensive understanding of the variables which significantly affect entrainment and the mechanisms involved. The research has been fruitful, providing flotation operators with information and understanding they can use to minimise entrainment in their flotation operations as well as providing relationships that could be used to develop an improved predictive entrainment model. This chapter summarises the main findings from these investigations and presents the insights gained.

9.2 Conclusions

9.2.1 Identification of operational variables and particle properties that affect the degree of entrainment significantly

There was no correlation observed between the measured water recovery and the degree of entrainment, which indicated that the key drivers of water recovery were not the same as those that affected the degree of entrainment.

In addition to particle size, particle density and the interaction between gas flowrate and froth height were found to have a significant effect on the degree of entrainment. The degree of entrainment also changed significantly with flotation time throughout each experiment. Though the reason for this change remains unsolved, additional testing suggested that frother concentration could significantly influence the degree of entrainment.

Interestingly, froth residence time showed no relationship with the degree of entrainment. It indicates that the drainage of gangue mineral particles relative to the water in the froth phase is not time dependent. Froth residence time did, however, have a significant effect on entrainment because of its significant effect on water recovery. This finding implies that the classification of entrained gangue solids in a froth may mainly occur at the pulp/froth interface region.

9.2.2 Identification of mechanisms by which the significant operational variables and particle properties affect the degree of entrainment

As mentioned above, it was identified that the degree of entrainment was significantly affected by the particle density, frother type and concentration, gas flowrate and froth height, apart from particle size. Experimental testing suggests that these effects were caused by these variables changing the forces acting on the particles at the pulp/froth interface, the apparent immersed weight of the particles and the drag force acting on the solid due to the upward flow of water in this region.

A power law relationship was found between the degree of entrainment and particle settling rate, suggesting that the balance of drag force and particle apparent immersed weight in the water determines the extent to which particles overcome the effects of gravity and report to the concentrate due to entrainment. Particle size and particle density are two variables that affect the drag force and apparent immersed weight of gangue solids, resulting in changes in the particle settling rate which alters the degree of entrainment. This explains why the degree of entrainment decreases with increases in the particle size and/or particle density.

A predictive empirical equation was proposed to relate the degree of entrainment to particle settling rate (see Equation 5-3). The empirical equation provided a reasonable fit to both laboratory and industrial experimental data sets, which indicates that it can be used for predicting the degree of entrainment for minerals with different particle densities.

$$ENT_i = \frac{1}{1 + p \cdot \left((\rho_s - \rho_w) \cdot d_i^2 \right)^q} \qquad 5\text{-}3$$

where p (in $(g \cdot cm^{-3} \cdot \mu m)^{-q}$) and q are constants which are specific to a particular flotation environment.

The effect of frother on the degree of entrainment was found to be due to its effects on the liquid velocity at the pulp/froth interface. The measured liquid velocity at the pulp/froth interface in tests performed using different types of frother at different concentrations was found to be linearly correlated with degree of entrainment, suggesting that the observed changes in the degree of entrainment was due to changes in the drag force acting upon the particles at the interface region. When frother concentration was increased, bubble size was shown to decrease significantly at the pulp/froth interface, which resulted in an increase in the gas holdup (i.e. a decrease in the liquid

holdup) and thus less volume for water to pass through. The water superficial flowrate was also found to increase. These changes produced a higher liquid velocity through the Plateau borders at the interface, and therefore increased the drag force acting on the solids. Therefore the amount of particles that could overcome the downward gravitational force to be recovered by the entrainment mechanism to the final concentrate increased, and the degree of entrainment increased. The results suggest that the liquid velocity at the interface is a key factor which affects the degree of entrainment.

This same mechanism was found to be responsible for the effect of the gas flowrate and froth height on the degree of entrainment. Gas flowrate and froth height was found to change the liquid velocity at the interface and again a linear correlation was observed between liquid velocity and the measured degree of entrainment. When gas flowrate and froth height was varied, bubble size was shown to be similar at the pulp/froth interface but water content between adjacent bubbles changed significantly. This resulted in a variation in liquid holdup and water superficial flowrate, which changed liquid velocity through the Plateau borders at the interface. Thus, the drag force acting on the solid particles was altered, resulting in different degree of entrainment values. This mechanism is further substantiated by the fact that the relationship between liquid velocity and the degree of entrainment was independent of whether gas rate, froth height, type of frother or frother concentration was varied in the experiments.

These findings suggest that the balance of the apparent immersed weight of solid particles and drag force at the pulp/froth interface determines the degree of solid rejection at the pulp/froth interface region and therefore the solid concentration in the water in the froth, as shown in Figure 9-1.

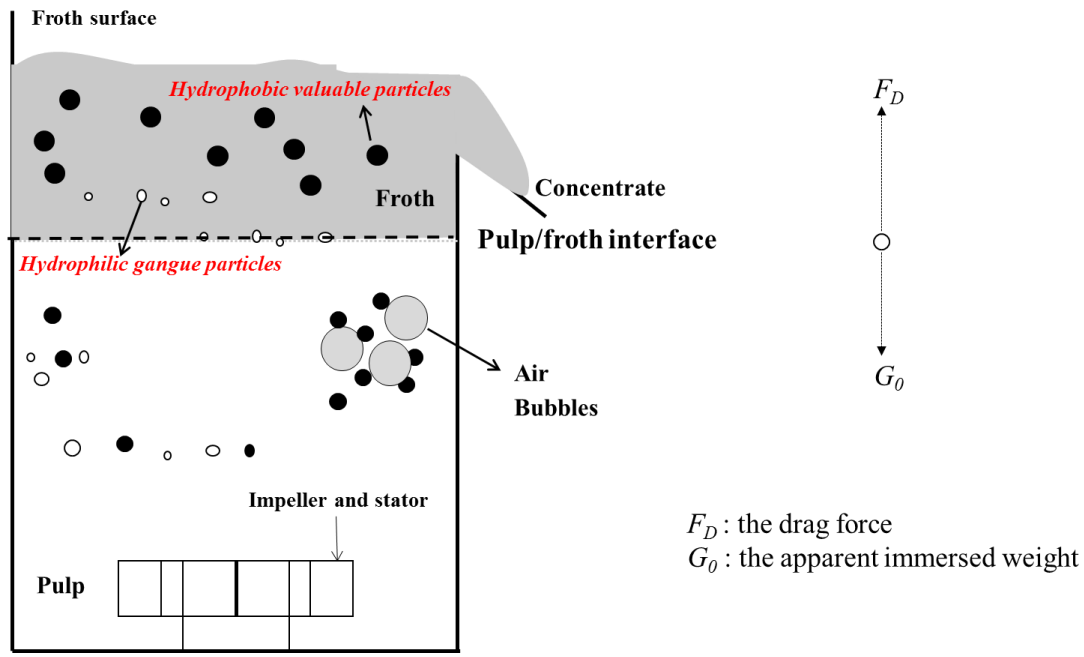

Figure 9-1 Schematic of forces acting on a solid particle at the pulp/froth interface.

This mechanism does not explain the high degree of entrainment observed when deep froths and low gas flowrates were used. It is speculated that this change is a result of entrapment occurring when the froth becomes "dry". This suggests that in some cases there is a region in the upper portion of froth where the solid concentration increases with froth height due to the solid particles being entrapped in the bubbles whilst water continues to drain.

9.2.3 Identification of operational variables and particle properties that significantly affect water recovery

Statistically, in the batch flotation factorial experiments, water recovery was significantly affected by (1) froth height, (2) gas flowrate, (3) impeller speed, (4) the interaction between impeller speed and particle density and (5) the interaction between gas flowrate and froth height (in the order of reducing effects). This verifies that the variables that affect water recovery do not necessarily affect the degree of entrainment.

Froth height and gas flowrate were the primary drivers as they affected the froth residence time in the froth and therefore the time available for drainage. Generally, the recovery of gangue minerals by entrainment does decrease with an increase in froth residence time, but this is because of a drop in water recovery instead of a significant change in the degree of entrainment.

There was an increase in the amount of water entering the froth phase when the impeller speed increased, especially when the gangue had a high specific gravity. For conditions under which the water flow to the froth from the pulp was kept relatively constant (i.e. constant impeller speed and particle density), water recovery could be modelled effectively as an exponential function of froth residence time:

$$R_w = R_w^{pulp} \cdot \exp(-b \cdot \lambda_{air}) \qquad 8\text{-}3$$

where R_w^{pulp} is the water recovery entering the froth from the pulp, and b is the rate of water drainage out of the froth to the pulp.

The results of this work highlight the need, when modelling water recovery, to consider both the water flow into the froth and water drainage in the froth as separate mechanisms.

9.3 Future work

9.3.1 Extension of experimental conditions

Based on the findings of this thesis, rheology, particle shape and solids percentage should also significantly affect entrainment because they will affect the magnitude of the drag force and the apparent immersed weight of the particles. Further work is recommended to investigate the extent of these effects on entrainment.

More work is also required to identify the mechanism by which the degree of entrainment changes during the time of a flotation test. It is hypothesised that this is due to a combination of factors changing either the apparent immersed weight and/or the drag force acting on particles at the interface, the key drivers identified in this study which affect the degree of entrainment. In addition, in the presence of valuable minerals, other mechanisms like “entrapment” may be involved. “True flotation” of the gangue contained in the valuable minerals is also possible when the gangue is not fully liberated. These would potentially overestimate the degree of entrainment. This, however, need to be studied in future testwork.

9.3.2 Modelling of entrainment

The degree of entrainment was found to be related to both the predicted solid settling velocity and the measured liquid velocity at the pulp/froth interface in this work, but these mechanisms have not yet been combined together to develop a more sophisticated model for the degree of entrainment. This should be the focus of future work.

In this study, the effect of particle size and density was considered and incorporated in the empirical equation for the degree of entrainment (i.e. Equation 5-3). This effect is a consequence of the degree of entrainment being affected by drag force and apparent immersed weight. It is likely that the constant parameters in Equation 5-3 can be related to the liquid velocity at the interface. Future work should involve investigating whether relationships between the parameters in the equation and the liquid velocity at the interface can be developed. Furthermore, the net liquid velocity at the interface region will be related to net water flow into the concentrate as well as the bubble size and coalescence rates at the interface. There is a need to be able to predict this liquid velocity as it changes with frother dosage/type, valuable mineral recovery, gas flowrate, froth height and impeller speed to enable the development of a model suitable for the estimation of the degree of entrainment in various applications.

This study also confirms that water recovery is more effectively modelled by considering the water flow into the froth and water drainage in the froth phase as independent mechanisms (see Equation 8-3). In the flotation system studied in this thesis, the drainage parameter in the froth phase remained the same despite changes in the water flow into the froth from the pulp. Future work is required to examine the degree to which this parameter responds to other flotation operational variables (e.g. particle size and hydrophobicity). To determine the water flow into the froth, more work is required to establish the mechanisms involved. In this study water flow varied with impeller speed but it is also likely to change with gas flowrate, feed solids concentration, frother type and dosage as well as the degree of bubble loading which will vary with feed grade and collector dosage. Further work is required to categorically determine the influencing variables and the mechanisms and relationships involved. When the models for the degree of entrainment and water recovery are developed, the entrainment flow into the concentrate can be effectively predicted using Equations 2-7 and 2-8 (Savassi, 1998).

It should be noted that the ultimate goal of this thesis work is to provide new understanding and assistance with plant simulation and modelling. It is well known that the scale-up of laboratory

flotation results to full plant performance is a difficult challenge. The kinetic flotation in the lab simulates the rougher and scavenger flotation in the plant where flotation conditions also vary down the bank with flotation time. The results from this work can be directly used for applications in industry. This is evidenced by the model developed in **Chapter 5**.

Even when the conditions under which the flotation cells are operated are different, the mechanism identified from the laboratory testwork should be able to guide the development of entrainment models for plant practice. For instance, drag and apparent immersed weight were identified as the key drivers in determining the degree of entrainment in the batch flotation tests in this thesis, and it is believed that they will play an important role in affecting entrainment in the full scale industrial cells as well. It is known that apparent immersed weight is determined by the weight of particles and their buoyancy, and drag force is a function of actual liquid velocity at the interface region. Both of these forces will apply, regardless of the size of the flotation cell or whether it is operating continuously or in batch operation and therefore should be the mechanisms affecting entrainment in laboratory and plant machines.

9.3.3 Entrainment control

Both the degree of entrainment and water recovery are of great importance in flotation as they influence the amount of entrained gangue mineral particles reporting to the concentrate and thus the concentrate grade. In this study, trends between the degree of entrainment and water recovery and flotation conditions have been identified and it is recommended that the developed knowledge be used to better control and minimise entrainment recovery in flotation plants. For instance, to avoid excessive water reporting to the concentrate reducing the concentrate grade, excessive impeller speeds should be avoided. The presence of a quiescent zone in the upper regions of most large scale industrial flotation cells should be maintained in order to minimise entrainment effects. Dilution of the flotation pulp to decrease the feed percent solids also should be utilised when entrainment becomes excessive. The feed size to flotation should be kept as coarse as possible and classification in grinding circuits should be optimised to prevent overgrinding and to sharpen the grind size product distribution curves to minimise the amount of ultra-fines present in the flotation feed. It should be recognised that the presence of large quantities of low density gangue in the feed may prove more problematic to treat than a high density gangue because of the greater entrainment recoveries which will result. Similarly, caution is required, especially when adjusting gas flowrate and froth height as well as frother dosage to improve true flotation, as all these variables may result

in high entrainment of gangue materials, thus making it more difficult to achieve flotation selectivity.

Optimising conditions to achieve optimal true flotation whilst minimising entrainment recovery is clearly a difficult exercise because of the competing mechanisms involved. It's the reason why effective flotation control will only be possible when models of both true flotation and entrainment are developed which respond to all the key variables which affect both processes. This should continue to be the focus of future work.

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Appendix A

Calculation of gas/liquid holdup at the pulp/froth interface

The effective bubble diameter consists of two parts: the air bubble and its surrounding water, as demonstrated in Figure 3-7 in **Chapter 3**. The average radius of the effective bubble, R, can be approximately obtained by:

$$R \approx r + d \tag{A-1}$$

The cross-sectional area of the flotation cell, A_c, can be expressed as:

$$A_c = l^2 \tag{A-2}$$

According to Eqs. (1) and (2), the number of bubbles, N_t, at the pulp/froth interface layer, can be approximated by:

$$N_t \approx \frac{l}{2R}\left(\frac{1-2R}{\sqrt{3}R}+1\right) \tag{A-3}$$

The volume of each bubble located at the interface, V_b, is givene by:

$$V_b = \frac{4}{3}\pi r^3 \tag{A-4}$$

The volume of air occupied within the interface layer, V_g, can be approximated by combining Eqs. (3) and (4):

$$V_g \approx N_t \cdot V_b \tag{A-5}$$

Thus, the gas holdup at the inferface layer, ε_g, can be approximately obtained by:

$$\varepsilon_g \approx \frac{V_g}{2R \cdot A_c} \tag{A-6}$$

The liquid holdup, ε_l, can therefore be obtained by difference:

$$\varepsilon_l = 1 - \varepsilon_g \tag{A-7}$$

Appendix B

Data for the degree of entrainment calculation in Chapter 5

B.1 Entrainment data for quartz, ilmenite and hematite described in Table 5-1.

Table B.1-1 Entrainment data for quartz from the laboratory entrainment test described in Chapter 5.

Mass of solids (g) in a given Size Interval				Mass of water (g)	
Size Interval	**Feed**	**Concentrate**	**Tailings**	**Concentrate**	**Tailings**
+150 μm	44.53	0.04	44.49	1110.90	2584.60
-150 +106 μm	135.58	0.19	135.39		
-106 +75 μm	231.24	0.67	230.57		
-75 +53 μm	226.57	1.78	224.79		
-53 +38 μm	157.41	3.41	154.00		
-38 +20 μm	262.90	14.37	248.53		
-20 μm	469.68	76.14	393.54		

Table B.1-2 Entrainment data for ilmenite from the laboratory entrainment test described in Chapter 5.

Mass of solids (g) in a given Size Interval				Mass of water (g)	
Size Interval	**Feed**	**Concentrate**	**Tailings**	**Concentrate**	**Tailings**
+150 µm	42.26	0.00	42.26	1141.05	2785.00
-150 +106 µm	95.17	0.03	95.14		
-106 +75 µm	193.35	0.04	193.31		
-75 +53 µm	202.83	0.20	202.63		
-53 +38 µm	230.80	1.08	229.72		
-38 +20 µm	262.13	4.89	257.24		
-20 µm	503.13	62.54	440.59		

Table B.1-3 Entrainment data for hematite from the laboratory entrainment test described in Chapter 5.

Mass of solids (g) in a given Size Interval				Mass of water (g)	
Size Interval	**Feed**	**Concentrate**	**Tailings**	**Concentrate**	**Tailings**
+150 μm	47.42	0.00	47.42	1071.00	2760.80
-150 +106 μm	86.51	0.01	86.50		
-106 +75 μm	176.92	0.06	176.86		
-75 +53 μm	216.99	0.24	216.75		
-53 +38 μm	194.05	0.71	193.34		
-38 +20 μm	318.99	6.18	312.81		
-20 μm	486.93	45.50	441.43		

B.2 Entrainment data for quartz and magnetite described in Table 5-2.

Table B.2-1 Entrainment data for fully-liberated quartz at the Ernest Henry copper mine described in Chapter 5.

Mass flowrate of solids (kg/h) in a given Size Interval						
Size Interval	**Feed**		**Concentrate**		**Tailings**	
	Exp	**Bal**	**Exp**	**Bal**	**Exp**	**Bal**
-75 +53 µm	4930.6	5157.8	2.6	2.5	7453.8	2155.3
-53 +38 µm	3005.7	3005.6	2.3	2.3	5428.2	3003.3
-38 +20 µm	10575.0	9872.0	9.1	8.9	7154.8	9863.1
-20 µm	9247.6	9670.4	43.9	42.8	11151.7	9627.6
Mass flowrate of water (tph)						
	Feed		**Concentrate**		**Tailings**	
	Exp	**Bal**	**Exp**	**Bal**	**Exp**	**Bal**
	2157.5	2164.9	26.4	38.3	2131.2	2126.6

Table B.2-2 Entrainment data for fully-liberated magnetite at the Ernest Henry copper mine described in Chapter 5.

Mass flowrate of solids (kg/h) in a given Size Interval						
Size Interval	**Feed**		**Concentrate**		**Tailings**	
	Exp	**Bal**	**Exp**	**Bal**	**Exp**	**Bal**
-75 +53 µm	28773.4	29766.2	14.0	13.7	40846.9	29752.5
-53 +38 µm	19432.3	20500.0	9.7	9.4	45384.7	20490.5
-38 +20 µm	51592.7	54454.0	17.8	17.4	63749.4	54436.5
-20 µm	50111.8	53111.0	97.2	94.9	83920.0	53016.2
Mass flowrate of water (tph)						
	Feed		**Concentrate**		**Tailings**	
	Exp	**Bal**	**Exp**	**Bal**	**Exp**	**Bal**
	2157.5	2164.9	26.4	38.3	2131.2	2126.6